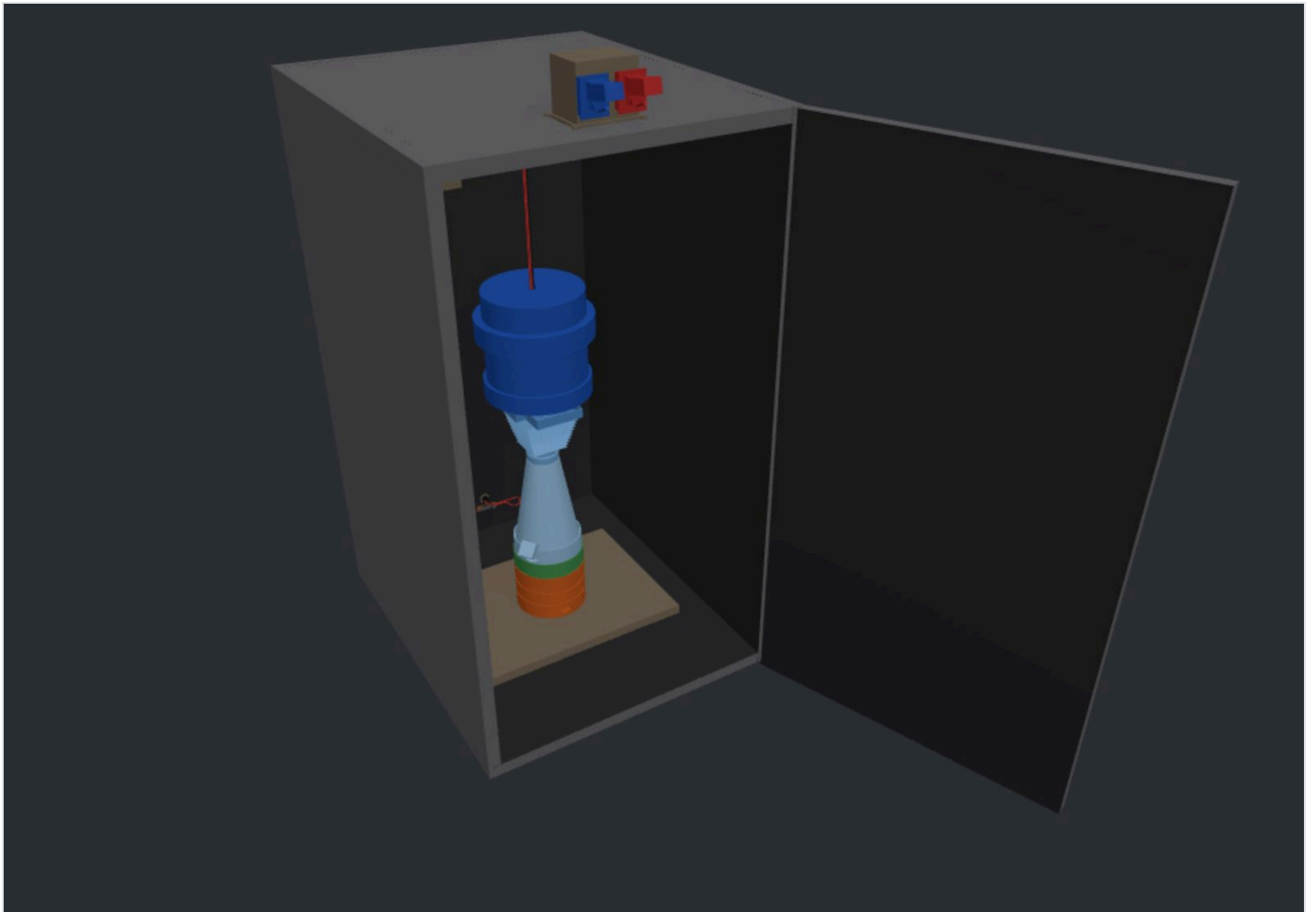


Light Source

Build guide

BETA 1.0



Before you start

Three rules. The first is the reason the others exist.

Cut the rail first — every time

The control board's off-state depends on the Pi's +5 V. If you unplug the Pi while the rail is live, **the LEDs come ON**. SW1 and the door interlock on the supply side are the only guarantee of darkness. Cut the rail before you touch anything.

Software off is not dark

A script that stopped is not a circuit that is open. After any run, cut SW1 or pull the supply.

Blue light: proxy on the bench, goggles otherwise

Bench work uses the white XP-E2 proxy. The 470 nm blue star runs only enclosed, with the interlock armed and OD6+ goggles on. 470 nm at close range damages vision without feeling bright enough to look away.

Nothing here is energised

This chapter is a tour. You are not asked to connect, power or touch anything — the safety rules arrive with the stage that needs them.

What you will be able to do

1. Identify every part of the light source on sight, and tell a purchased part from one you print.
2. Print the parts that are yours to print, in PETG, with the dimension on each that has to be right.
3. Explain why the emitter runs on a constant-current driver and what its CTRL pin does.
4. Assemble the lamphouse stack in the one order that works.
5. Prove a circuit on a breadboard before you solder it.
6. Transfer a proven circuit to perfboard: connectors, components, bridges, headers — in that order.
7. Wire a device into a machine, and dress the harness so a door can open through it.
8. Gate power with a switch and a door interlock, and prove both fail dark.
9. Bring the device up from a clean software install and read a result as pass or fail.

What you need

Check this before a build session, not during one. A missing 18 AWG spool stops a session as surely as a missing driver.

Power

PSU15 — 15 V DC, ≥ 0.75 A, centre-positive barrel	15 V is a choice, not just a legal value. The driver runs on 13–24 V, but 24 V burns about 5 W as heat where 15 V burns about 2 W for the same light. Check polarity before it goes anywhere near the jack.
--	---

Purchased parts

Cree XP-E2 on a 20 mm star	The emitter. White proxy for bench work, blue 470 nm only enclosed.
Carclo 10140 — 20 mm wide diffuse optic	Spreads the emitter into an even field rather than a spot.
DLH-1UpEH aluminium housing	Carries the star and sinks its heat.
BuckPuck 3023-D-E-350	Constant-current driver, 350 mA. Its CTRL pin is what the Pi drives.
NS-04 perfboard, 20 × 14	The control board you will build.
1 kΩ · 100 Ω $\frac{1}{2}$ W · 2N3904	The board's three components.
WAGO 221-413 (3-way) · 221-2401 (2-way)	The LED ground star and the supply inlet splice.
SW1 guarded toggle · ME-8108 door interlock	The two gates. Neither is optional.
DC barrel jack + panel bushing	The inlet, and the breakaway point.
40-pin GPIO breakout and ribbon	Pi to breadboard and board.
Lamp rod — 1/8 IP threaded hollow, with nut and washer	Standard lamp-kit hardware. Mounts the light element to the ring, and the emitter leads run up its hollow centre.

Printed parts — PETG, all six

Light ring · circuit hat · large spacer · lower ring	The lamphouse stack.
Switch box · inlet bushing	The enclosure's printed furniture. An OEM electrical bushing or heat-shrink will stand in for the bushing.
PETG, not PLA — proximity to heat, not contact with it	The emitter and its heat sink run warm. PLA softens at a temperature the lamphouse reaches, and a spacer that creeps changes the optical distance — a dimension the exposure depends on.

Cabinet and door — NOT printed	Fabricated. Plywood for the first prototype, hinged to close light-tight; plexiglass, sheet metal, a modified cabinet or even cardboard are all legitimate later.
Wire and consumables	
18 AWG stranded — power path	The gauge rule: 18 AWG power, 22 AWG logic, never below 24 AWG in a WAGO.
22 AWG stranded — logic and control	CTRL, GPIO and sense lines.
Dupont jumpers, M–F and F–F	Pi header and breadboard work.
Ferrules, heat-shrink, sleeving	Stranded wire into a WAGO wants a ferrule. Size printed pass-throughs for the SLEEVED outside diameter and deburr them.
Tools	
Soldering iron, solder, flush cutters	The star is an aluminium board: it pulls heat away from the iron, so a joint that looks fine can be cold. Give it time to wet.
Wire strippers for 18–24 AWG	
Multimeter with continuity	The only way to prove a rail is dead. “It looked right” is not a proof.
Helping hands	The star and the board are both awkward to hold.
OD6+ blue-light goggles	For any work with the 470 nm emitter.

02.0 The Path of Light

ONLINE: ACTIVITY 02.0 · 3D VISUALIZER

The device is one beam and a series of things done to it. Each step adds the next stage, in the order the light meets them.

02.0.1



The Whole Machine

Everything, assembled. A column of light with a living plate at the bottom of it: the source at the top, a lens to straighten the beam, a screen to put an image on it, a lens to project that image down, a dark chamber to carry it, and a warm stage to land it on. Every step after this one takes a stage away until only the source is left, then puts them back in the order the light meets them.

NOTE · Nothing here is energised. This chapter is a tour.

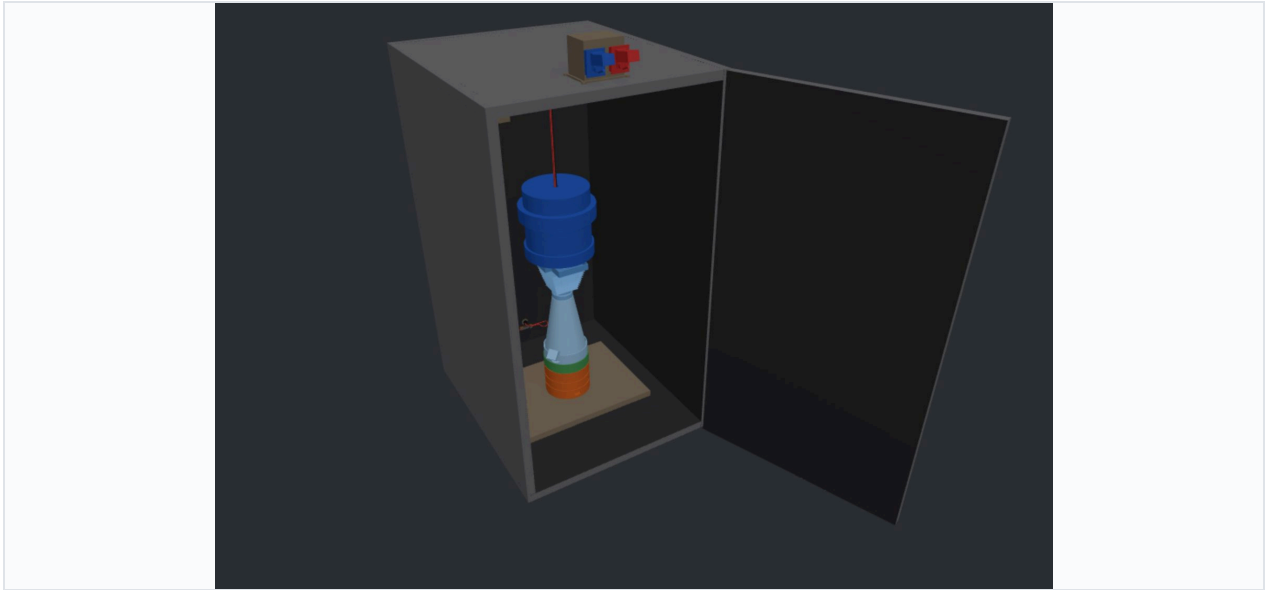


Fig 02.0-1 · 3D Visualizer ·

02.0.2



Light Source

Produce controllable, attenuable 470 nm light. This is where the photons start.

NOTE · Origin. Built in 02-light-source — Light Source.

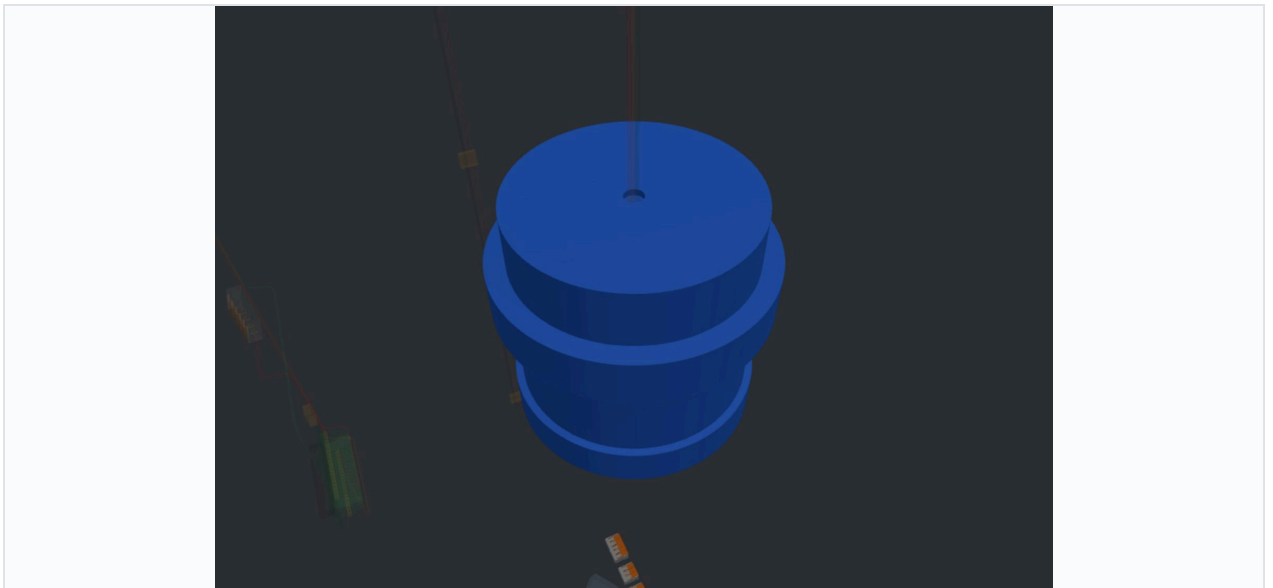


Fig 02.0-2 · 3D Visualizer ·

02.0.3



Condenser Lens

bogen-condenser

Turn a point-like emitter into an even, parallel field. Without this the projection is bright in the middle and dark at the edges, and the exposure is uncalibratable.

NOTE · Collimation. Built in 04-image-mask — Image Mask.



Fig 02.0-3 · 3D Visualizer · bogen-condenser

02.0.4



Image Mask

Impose the image on the beam. The mask sits in the enlarger's negative-carrier position and works subtractively, like a photographic negative.

NOTE · Modulation. Built in 04-image-mask — Image Mask.

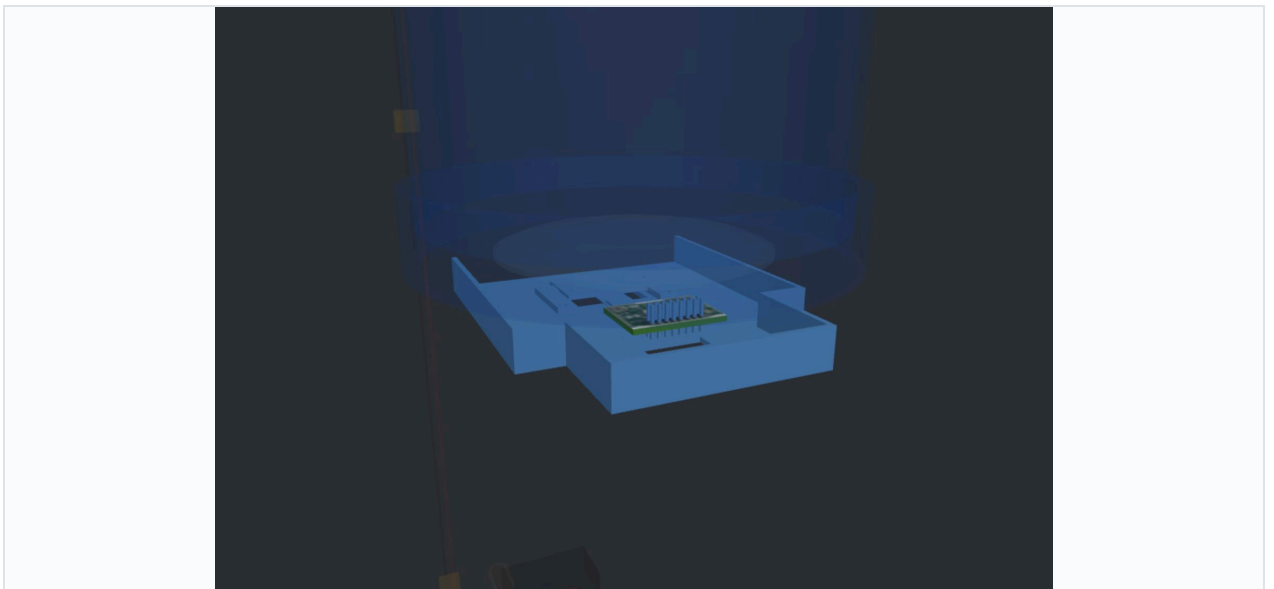


Fig 02.0-4 · 3D Visualizer ·

02.0.5



Enlarger Lens

Project the mask onto the substrate, and provide the coarsest attenuation control the platform has. The f-stop is a $(2.8/N)^2$ term in the exposure model.

NOTE · Projection + attenuation. Built in 04-image-mask — Image Mask.



Fig 02.0-5 · 3D Visualizer ·

02.0.6



Cone and Spacers

Enclose the beam between lens and plate. The cone is the light-tight envelope of the optical path; the spacers set the throw in 1 in increments.

NOTE · Dark chamber. Built in 07-plate-stage — Plate Stage.

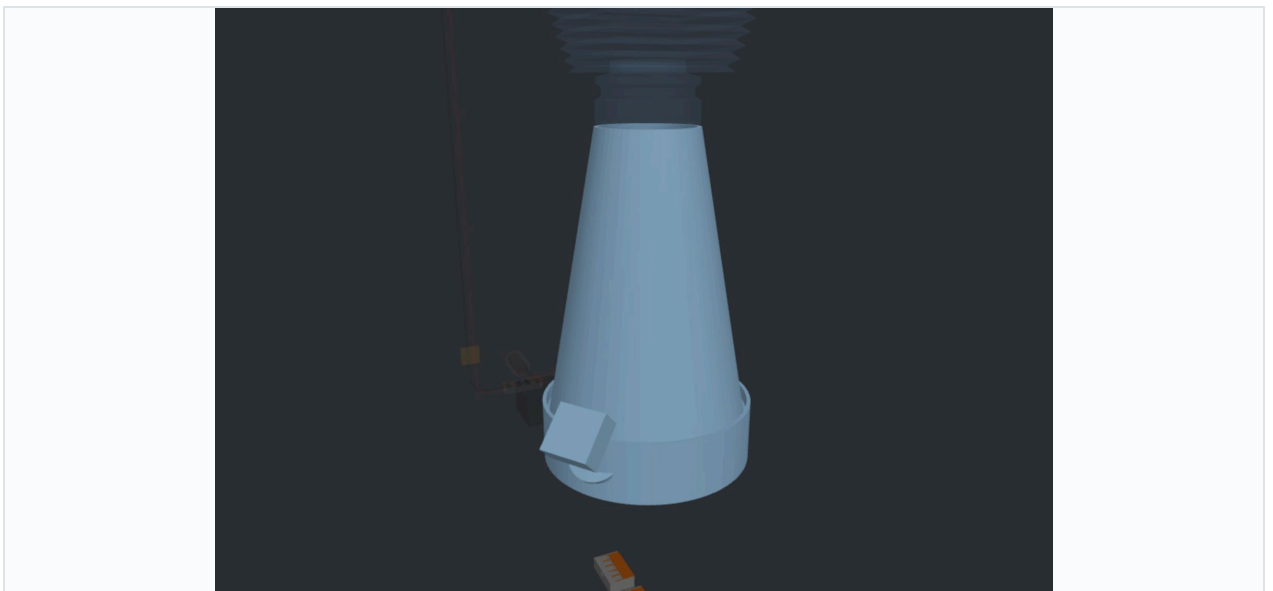


Fig 02.0-6 · 3D Visualizer ·

02.0.7



Plate Stage

Hold the substrate at the focal plane, with even airflow — and accept the spectral sensor in its place, at exactly the same height.

NOTE · The stage. Built in 07-plate-stage — Plate Stage.

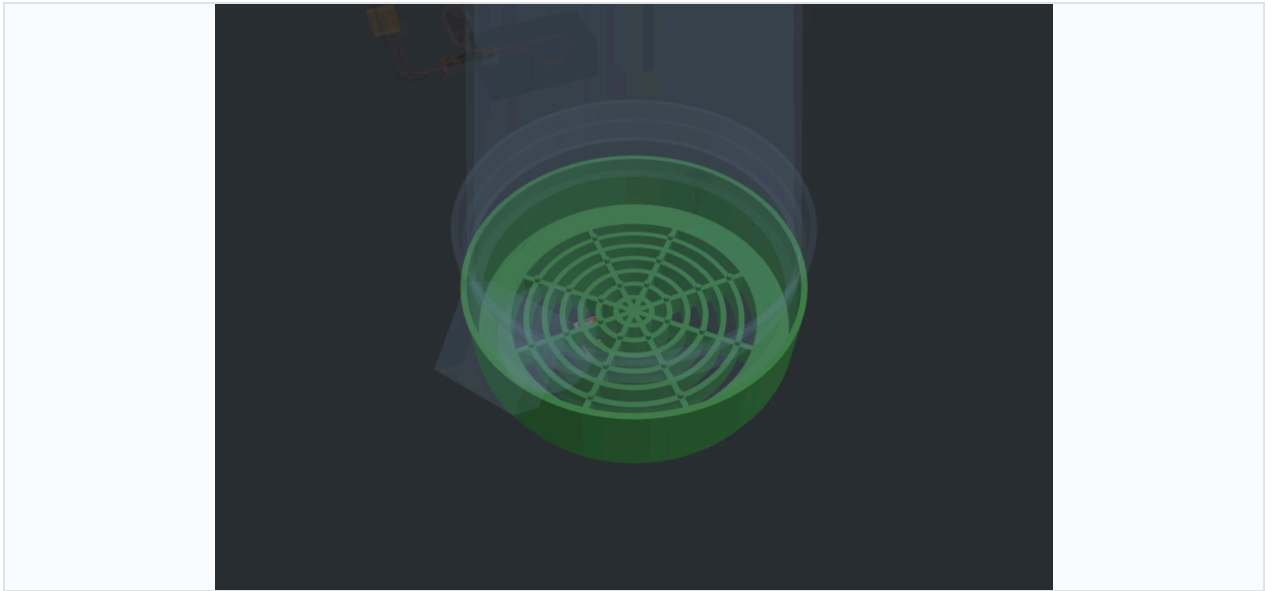


Fig 02.0-7 · 3D Visualizer ·

02.0.8



Spectral Sensor

Measure what actually arrives at the substrate. Every irradiance number in the exposure model traces back to this sensor, mounted where the plate would be.

NOTE · Built in 07-plate-stage — Plate Stage.

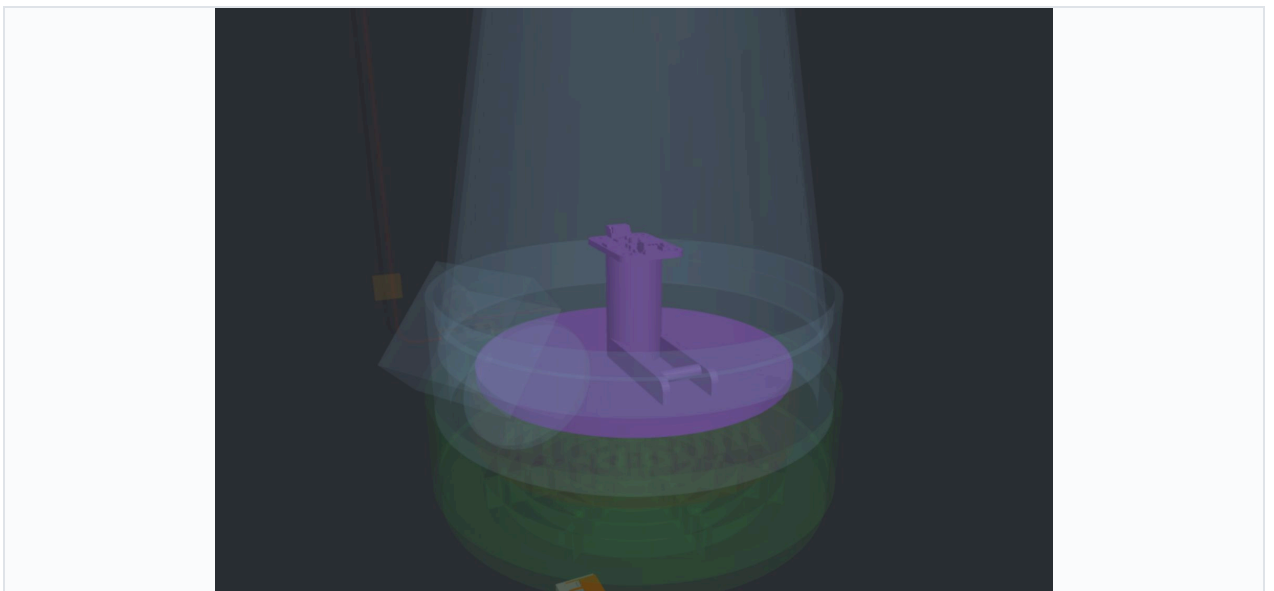


Fig 02.0-8 · 3D Visualizer ·

02.0.9



Incubation Heater

Hold the substrate at 37 °C safely, for hours, with three independent hardware protections and a closed feedback loop.

NOTE · Documented in 10-incubation-heater; no chapter yet.

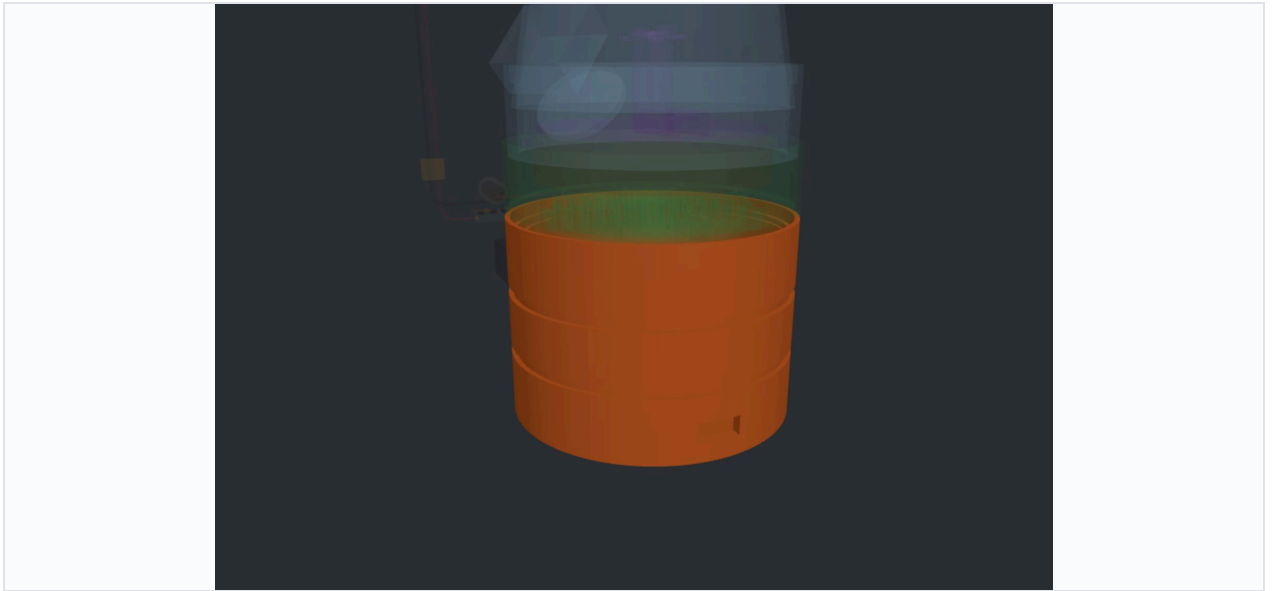


Fig 02.0-9 · 3D Visualizer ·

02.0.10



The Central Brain

The electrical layer under everything: the Pi, the GPIO breakout, the two shared stars, and the returns that must never bond to them.

NOTE · Documented in 11-circuit-and-power; no chapter yet.

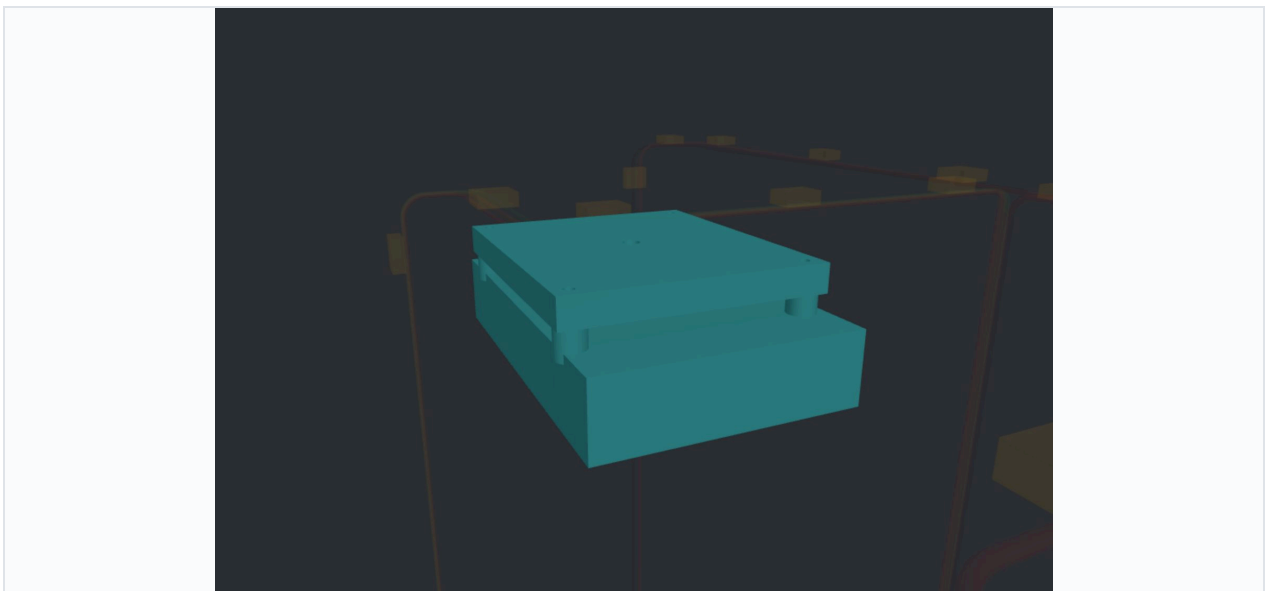


Fig 02.0-10 · 3D Visualizer ·

02.0.11



The Whole Machine, Again

Back where you started, and now every part of it has a name. The same cabinet, the same column — but you have seen what each stage is for, which ones the light passes through, which ones exist so that it can, and which chapter builds each of them.

NOTE · Next: 02 Light Source, where the photons start.

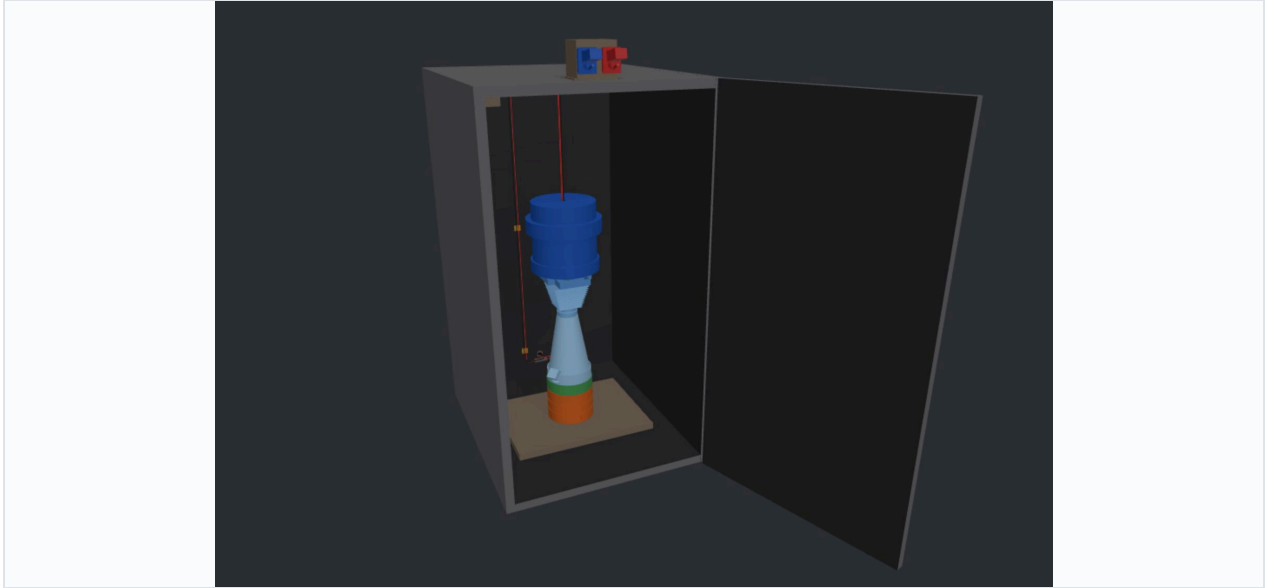


Fig 02.0-11 · 3D Visualizer ·

02.1 Parts, Tools and Consumables

ONLINE: ACTIVITY 02.1 · 3D VISUALIZER

Everything this chapter asks you to touch, laid out and recognised BEFORE a session starts. A missing driver found now costs a click; found at the bench it costs the evening. Nothing here is energised and nothing is wired -- this is identification.

02.1.1 The supply, and why 15V



psu15

A 15V DC brick, 0.75 A or better, centre-positive barrel. The driver will run anywhere from 13 to 24V, so 15 is a choice rather than the only legal value: at 24V the driver burns about 5 W as heat to make the same light 15V makes for about 2.

HAZARD · Check the polarity marking before the plug goes anywhere near the jack. Centre-positive is what this build expects.

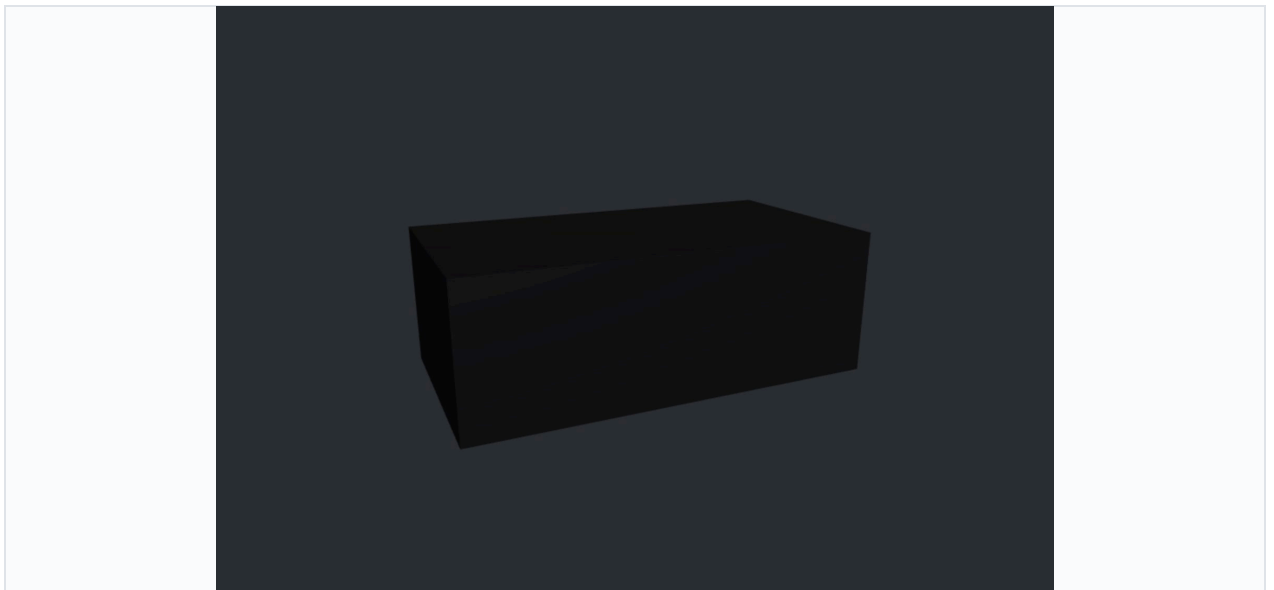


Fig 02.1-1 · 3D Visualizer · psu15

02.1.2



The inlet, and the breakaway point

barrel-jack

A DC barrel jack and its panel bushing. The jack mounts inside the bushing, so the supply plug pushes in from outside and the cabinet never has to be opened to power the build.

NOTE · This is the deliberate breakaway: a tug on the supply lead is taken by the cabinet, not by the jack's solder tails.



Fig 02.1-2 · 3D Visualizer · barrel-jack

02.1.3



SW1, the first gate

sw1-light

A guarded toggle, panel legend LIGHT. It breaks the +15 V rail upstream of everything else, which is the only reason any later step is safe to carry out.

HAZARD · The rail defaults ON. The BuckPuck's CTRL is inverted, so a Pi that is off or unplugged does not make the lamp dark -- SW1 does.

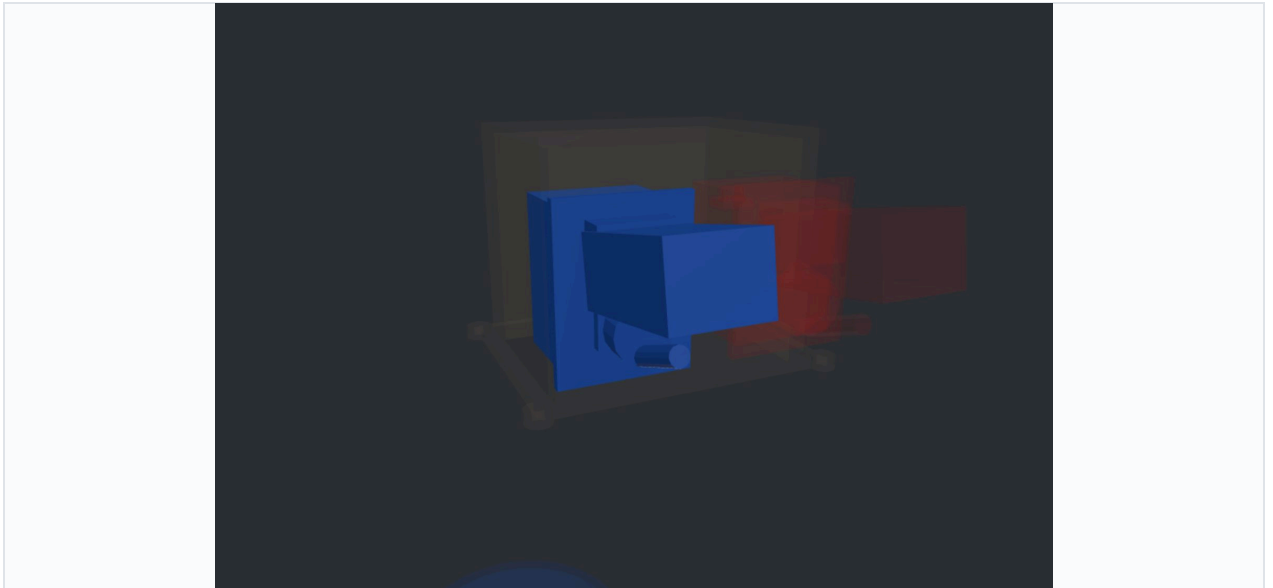


Fig 02.1-3 · 3D Visualizer · sw1-light

02.1.4



ILK1, the second gate

ilk1

An ME-8108 door interlock. Four terminals in two isolated pairs; only the NO pair, 3 and 4, is used, so the contacts close when the door is shut and open the rail the moment it is not.

HAZARD · Wired on 1 and 2 instead -- the NC pair -- it does the exact inverse and would pass any casual test. It is an electrical safety device, not a sensor.



Fig 02.1-4 · 3D Visualizer · ilk1

02.1.5



The emitter, in two versions

cree-xpe2-star

A Cree XP-E2 on a 20 mm hex star. Two parts are used across the project, one at a time: a WHITE star for bench work, where the point is to watch the optics behave, and a BLUE 470 nm star for calibration and exposure.

NOTE · Bring-up is done on the white proxy. The blue part is fitted once, at final integration, and only inside a shut enclosure.

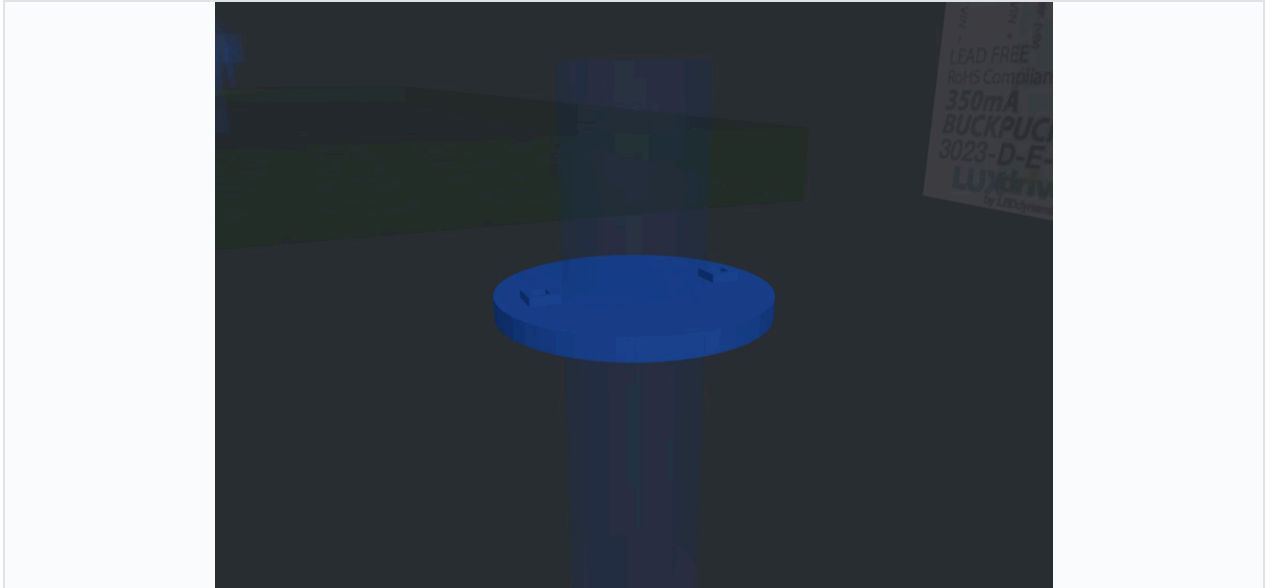


Fig 02.1-5 · 3D Visualizer · cree-xpe2-star

02.1.6



The optic is a diffuser, not a collimator

carclo-optic

A Carclo 10140, 20 mm wide diffuse. It spreads the emitter into an even field rather than narrowing it into a spot, which is the direction this column needs: the plate wants uniform illumination, and a collimated spot would expose the middle and starve the edges.

NOTE · Carclo make several 20 mm optics that look alike in the hand. The 10140 is the wide DIFFUSE one -- check the part number on the bag, not the shape.

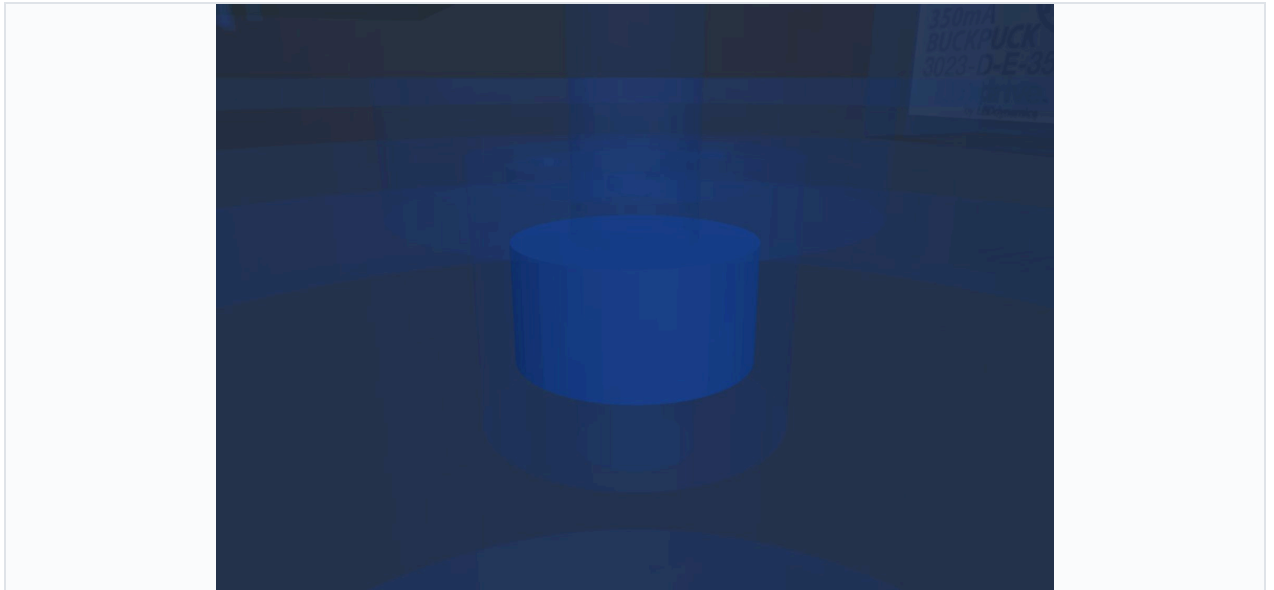


Fig 02.1-6 · 3D Visualizer · carclo-optic

02.1.7



The housing is the heat sink

dlh-1upeh

A DLH-1UpEH aluminium housing. It carries the star and takes its heat away; the 20 mm optic threads into the front of it.

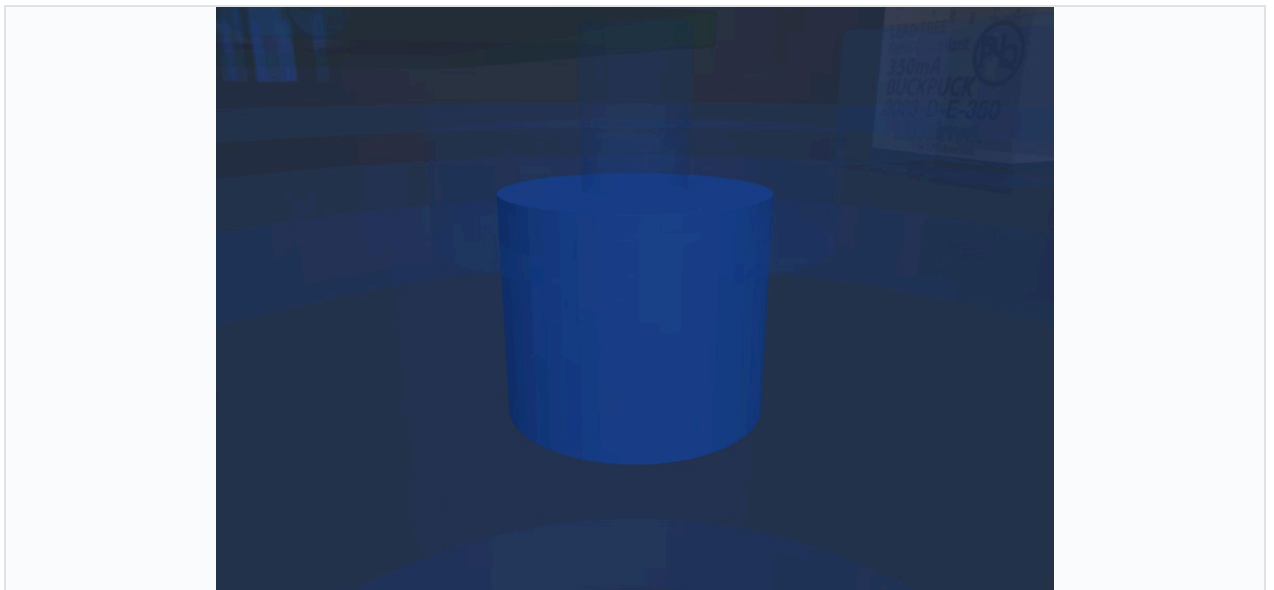


Fig 02.1-7 · 3D Visualizer · dlh-1upeh

02.1.8



The driver holds current, not voltage

buckpuck

A LuxDrive BuckPuck 3023-D-E-350: constant current at 350 mA. Its CTRL pin is the one the Pi drives, and it is the only thing in the build that decides how hard the emitter runs.



Fig 02.1-8 · 3D Visualizer · buckpuck

02.1.9



The board, and its three components

ns04-perfboard

An NS-04 perfboard, 20 × 14. Onto it go exactly three parts: a 1 k Ω resistor, a 100 Ω $\frac{1}{2}$ W resistor and a 2N3904 transistor. One sets how hard the transistor is driven; the other limits what flows through it.

NOTE · Perfboard holes are M2. Standoffs that carry this board are M2 as well, not the M3 the rest of the build defaults to.

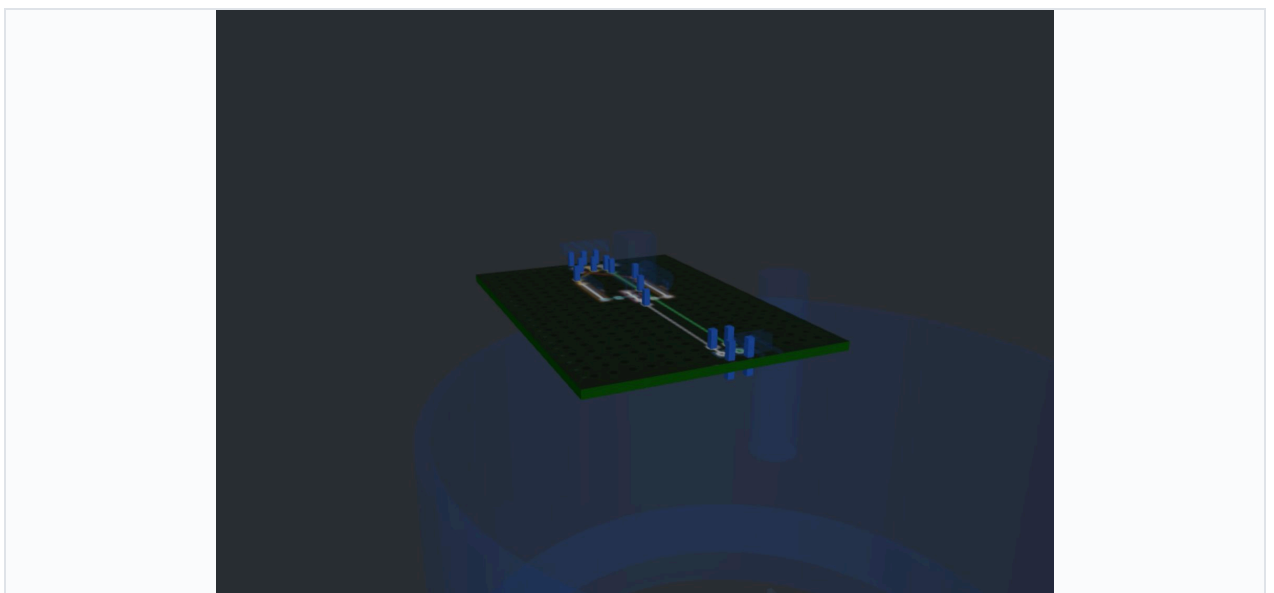


Fig 02.1-9 · 3D Visualizer · ns04-perfboard

02.1.10



The connectors that make the joins

wago-led-gnd

WAGO 221-413 for the LED ground star and 221-2401 for the supply inlet splice. Stranded wire into a WAGO wants a ferrule.

NOTE · The gauge rule for this chapter: 18 AWG on the power path, 22 AWG for logic and control, and never below 24 AWG into a WAGO.



Fig 02.1-10 · 3D Visualizer · wago-led-gnd

02.1.11



The breakout, and the mistake it invites

gpio-bridge

A 40-pin GPIO breakout and its ribbon, which is how the Pi reaches the breadboard and the board.

HAZARD · Check the ribbon is not 180° reversed before anything is harnessed to it. A flipped ribbon gives a pinout that looks entirely plausible and is wrong end to end -- it has already cost one full false bring-up here.

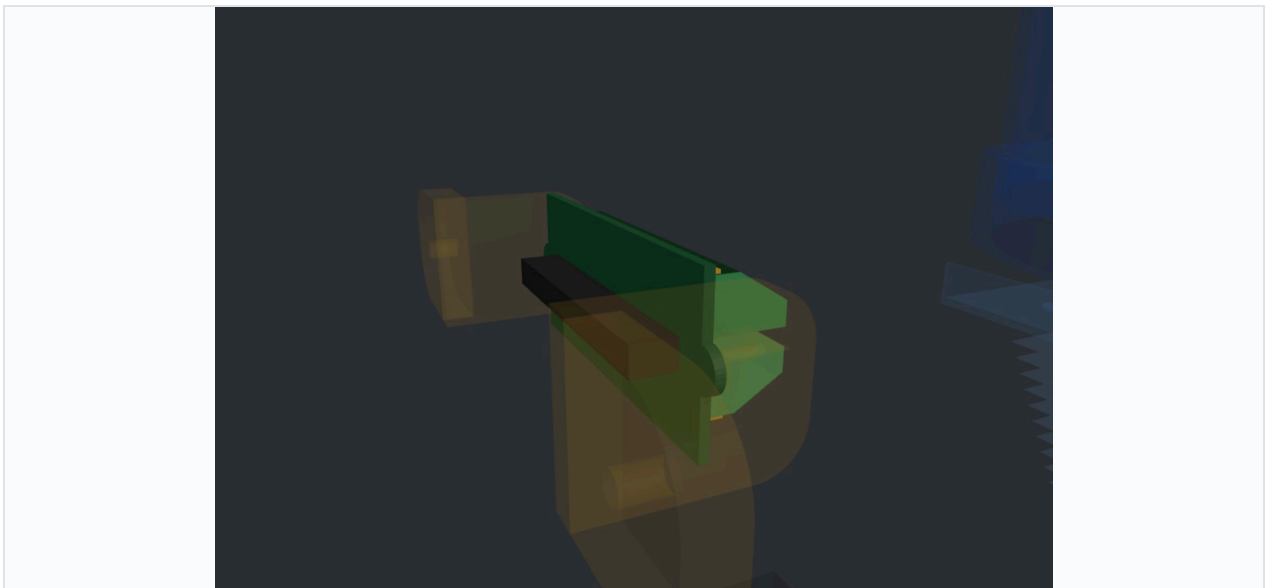


Fig 02.1-11 · 3D Visualizer · gpio-bridge

02.2 Printed Parts

ONLINE: ACTIVITY 02.2 · 3D VISUALIZER

Six parts you print, and two you do not. Each one is identified by the job it does and by the dimension or feature that has to be right for it to do that job. PETG for all six: the emitter and its housing run warm, PLA softens at a temperature the lamphouse reaches, and a spacer that creeps changes an optical distance the exposure depends on.

02.2.1 **Light ring -- seats the star on axis**



cree-lightring

Holds the single emitter star on the optical axis and on a known plane, so the die sits where the diffuser and the column below expect it.

NOTE · A circular ring with one centred LED. The ring is the carrier; the single star sits at its centre, on the axis the whole column is measured from.

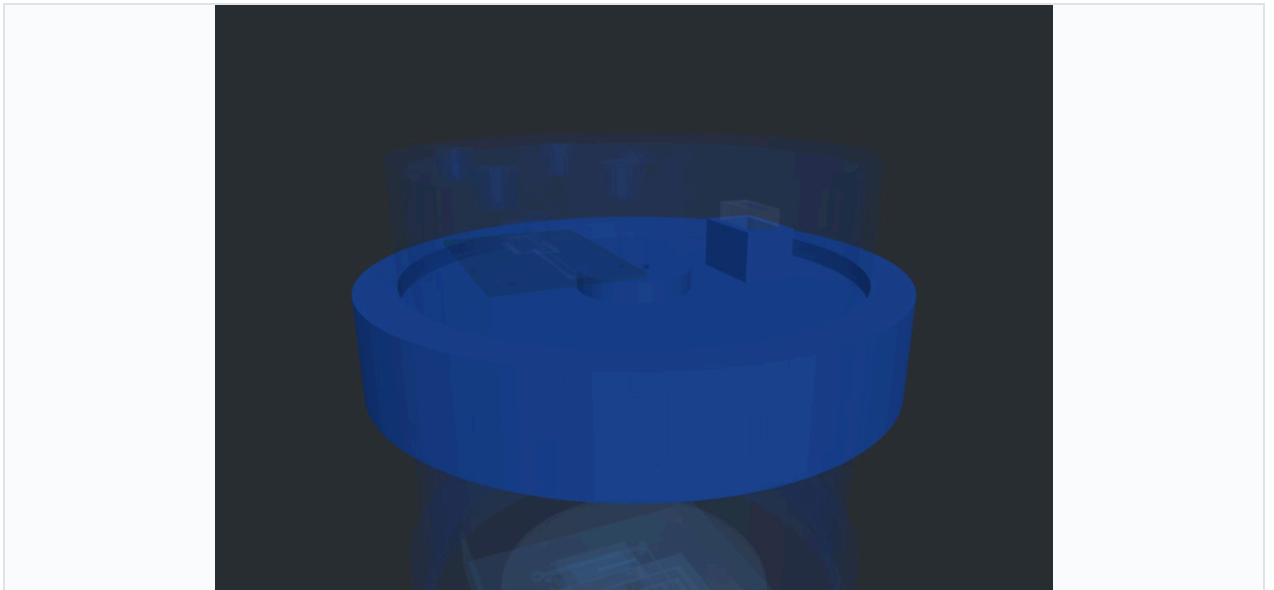


Fig 02.2-1 · 3D Visualizer · cree-lightring

02.2.2



Spacer seat -- the 1° draft is the part

lightspacer-lower-ring

Seats the spacer stack. The 1° draft is what holds it: a seat with a draft angle grips the stack above it, where a flat one lets it slip.

NOTE · Check the fit before you print a stack of them: the seat's outer diameter has to land ON the enlarger body opening, not drop through it.

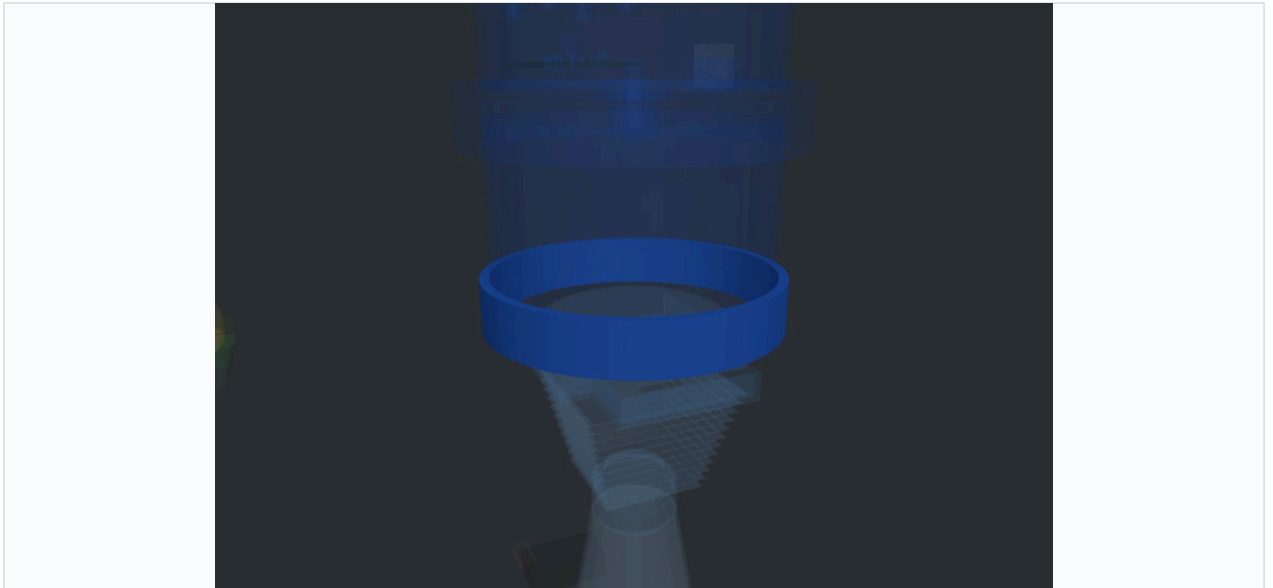


Fig 02.2-2 · 3D Visualizer · lightspacer-lower-ring

02.2.3



Light spacer -- sets the drop

lightspacer-large

Sets the distance from the emitter to the optics below. That is an OPTICAL dimension, not a mechanical one: its height is part of the exposure, so print it in PETG and print it accurately. A spacer that creeps moves the light.

NOTE · Print it in gray matte PETG. Inside a 470 nm lamphouse the inner surface is working: anything it bounces back up the column is stray light in the image.

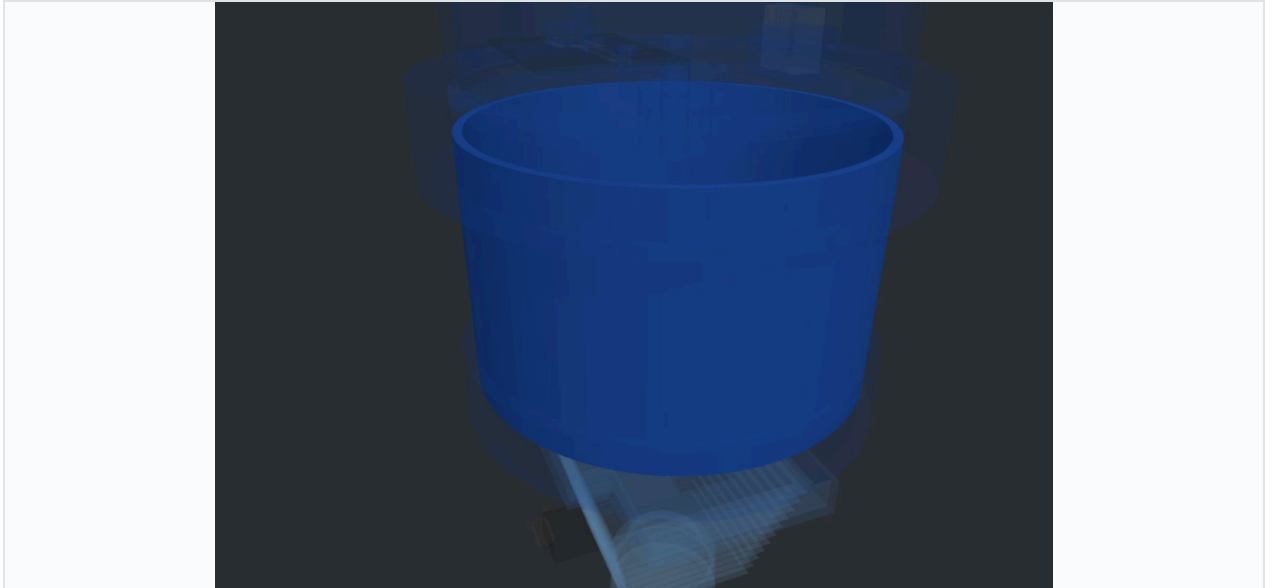


Fig 02.2-3 · 3D Visualizer · lightspacer-large

02.2.4



Circuit lid -- closes the lamphouse, carries the board

light-circuit-hat

Closes the top of the lamphouse and gives the NS-04 board its four standoffs. The four Ø4.0 bores run through the top face and are closed by the hardware that goes into them: bolting the perfboard down fills each one, so the lid is dark once the board is on it.

NOTE · Perfboard holes are M2, so the standoffs and bolts here are M2 -- not the M3 the rest of the build defaults to. Fit the board before you trust the box to be dark.

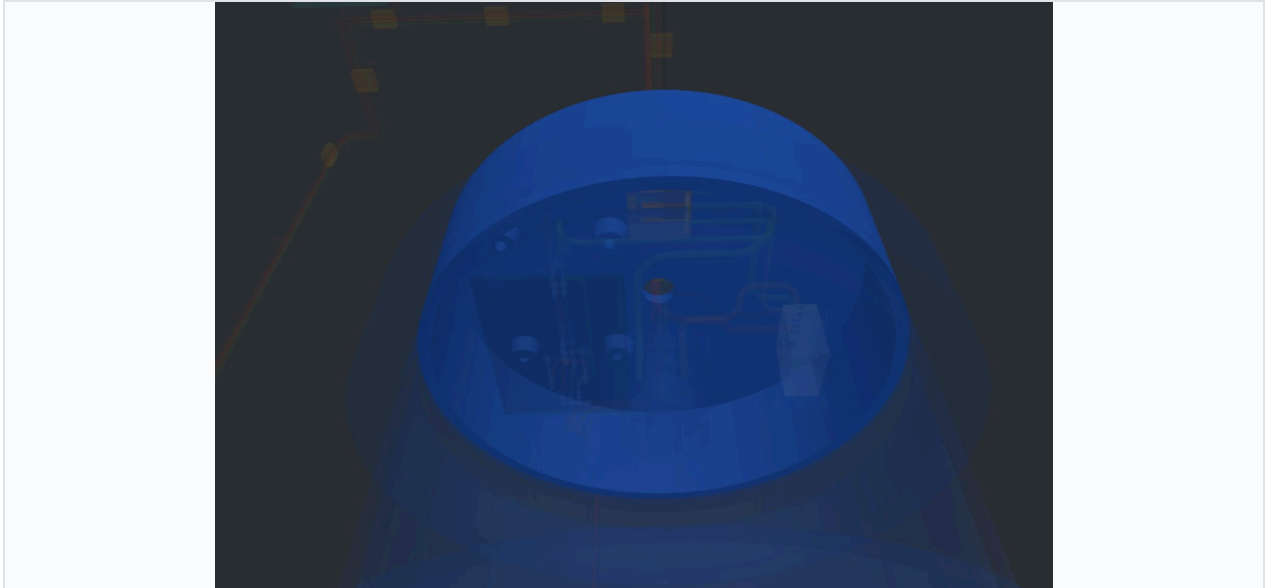


Fig 02.2-4 · 3D Visualizer · light-circuit-hat

02.2.5



Switch panel -- both rail switches, outside the lid

switchbox

Carries both rail switches on the outside of the lid, where a hand reaches them with the door shut. That is the whole point of it: the gates must be usable without opening the thing they make safe.

NOTE · Size the two switch holes Ø12.5 mm for the KN3C-101AA. It carries SW1 (LIGHT, +15 V, with ILK1) and SW2 (HEAT, +12 V, with the fuse and the KSD9700) -- two switches on two separate rails, in one panel.

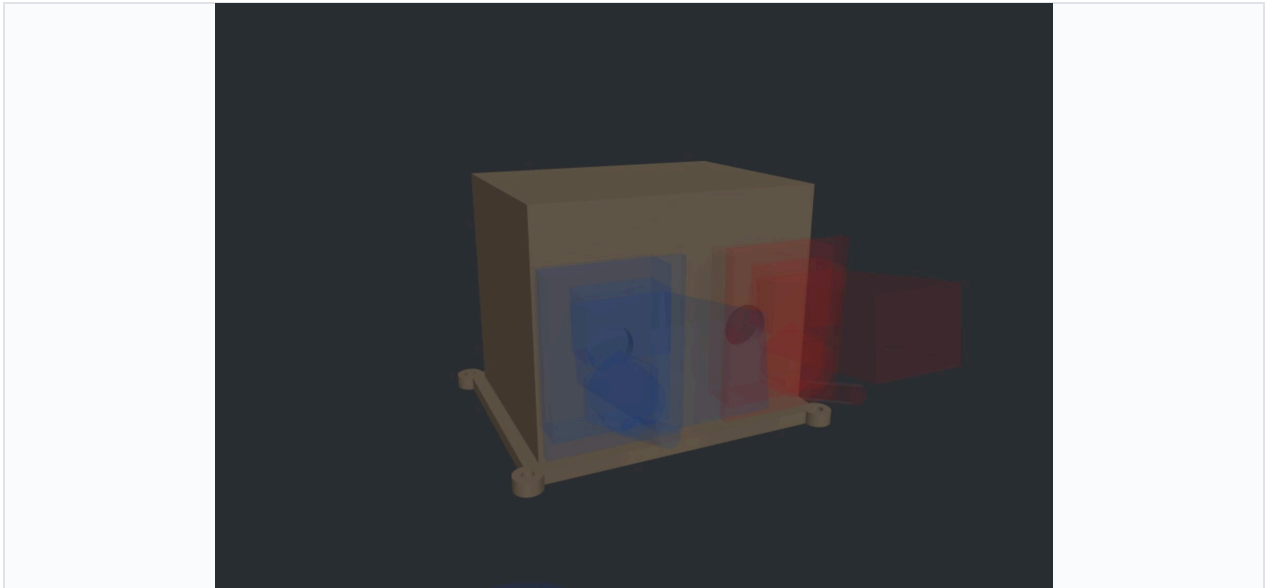


Fig 02.2-5 · 3D Visualizer · switchbox

02.2.6



Inlet bushing -- three ways to have one

cable-retainer

Fills the hole in the back wall and holds the barrel jack in it, so a pull on the supply lead is taken by the cabinet rather than by the jack's solder tails. An OEM electrical bushing sized to the hole does this job and is the quickest route: it is a standard part, it is cheap, and it is already designed.

HAZARD · CHALLENGE -- design it yourself. This is the smallest real part in the build and the best one to learn on: a bore to suit the jack body, a shoulder that lands on the wall, and some way of retaining it -- a nut, a snap, or printed-in threads. One feature at a time, each one testable by whether the jack sits square and the lead cannot be pulled out.

NOTE · RAPID ROUTE -- heat-shrink tubing. Shrunk over the lead where it passes the wall it gives you strain relief and a light seal in one, with no printing and no waiting. Good enough to prototype the whole inlet and find out where the hole should actually be.

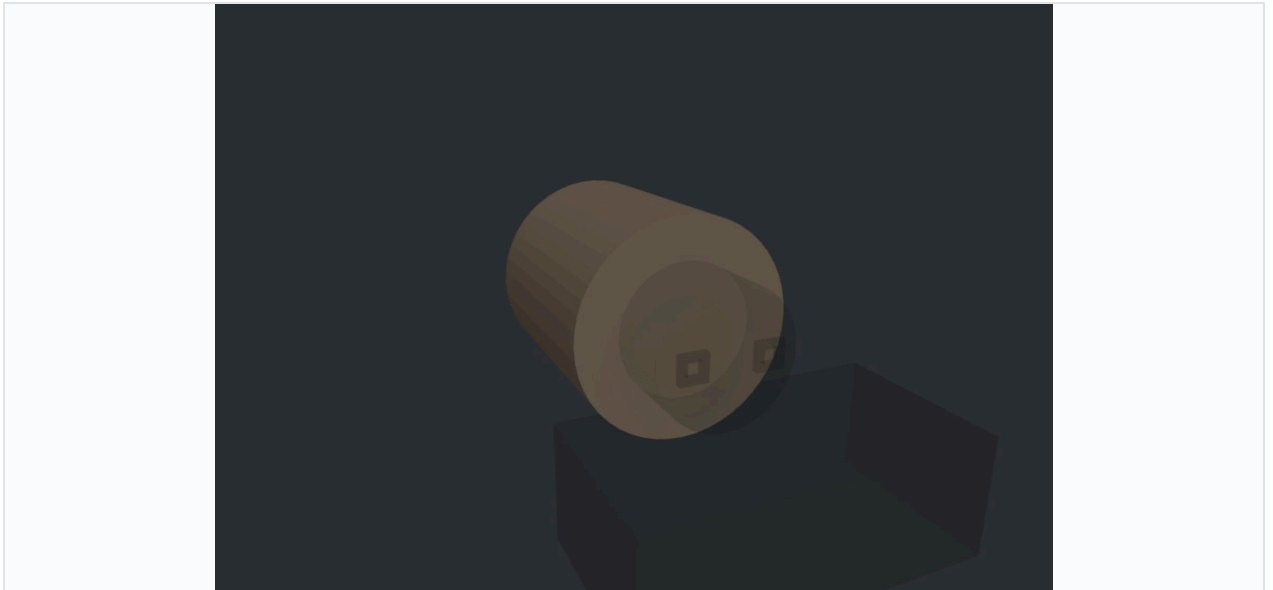


Fig 02.2-6 · 3D Visualizer · cable-retainer

02.2.7



The cabinet is made, not printed

enclosure-shell

The shell is fabricated rather than 3D printed. The first prototype is plywood throughout, which makes it easy to mount and position every component and to move one when it turns out to be in the wrong place. Later iterations have room to be cut and assembled opaque plexiglass, or a sheet-metal fabrication; an existing structural cabinet can be modified instead, and a cardboard box is a perfectly honest enclosure for an early iteration.

NOTE · The floor is the datum. Every position in this build is measured from it, so whatever you make the cabinet from, build it to the interior dimensions first and place the parts into it -- not the other way round.

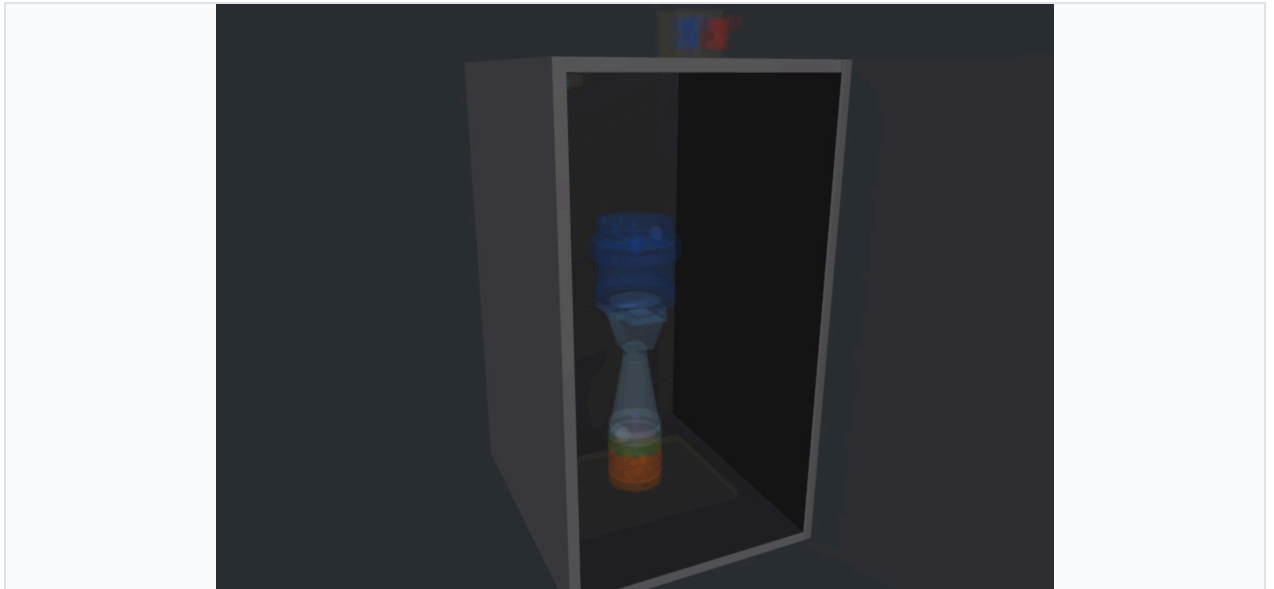


Fig 02.2-7 · 3D Visualizer · enclosure-shell

02.2.8



The door, and the seal that makes it dark

enclosure-door

Also fabricated, in plywood for the first prototype. What matters is the hinging: the door and the shell have to close to a light-tight seal, so the hinges are set to pull the door flat against the opening rather than merely hold it there.

HAZARD · The door carries ILK1. Opening it must CUT the 470 nm rail -- not be noticed by software, not be logged. That is an electrical requirement on the door, and it is why the interlock sits under the ceiling with its roller facing this panel.

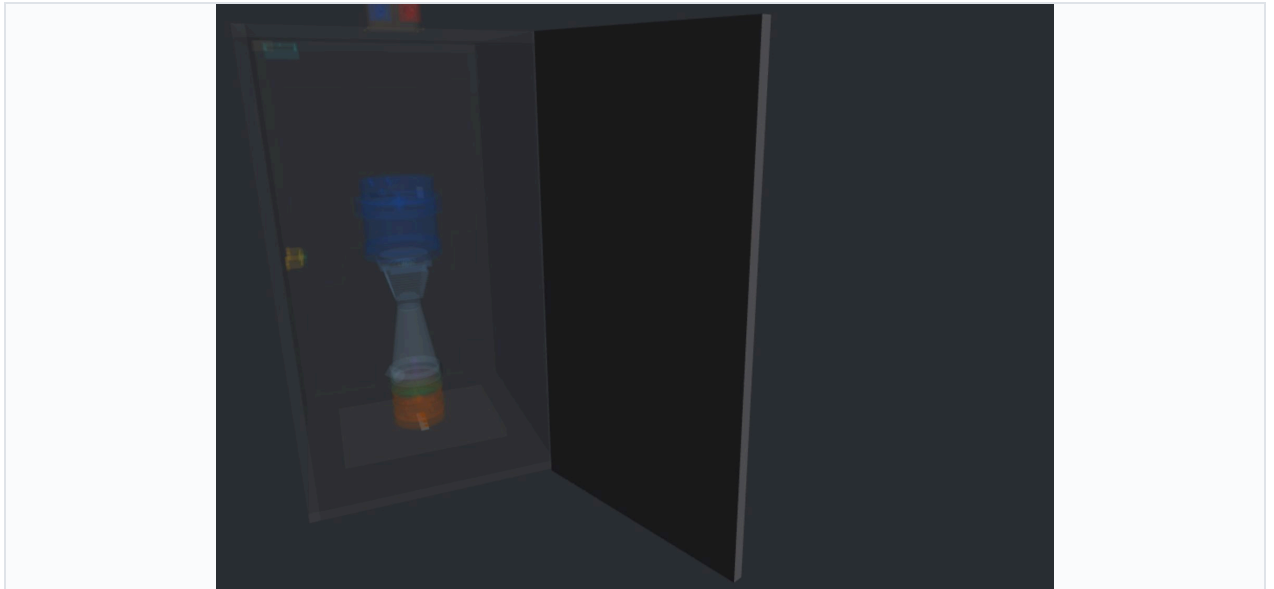


Fig 02.2-8 · 3D Visualizer · enclosure-door

02.3 Door Switch and Power Cutoff

ONLINE: ACTIVITY 02.3 · 3D VISUALIZER

SW1 on the supply side and the door interlock. The rail defaults ON, so these two are the only things that make it dark — everything after this section assumes you have used them.

02.3.1



Cut the rail before anything else

sw1-light

Open SW1 (panel legend LIGHT), shut the Pi down, then unplug the supply brick. Do it in that order.

HAZARD · The BuckPuck's CTRL is **INVERTED**: with nothing driving it, the rail is ON. A Pi that is off does not make the lamp dark — SW1 and the door interlock are the only things that do.

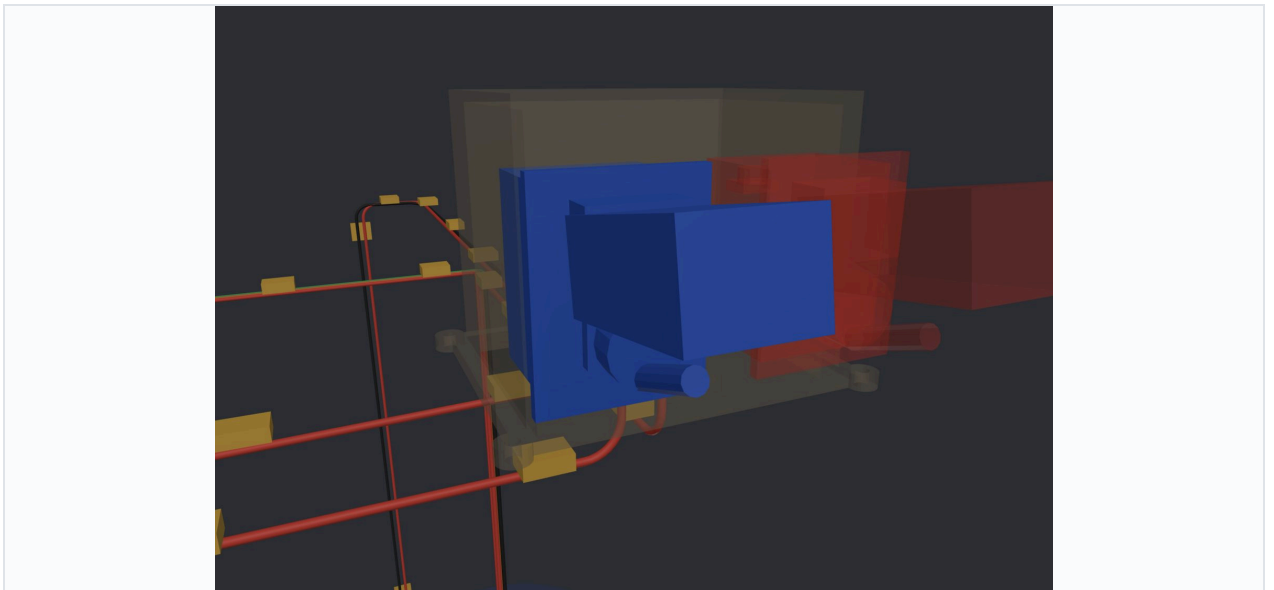


Fig 02.3-1 · 3D Visualizer · sw1-light

02.3.2



Find the second gate

ilk1

Locate the ME-8108 door interlock on the inside top of the enclosure, just under the ceiling, with its roller facing the door. Shutting the door presses that roller and engages the switch. Check that it is wired on terminals 3 and 4 — the NO pair — so the contacts close only while the door is shut.

HAZARD · Wired on 1 and 2 instead, the NC pair, it does the exact inverse: dark when open, live when closed. It would pass any casual test.

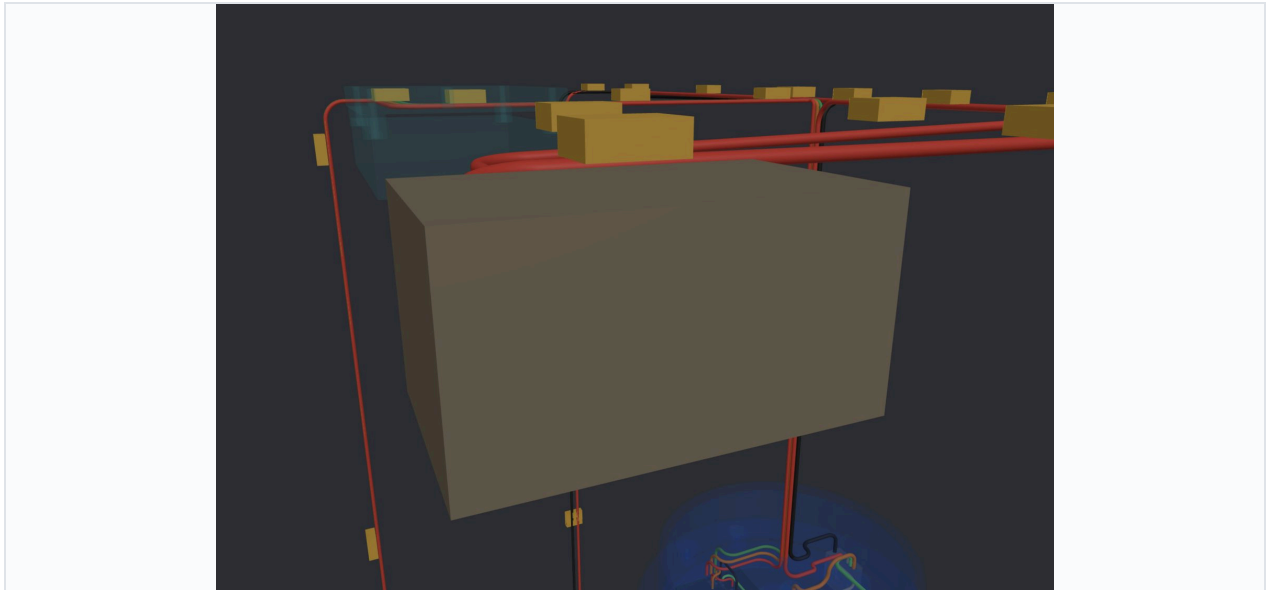


Fig 02.3-2 · 3D Visualizer · ilk1

02.4 The Light Element and Its Driver

ONLINE: ACTIVITY 02.4 · 3D VISUALIZER

What makes the light, what shapes it, and what holds it at 350 mA. No tools and nothing energised -- this is the activity where you learn what you are about to wire.

02.4.1



Find the emitter and its board

cree-xpe2-star

Identify the 20 mm hex metal-core board with a single Cree XP-E2 on it. The board spreads the die's heat into the aluminium housing and gives the emitter something to be wired and mounted by.

NOTE · Two stars are used across the project, one at a time: a WHITE part for bring-up, where the point is to see the optics work, and a BLUE 470 nm part for calibration and exposure.

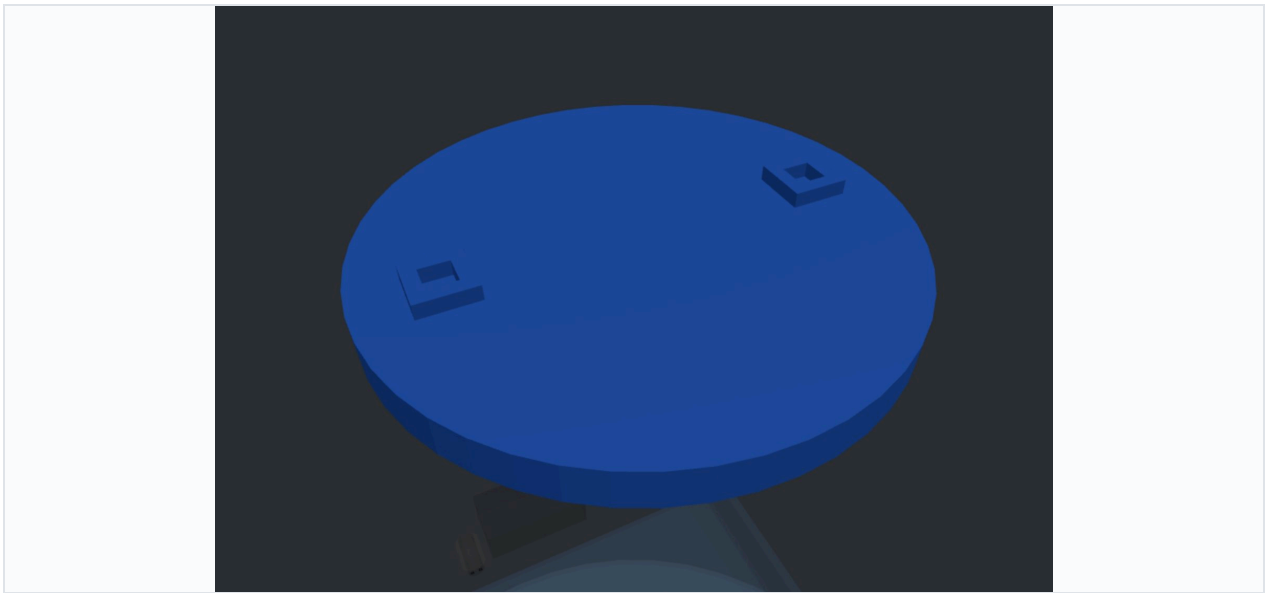


Fig 02.4-1 · 3D Visualizer · cree-xpe2-star

02.4.2



The die is the source

cree-xpe2-emitter

The 3.5 mm die sits ON the star's metal board rather than in place of it. One die, not an array.

NOTE · The wavelength is what the biology responds to, so the white part proves the light path and the blue part is the instrument.

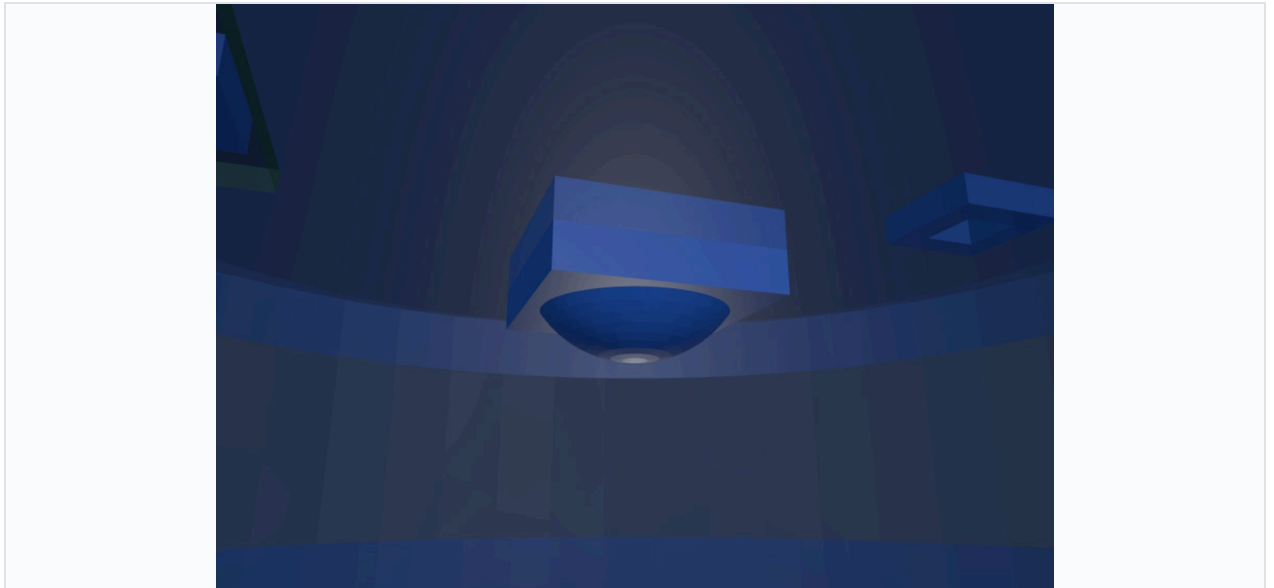


Fig 02.4-2 · 3D Visualizer · cree-xpe2-emitter

02.4.3



The housing is the heat sink

dlh-1upeh

Identify the DLH-1UPEH aluminium housing. The 20 mm optic threads into its front and the 20 mm star seats behind it, so unscrewing the housing is how you reach the die.

NOTE · It does two jobs at once: holding the star and the lens in register, and carrying heat out of a 350 mA emitter. That is why it is aluminium and not printed.

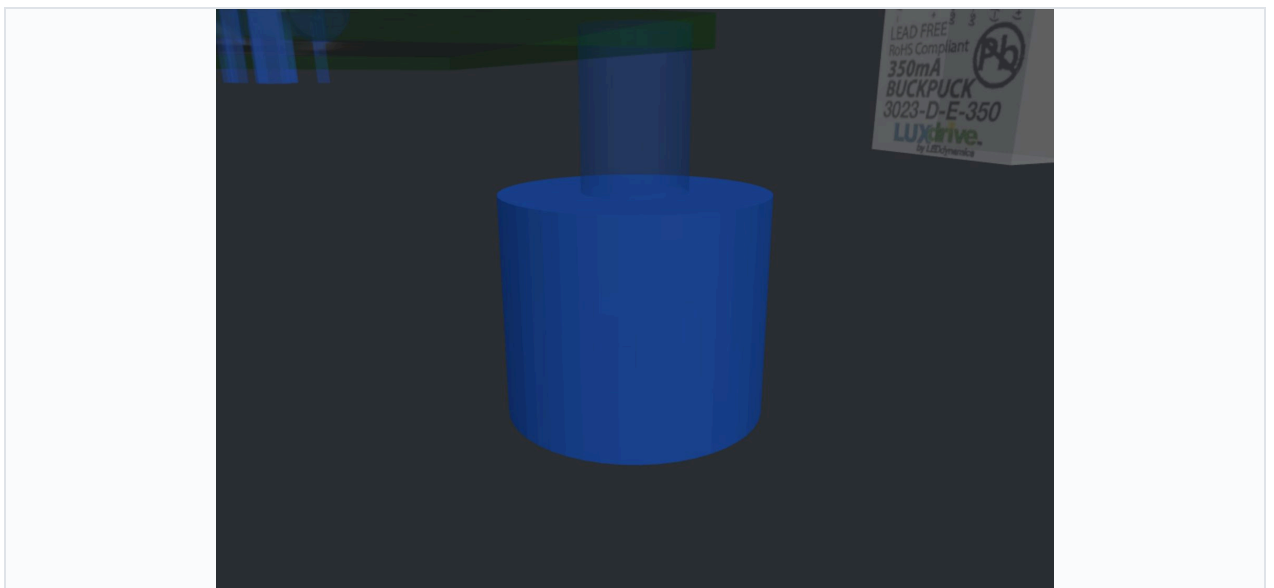


Fig 02.4-3 · 3D Visualizer · dlh-1upeh

02.4.4



The optic spreads, it does not focus

carclo-optic

Identify the Carclo 20 mm optic. It is WIDE DIFFUSE, not a collimator: it spreads the emitter's output into an even field rather than narrowing it into a beam.

NOTE · The plate wants uniform illumination across its whole area. A collimated spot would expose the middle and starve the edges.



Fig 02.4-4 · 3D Visualizer · carclo-optic

02.4.5



The ring seats the star on axis

cree-lightring

Identify the printed carrier that holds the star on the optical axis and on a known plane, so the die sits where the diffuser and the column expect it.

NOTE · A circular ring with one centred LED — the ring carries, the single star sits at its centre.

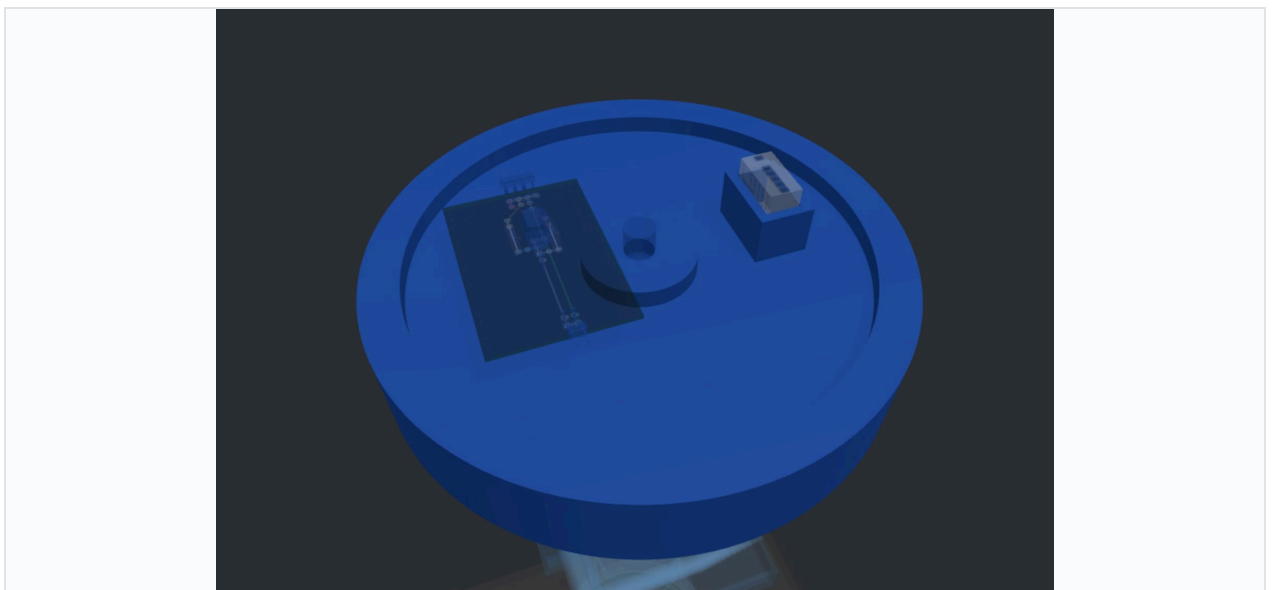


Fig 02.4-5 · 3D Visualizer · cree-lightring

02.4.6



The driver holds the current, not the voltage

buckpuck

Identify the LuxDrive BuckPuck 3023-D-E-350. It holds the emitter at 350 mA whatever the supply does, which is what makes an exposure repeatable. A resistor and a fixed supply would not: the same circuit would give a different brightness at 15 V than at 24 V.

HAZARD · Its CTRL input is INVERTED. With nothing driving CTRL the rail defaults ON — which is why the switch and the door interlock, not software, are what make the device dark.



Fig 02.4-6 · 3D Visualizer · buckpuck

02.4.7



Know its four leads before you wire them

buckpuck

The driver has four: Vin+ and Vin− to the gated supply, LED+ to the star's anode, and LED− to the ground star. A fifth, CTRL, is the grey lead the board drives.

NOTE · You will land every one of these in activity 02.8 and 02.9. Reading them now is what makes those steps short.

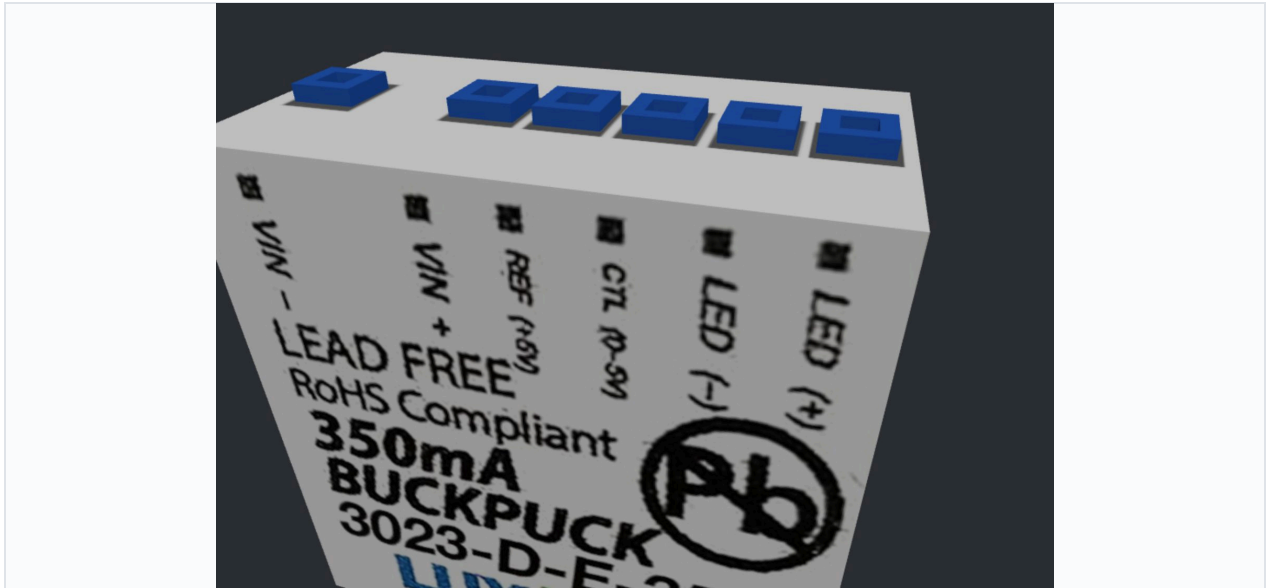


Fig 02.4-7 · 3D Visualizer · buckpuck

02.5 Breadboard the Circuit

ONLINE: ACTIVITY 02.5 · SPACEPLACER

NS-04 Cree Control — Breadboard (test before perfboard)

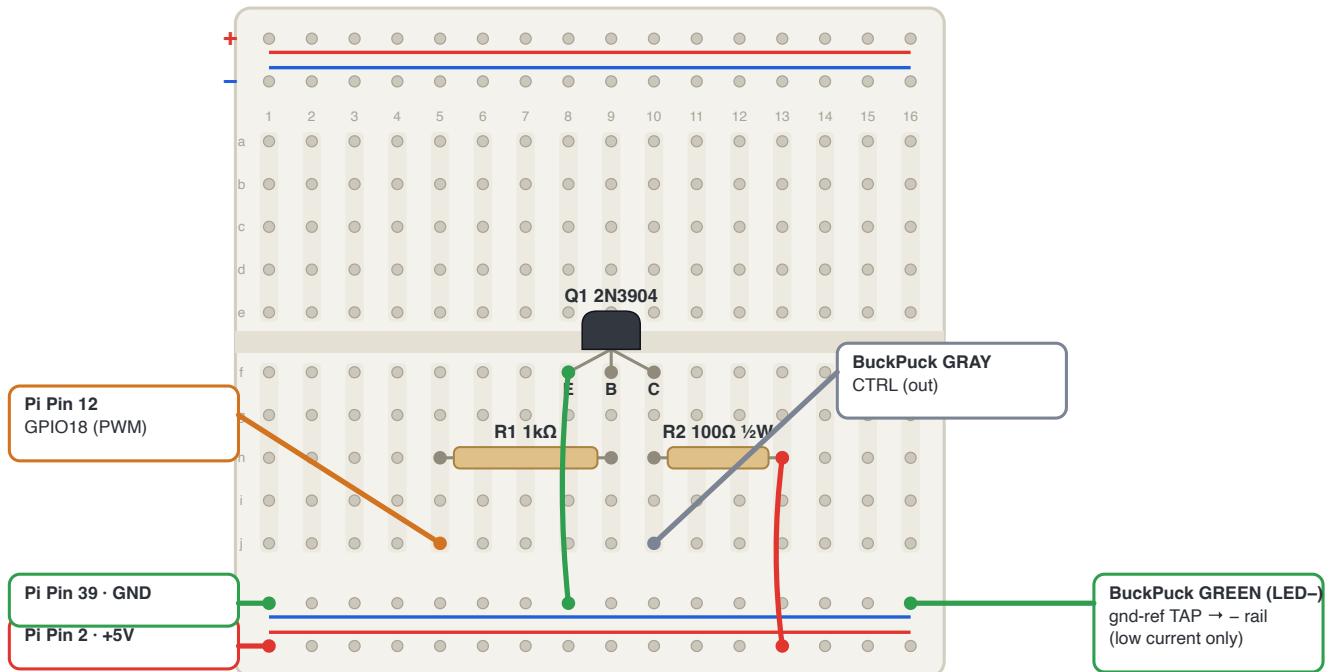


Fig 02.5-B1 · The breadboard, as it is when this activity is finished.

Off-board — the 350 mA loop

None of this touches the breadboard. The driver makes the current; the emitter uses it; the loop closes at a WAGO.

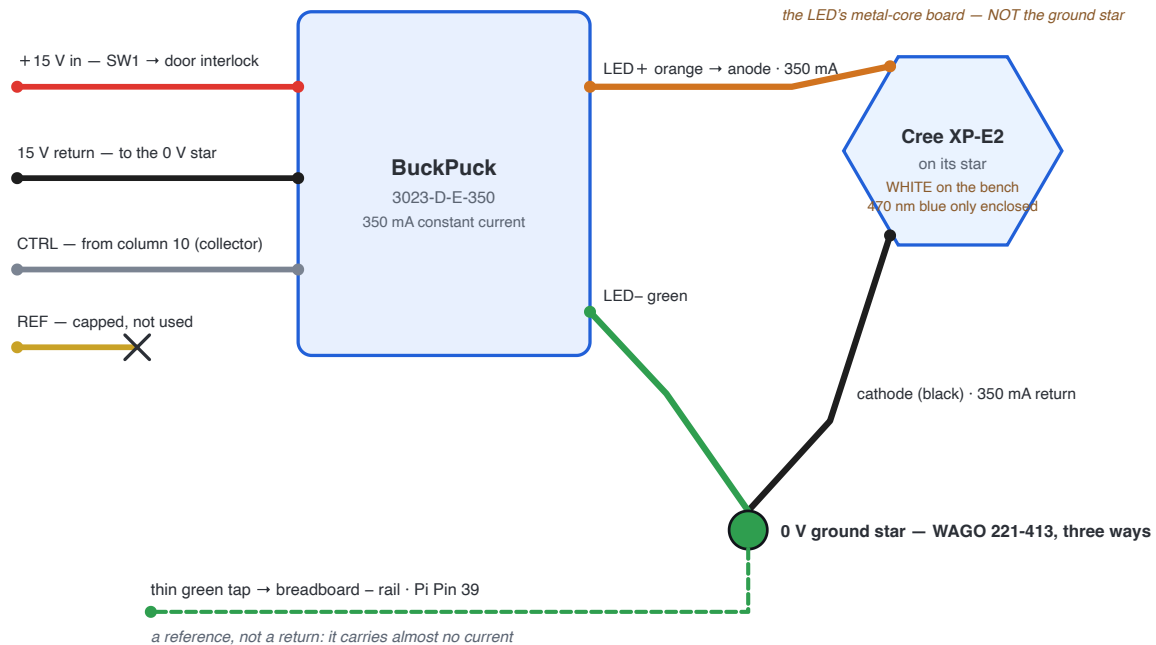


Fig 02.5-B2 · The off-board loop. It is NOT on the breadboard: the rails stay isolated.

Two things can be wrong before you start, and both look fine: the breakout can be on backwards, and the board can be read off by one column.

02.5.1 Find the five columns you will use



Locate columns 5, 8, 9, 10 and 13. Every part of this circuit lands on one of them: 5 is the GPIO input, 8·9·10 are the transistor's emitter, base and collector, and 13 is the +5 V tap. Nothing goes anywhere else.

NOTE · Within a column, the five tie points a–e are one node and f–j are another. The centre groove separates them — that is what lets a chip straddle it.

02.5.2 Meter the breakout before you trust its labels



pi-p12

With the Pi ON and the breadboard not yet connected, meter the breakout's pads against the Pi's own ground: the 5V pad should read +5 V, the GND pad 0 V, and the GPIO18 pad +3.3 V once you set it high. Confirm all three before continuing.

HAZARD · A ribbon fitted 180° out mirrors all 40 pins. Every voltage reads backwards and +5 V arrives on GPIO18 — a 3.3 V input. A full bring-up has already been lost to this once, and nothing about the assembled cable looks wrong.

Q1 is the only part with an orientation. Get it wrong and the circuit does nothing, or conducts when it should not.

02.5.3



Seat Q1 across columns 8, 9 and 10

Q1

Hold the 2N3904 with its FLAT face toward you and its legs down, then push it into row f so the legs land in columns 8, 9 and 10. Press until the shoulders of the legs meet the board.

NOTE · Flat face toward you puts the pins in the order emitter · base · collector, left to right. Turned around, the same three legs read collector · base · emitter and the circuit is backwards.

02.5.4



Confirm E · B · C, left to right

Check the three legs against the board: 8 is the emitter, 9 the base, 10 the collector. Say it out loud once — the next four steps are written in those terms and never in column numbers alone.

One sets how hard the transistor is driven; the other limits what flows through it. They are not interchangeable, and one of them has a wattage requirement.

02.5.5



Fit R1, 1 k Ω , from column 5 to the base

R1

Bend the legs of the 1 k Ω resistor and seat it in row h, from column 5 to column 9. This is the only path from the Pi's GPIO pin into the transistor's base, and it is what keeps the Pi from sourcing more current than it can.

NOTE · Row h is in the same column-node as row f, so landing on 9h connects it to Q1's base leg at 9f. You do not need a jumper between them.

02.5.6



Fit R2, 100 Ω , from the collector to column 13

R2

Seat the 100 Ω resistor in row h, from column 10 to column 13. It is the pull-up on CTRL: with Q1 off it lifts the collector to about 4.6 V, which is what holds the driver OFF.

HAZARD · R2 must be a ½ W part. It dissipates about 230 mW with the rail on — a ¼ W resistor runs at its limit and is acceptable only for a brief bench test, never in the assembled device.

Two wires connect the circuit you have built to the two power rails running down the edges of the board.

02.5.7



Jumper the emitter to the – rail

8f-8Bm

Run a jumper from 8f down to the – rail. This gives the emitter its return path. Use the – rail nearest you and push both ends fully in.

NOTE · Loose legs are the most common breadboard fault and the hardest to find: the circuit works, then stops when the bench is bumped.

02.5.8



Jumper column 13 to the + rail

13h-13Bp

Run a jumper from 13h up to the + rail. R2's high side now has +5 V waiting for it, and the circuit is complete except for what comes in from outside.

The circuit is now built and NOT yet connected to the Pi. This is the only moment at which the next test is possible, and it is the reason you breadboard at all.

02.5.9 Column 5 must read OPEN to the – rail



5

With the Pi off and nothing plugged into the breadboard, set your meter to resistance and measure from column 5 to the – rail. It must read OPEN — no continuity. If it does not, find the short before you go on.

02.5.10 Column 5 must read OPEN to the + rail



5

Measure from column 5 to the + rail. This must also read OPEN.

HAZARD · Column 5 is where the Pi's GPIO18 lands, and GPIO18 is a **3.3 V** input. If column 5 is connected to the + rail — a jumper one column off, a leg touching — then landing Pin 12 puts +5 V straight into it. Pass both measurements before you connect anything to the Pi.

Power first, ground second, signal last. That order means the signal pin never arrives at a circuit whose reference is still floating.

02.5.11 Pi Pin 2 (+5 V) to the + rail



pi-p2

Run the RED lead from Pi Pin 2 to the + rail at the left end of the board.

02.5.12 Pi Pin 39 (GND) to the – rail



pi-p39

Run the BLACK lead from Pi Pin 39 to the – rail at the left end. The Pi and the circuit now share a reference — without this the circuit cannot switch.

02.5.13 Pi Pin 12 (GPIO18) to column 5



pi-p12

Only now, with both gate measurements passed, run the ORANGE lead from Pi Pin 12 into column 5 — the free end of R1.

NOTE · GPIO18 is the hardware-PWM pin. Later, when the software dims the lamp, this is the wire the duty cycle travels down.

Two thin wires go to the BuckPuck. The 350 mA that actually lights the emitter does not come near this breadboard.

02.5.14 The collector out to the BuckPuck's CTRL



bp-ctrl

Run the GRAY lead from column 10 — the collector — to the BuckPuck's CTRL input. This is the circuit's one output.

HAZARD · CTRL is **INVERTED**: pulling it LOW runs the driver at full output, and R2 holds it near +5 V to keep the rail OFF. That off-state needs the Pi's +5 V — if the Pi loses power while +15 V is live, CTRL falls and the lamp comes ON.

02.5.15



The driver's LED– reference to the – rail

bp-ledm

Run the GREEN lead from the BuckPuck's LED– to the – rail at the right-hand end of the board. Land it at ONE point only.

NOTE · This is a reference tap, not a return path: it carries almost no current. Its job is to let the transistor and the driver agree on what 0 V means.

02.5.16



Know what is not on this board

Trace the 350 mA loop once: BuckPuck LED+ goes straight to the star's anode, and the star's cathode returns straight to LED–. The only two wires joining the two circuits are the GRAY CTRL wire and the thin green reference tap — both low current.

NOTE · The word "star" is used for two different things. The Cree star is the LED's metal-core board. The ground star is a WAGO junction. They are not the same node — the LED star only references the ground star through the thin green tap.

The circuit is built and the driver is landed. Now close the loop that actually carries the current, off the board, on the WHITE emitter.

02.5.17



Fit the WHITE XP-E2 star, not the blue one

cree-xpe2-star

Mount the white XP-E2 star. Every bring-up in this chapter is done on white; the 470 nm blue star goes in once, at final integration, after the enclosure is closed and the interlock is proven to fail dark.

NOTE · The two stars are physically identical, so nothing about the board will stop you fitting the wrong one. Check the part before it goes in.

02.5.18



LED+ from the driver to the star's anode

ledp

Run the ORANGE LED+ lead from the BuckPuck to the star's anode. This leg carries the full 350 mA and never comes near the breadboard.

02.5.19



Close the return through the ground star

cathode

Run the star's cathode into the 3-way WAGO, and the BuckPuck's LED– into the same WAGO. That junction is the one point where the 350 mA return and the board's reference tap meet.

NOTE · Three conductors land there: the star's cathode, the driver's LED–, and the thin green tap you ran to the – rail. One splice, one point.

02.5.20



Gate the +15 V before you connect it

vin

Run the supply into the BuckPuck's Vin+ through SW1, and through the door interlock if the enclosure is built. Open SW1 before you plug the brick in.

HAZARD · SW1 and the interlock are the only things that guarantee darkness. The driver's CTRL is inverted and its off-state depends on the Pi's +5 V, so cutting the rail is the only cut that does not depend on software.

The whole circuit, on and off the board, proving itself — and the power order that keeps it from lighting when you did not ask it to.

02.5.21 **Power the Pi FIRST, then close SW1**



Boot the Pi and let it settle, then close SW1. In that order.

NOTE · This is the single most important habit in the chapter, and it is the one a bench test teaches you badly if you get away with the wrong order once.

02.5.22 **Drive GPIO18 high — the white emitter lights**



Run `pinctrl set 18 op dh`. Q1 saturates, CTRL falls to about 0.2 V, the BuckPuck runs, and the white star lights at 350 mA. If it does not, open SW1 before you start looking.

NOTE · GPIO18 HIGH is the lamp ON. Two inversions cancel — the Pi's high turns Q1 on, and the driver runs at full output when CTRL is pulled low.

02.5.23 **Drive GPIO18 low — it must go out**



Run `pinctrl set 18 op dl`. Q1 turns off, R2 lifts CTRL to about 4.6 V, and the emitter goes dark. Both halves must pass before this circuit goes anywhere near copper.

NOTE · That is the whole point of breadboarding it: you have just proven the exact circuit you are about to solder, on the exact emitter, with nothing committed.

02.5.24 **Prove it fails dark**



With GPIO18 still high and the emitter lit, open SW1. It must go out and stay out. If the enclosure and its interlock are built, open the door with SW1 closed and confirm the same.

HAZARD · A lamp that only goes out when the software says so is not gated. The rail cut is the proof, and it is the one you repeat every time before you put your hands in.

NOTE · Then power down in order: SW1 open first, Pi second.

02.6 Build the NS-04 Board

ONLINE: ACTIVITY 02.6 · 3D VISUALIZER

Three components, two headers and five jumpers, in the order that keeps the board flat on the bench — lowest first. Do all of it before the board goes near the machine.

02.6.1



Orient the board

ns04-perfboard

Set the board down with the IN header at column A and the OUT header at column T, rows running 1–14.

Get this the right way round before you start: every instruction below names a pad by column and row.

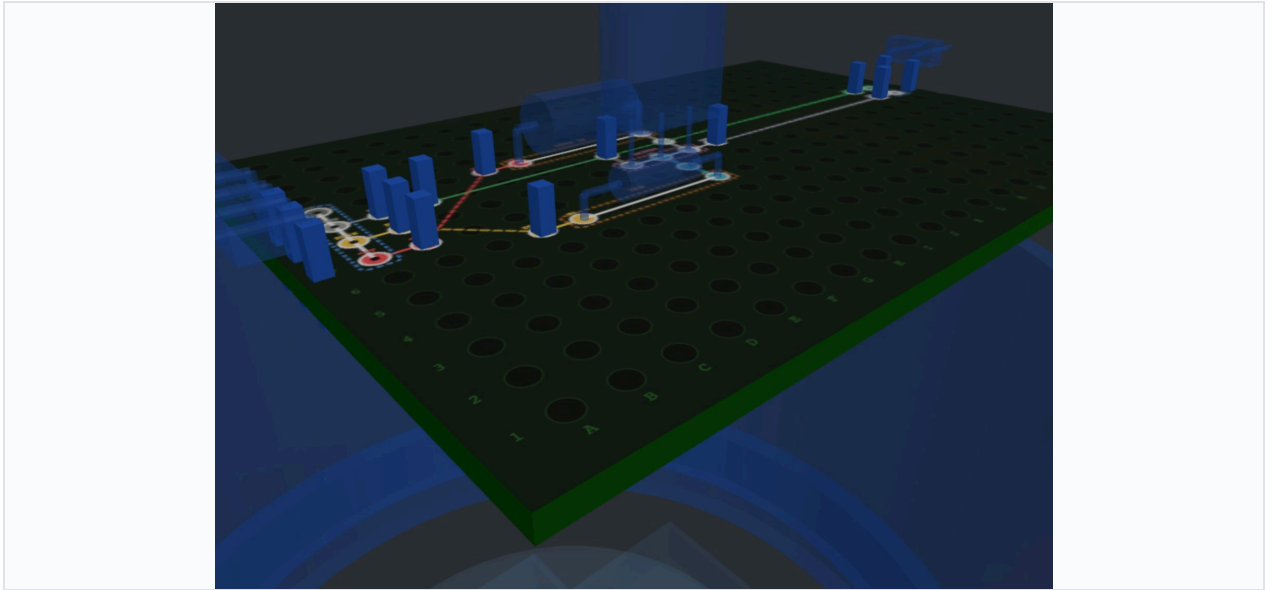


Fig 02.6-1 · 3D Visualizer · ns04-perfboard

02.6.2



Solder R1, 1 k Ω , into E6 and I6

ns04-r1

Form the 1 k Ω resistor's legs to 12 mm and drop them into E6 and I6, in row 6. Lay it flat on the board, solder from the back, then trim the legs flush.

NOTE · Resistors first, because they lie flattest. A board with nothing tall on it sits face-down on the bench while you solder, and every leg stays where you put it.

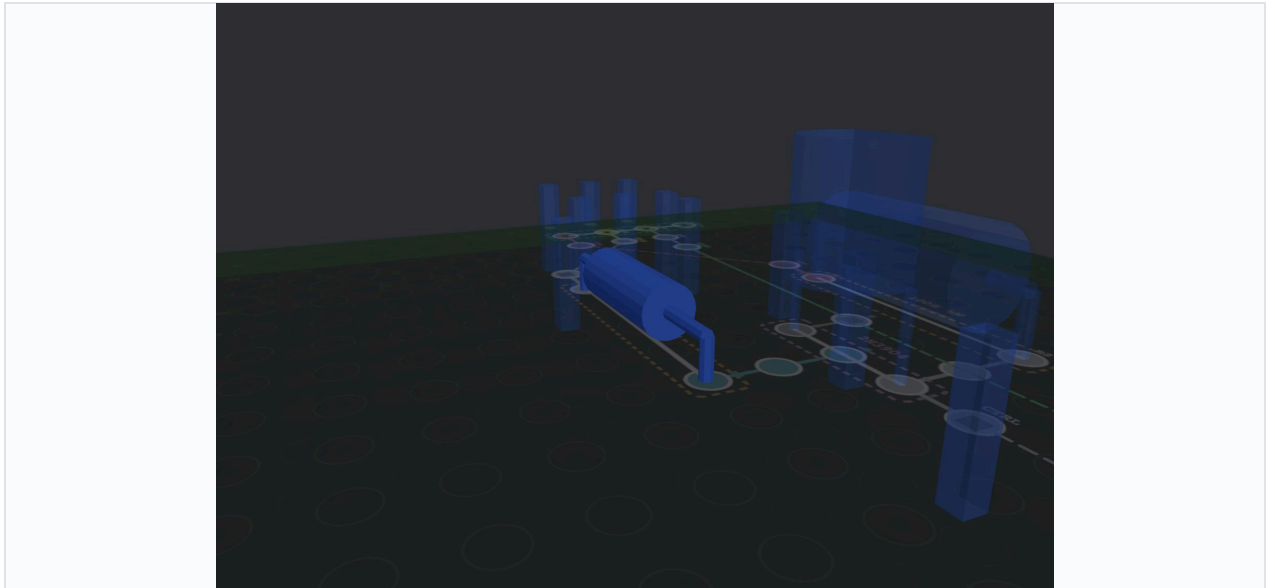


Fig 02.6-2 · 3D Visualizer · ns04-r1

02.6.3



Solder R2, 100 Ω , into F10 and J10

ns04-r2

Form the 100 Ω resistor the same way and fit it into F10 and J10, in row 10.

HAZARD · R2 must be a $\frac{1}{2}$ W part — it is visibly fatter than R1, about 8.5 mm long against 6.3. It dissipates ~ 230 mW whenever Q1 is conducting. A $\frac{1}{4}$ W resistor fits the same holes and runs at its limit.

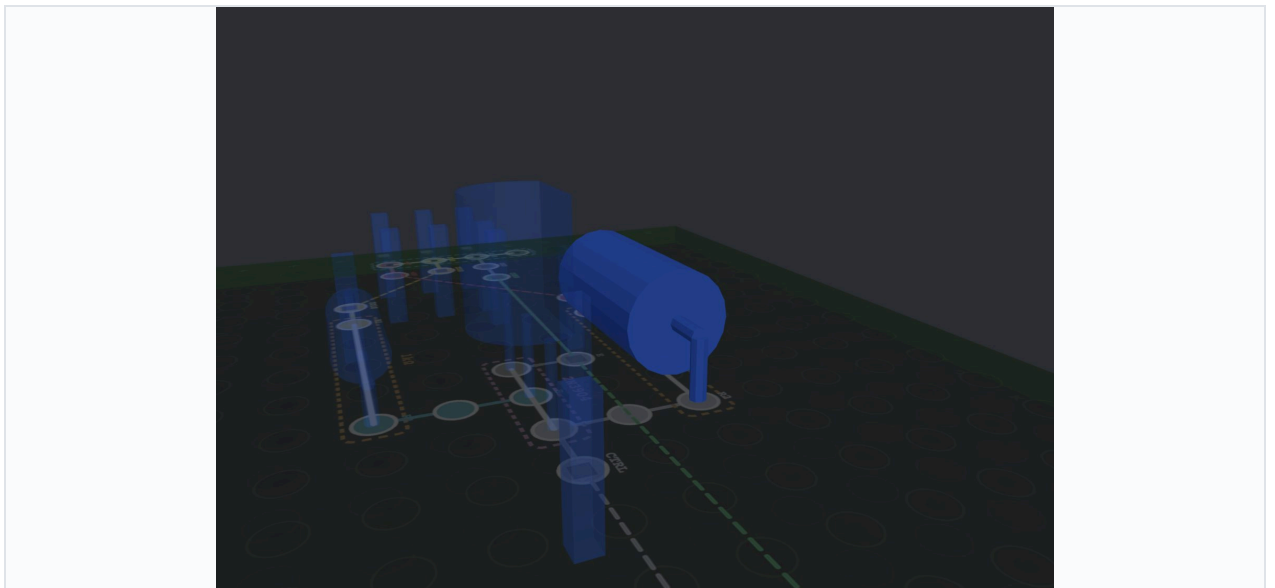


Fig 02.6-3 · 3D Visualizer · ns04-r2

02.6.4



Solder Q1, the 2N3904, into H8, I8 and J8

ns04-q1

Fit the transistor with its FLAT face toward the board's row-14 edge, legs into H8, I8 and J8. Stand it about 3 mm off the board so the legs can be cleaned up, and solder the middle leg first.

HAZARD · Orientation is the whole step. H8 is the emitter, I8 the base, J8 the collector — which is what you get with the flat facing that way. Turned round, the same three legs read collector, base, emitter, and the circuit conducts when it should not.

NOTE · Solder one leg, check the body is square and seated, then do the other two. Reheating one joint to straighten it is easy; reheating three is not.

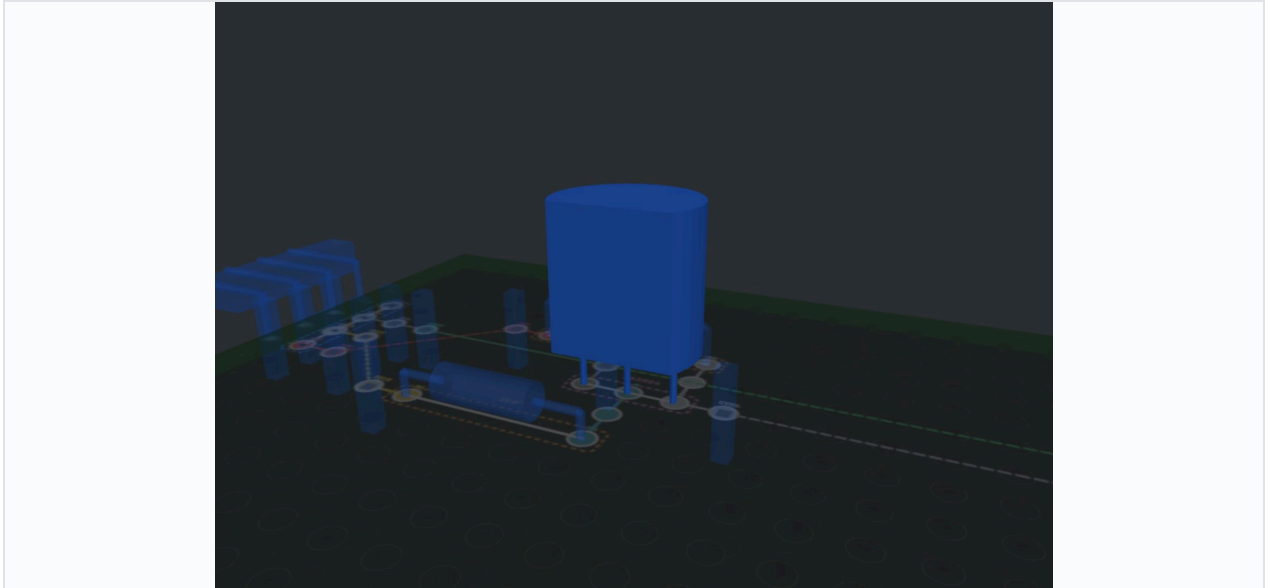


Fig 02.6-4 · 3D Visualizer · ns04-q1

02.6.5



Solder the ten male header pins

ns04-perfboard

Fit a male header pin into each of B7, D6, B8, E10, B9, H9, C9, S9, K8 and S8 — the two ends of each of the five jumpers — and solder each one into its own pad. They are not connected to anything yet; that comes next.

NOTE · The pins do two jobs. They give the solder a post to flow onto beside a component lead, and they give the female Dupont jumpers something to push onto — which is why the jumpers are fitted last and never soldered.

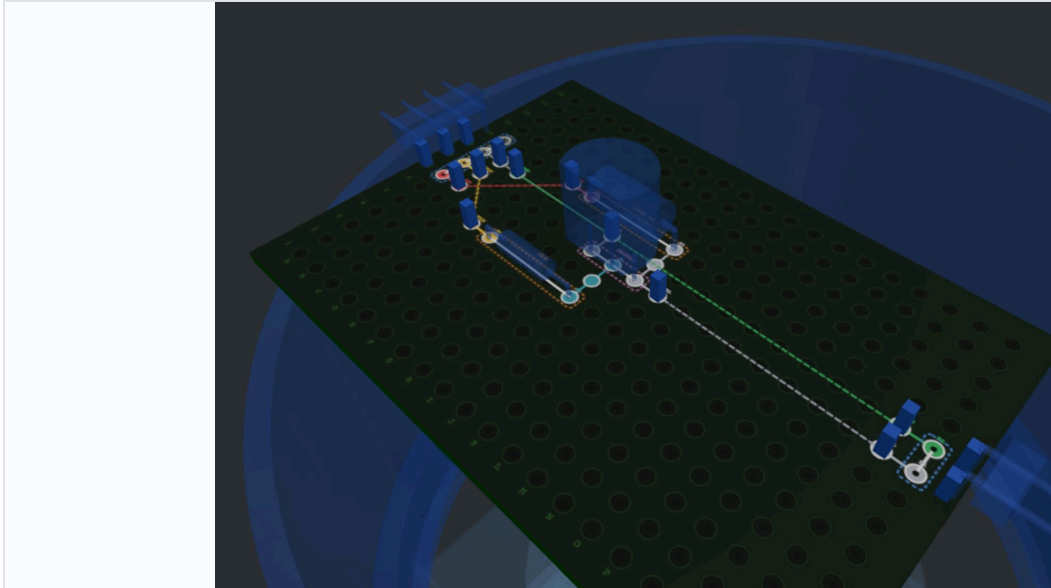


Fig 02.6-5 · 3D Visualizer · ns04-perfboard

02.6.6



Solder the 4-way angled header at J1

ns04-hdr-in

Fit the 4-way angled header into A7–A10, the board’s first row, with the pins leaving over the board’s edge. Solder one end pin, check the body is square against the edge, then do the rest.

NOTE · A7 is +5 V, A8 is GPIO18, A9 is logic ground, A10 is spare and connects to nothing. It is populated so the header has four posts to sit square on.

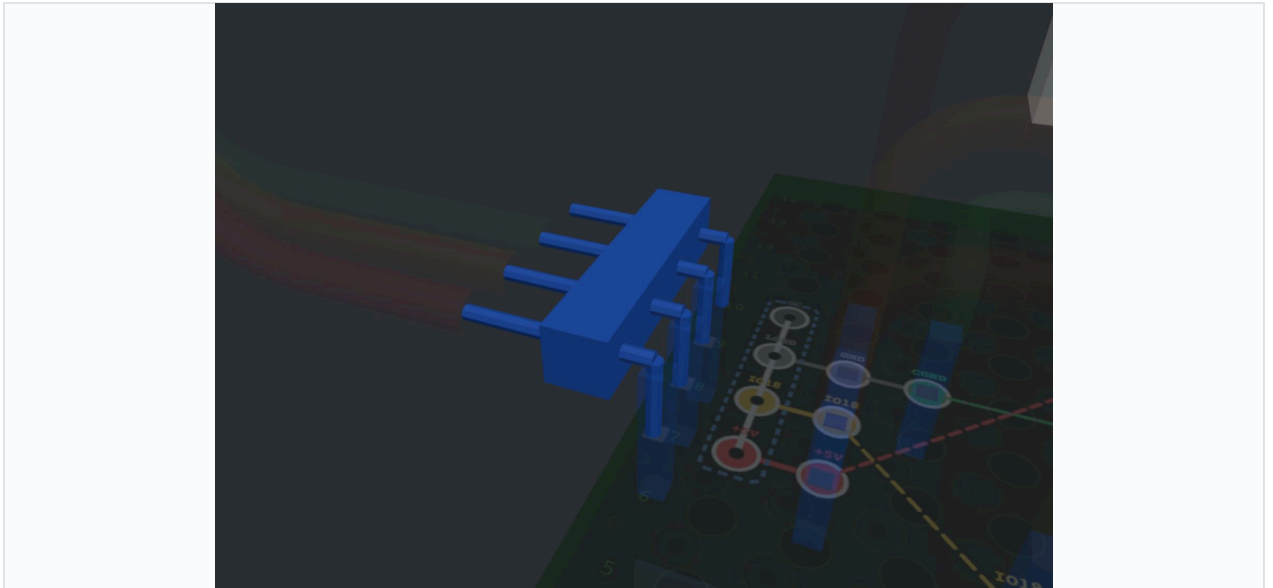


Fig 02.6-6 · 3D Visualizer · ns04-hdr-in

02.6.7



Solder the 2-way angled header at J2

ns04-hdr-out

Fit the 2-way angled header into T8–T9, the board’s last row, pins leaving over the opposite edge. T8 carries CTRL to the BuckPuck’s gray lead, T9 the control-ground reference to its green.

NOTE · Angled on both ends means the two harnesses leave the board flat and in opposite directions — input one side, output the other — so neither has to bend over the board to get out of the lamphouse.

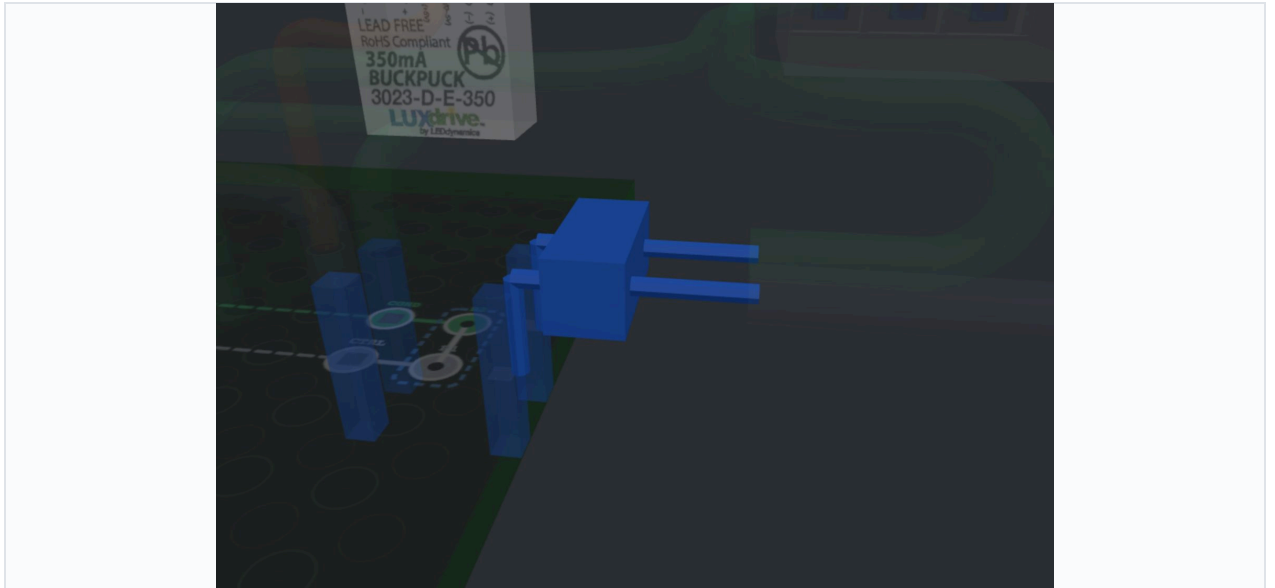


Fig 02.6-7 · 3D Visualizer · ns04-hdr-out

02.6.8



Solder the twelve joints that finish the circuit

ns04-perfboard

Everything is now in its own pad and none of it is connected. Work the solder map above, joint by joint: flow solder across each numbered pair until the two pads are one. Five of the twelve tie a male pin to a component lead, four tie a header pin to a male pin, two run across an empty pad to join two component leads, and one ties two male pins together.

HAZARD · Every pad on this board is **ISOLATED** — no strips, no copper between holes. Until a joint is made there is no circuit across it, and a board with every part fitted and no bridges flowed is an open circuit in twelve places. Nothing about it looks unfinished.

NOTE · Two of them — I6–I7–I8 and J10–J9–J8 — run across a pad that holds nothing. I7 and J9 exist only to be bridged over, carrying R1 into the base and R2 into the collector. Check each joint with a meter before you move on; a dry joint here looks exactly like a good one.

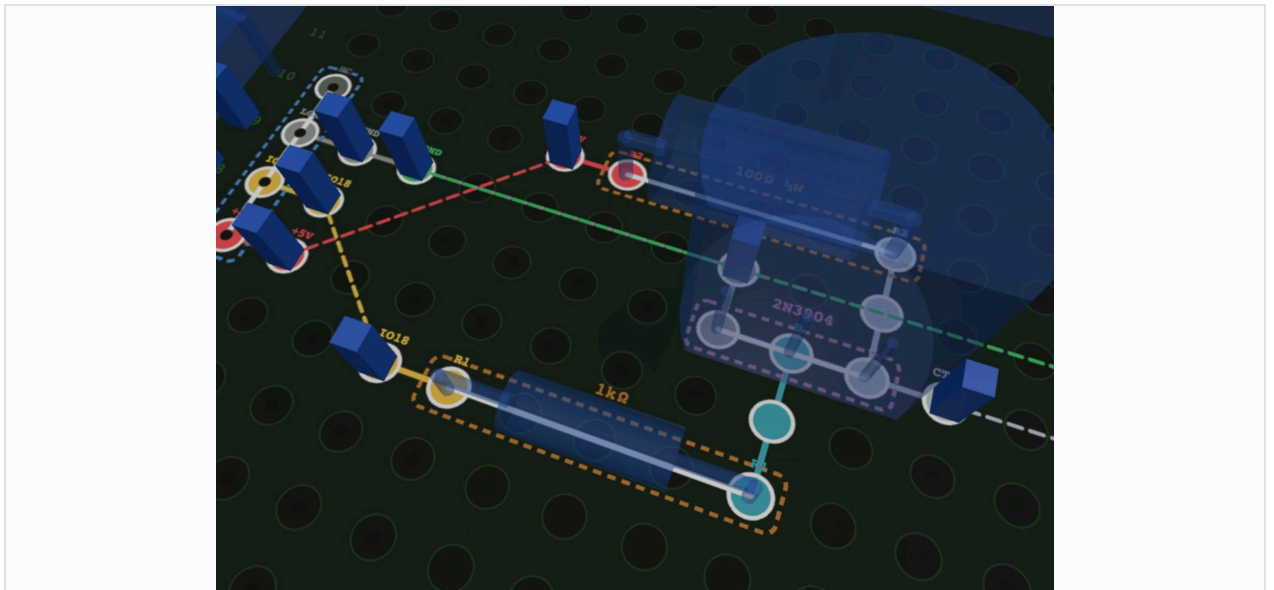


Fig 02.6-8 · 3D Visualizer · ns04-perfboard

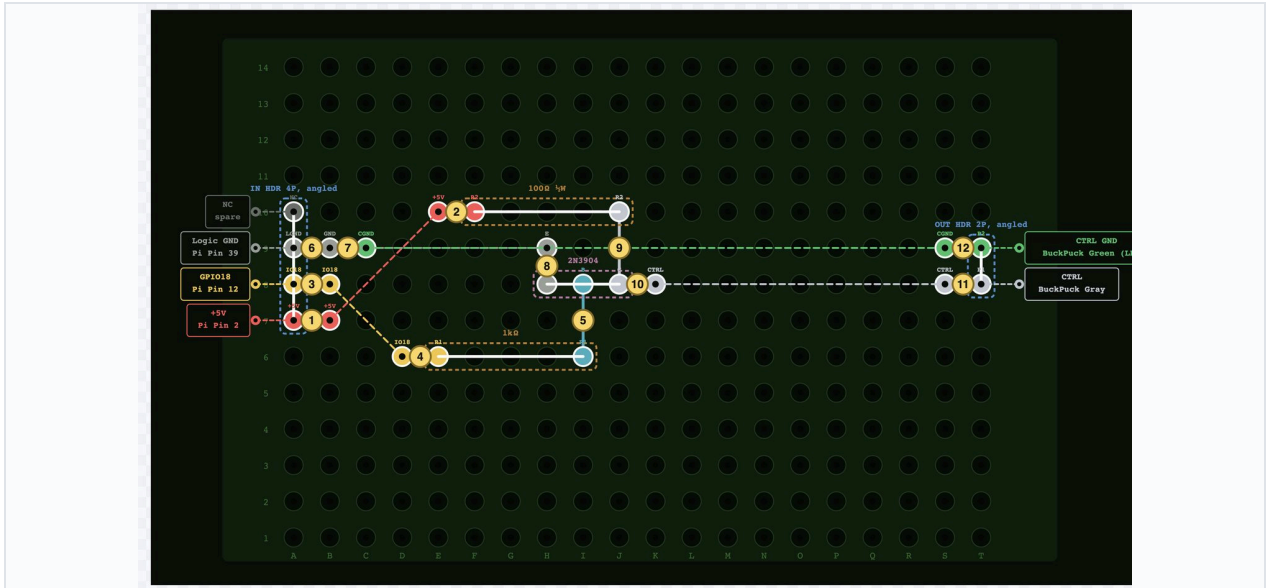


Fig 02.6-8d · The twelve joints on THIS board, numbered as the legend numbers them. Generated from the board file, so it cannot disagree with the pads the components were fitted to.

02.6.9



Jumper B9 → H9 — ground

W.027

Run a BLACK jumper from B9 to H9. A9 brings logic ground in from Pi Pin 39 and solders to B9; this wire carries it to H9, which bridges to H8 — Q1's emitter.

NOTE · B9 is the single point. Everything on this board that needs ground references it and nothing else, which is what makes the control reference trustworthy.

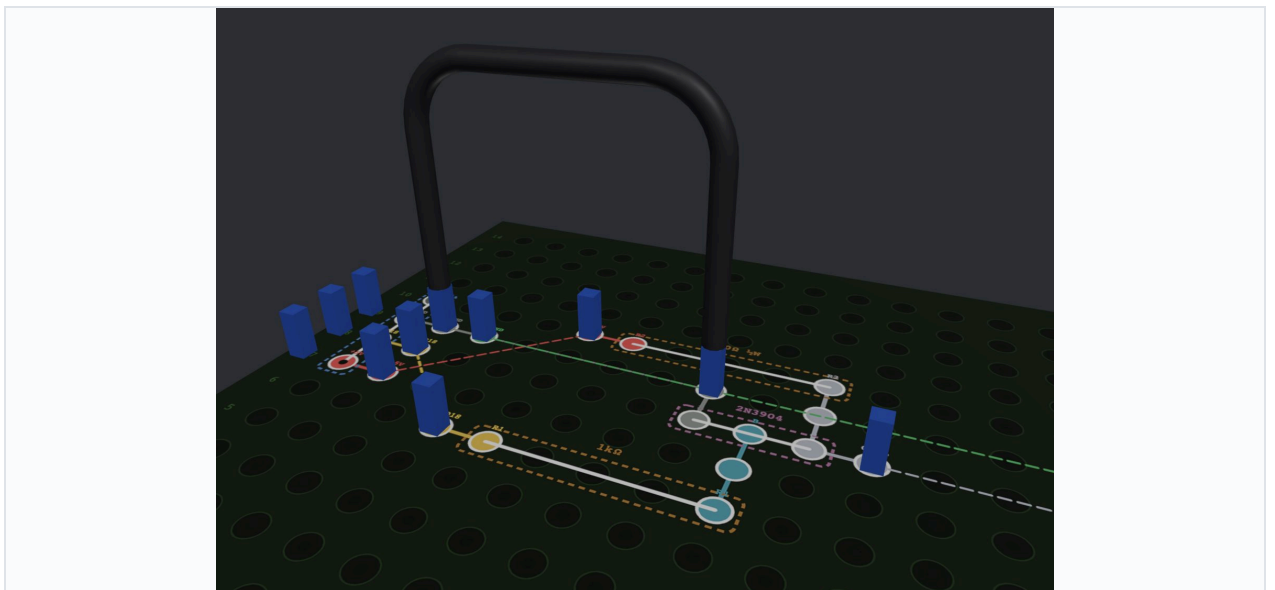


Fig 02.6-9 · 3D Visualizer · W.027

02.6.10



Jumper C9 → S9 — control ground reference

W.023

Run a GREEN jumper from C9 to S9. B9 bridges to C9; this wire takes the reference out to S9, which bridges to T9 on the OUT header.

NOTE · Low current, and it must stay that way. It shares a node with the 350 mA return but never a path.

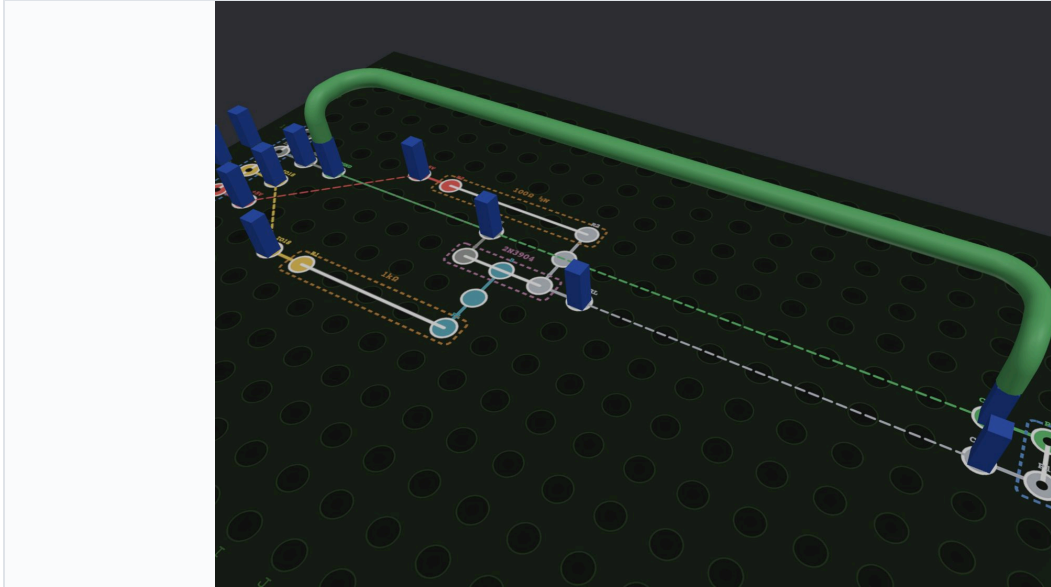


Fig 02.6-10 · 3D Visualizer · W.023

02.6.11



Jumper B7 → E10 — +5 V

W.024

Run a RED jumper from B7 to E10. A7 brings +5 V in from Pi Pin 2 and solders to B7; this wire carries it to E10, which bridges to F10 — R2's high side.

NOTE · That is the entire +5 V distribution on this board. R2 is the only thing drawing it.

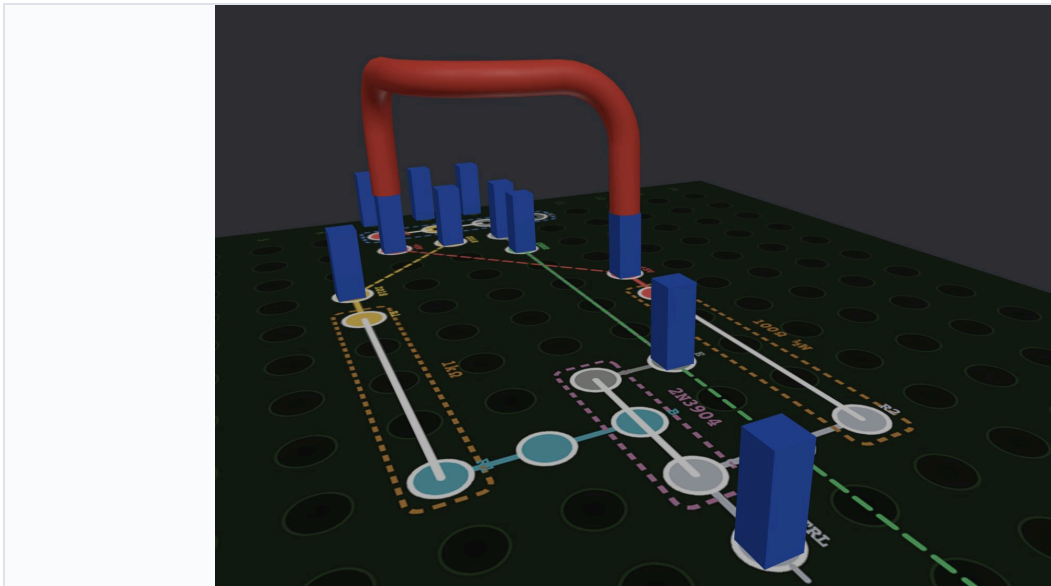


Fig 02.6-11 · 3D Visualizer · W.024

02.6.12



Jumper B8 → D6 — GPIO18

W.025

Run an ORANGE jumper from B8 to D6. A8 brings GPIO18 in from Pi Pin 12 and solders to B8; this wire carries it to D6, which bridges to E6 — R1's high side.

NOTE · This is the modulation. Everything the light does in time arrives on this one wire.

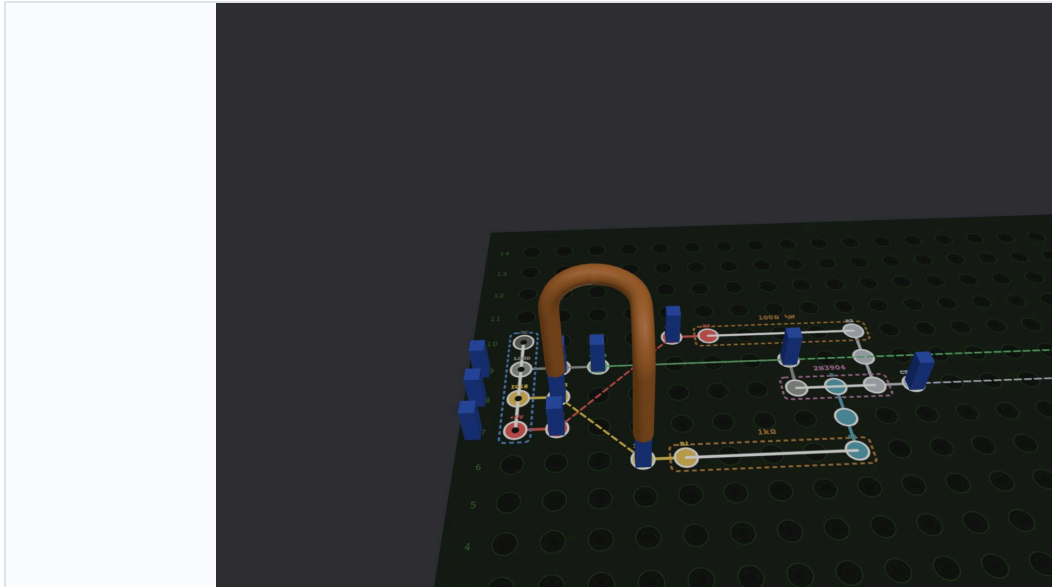


Fig 02.6-12 · 3D Visualizer · W.025

02.6.13



Jumper K8 → S8 — CTRL

W.026

Run a GRAY jumper from K8 to S8. Q1's collector bridges to K8; this wire carries CTRL to S8, which bridges to T8 on the OUT header.

HAZARD · CTRL is inverted — Q1 pulling it **DOWN** is what turns the LED on. An open circuit along here is a lamp stuck ON, not a lamp that stops working.

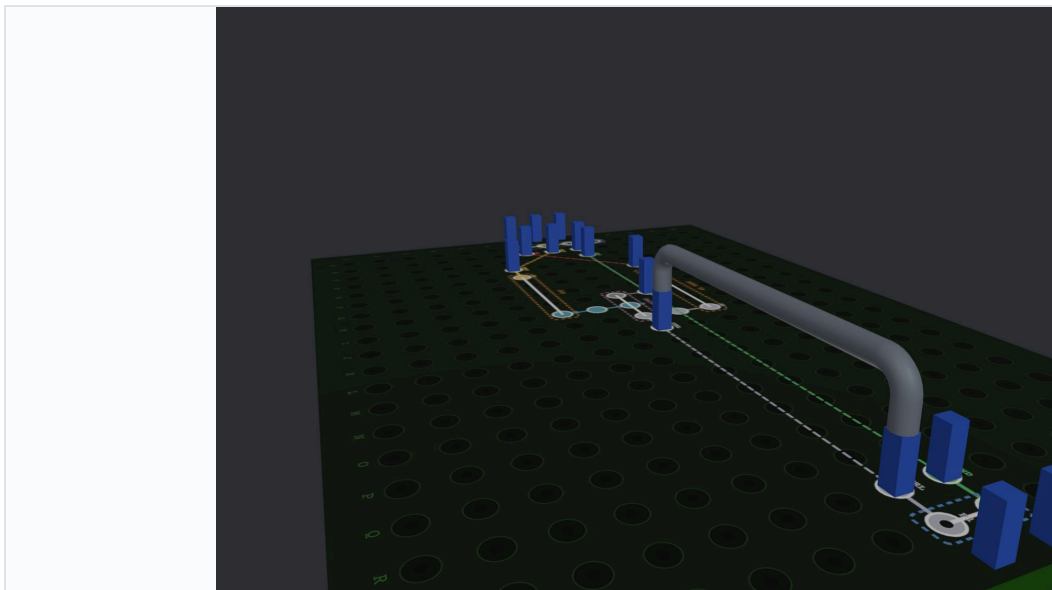


Fig 02.6-13 · 3D Visualizer · W.026

02.7 Mechanical Assembly — the Lamphouse Stack

ONLINE: ACTIVITY 02.7 · 3D VISUALIZER

The light element is built in your hand, the column is built on the enlarger, and the two are married at the end. Order is most of this activity: each piece closes access to the one under it, and two things -- the solder joints and the lead dressing -- are impossible to get at once the star is seated. What each part IS belongs to 02.2 and 02.4; this is how they go together.

02.7.1



Start at the star -- the heart of the module

cree-xpe2-star

Everything in this chapter exists to put current through one Cree XP-E2 on a 20 mm hex metal-core board, and everything in the chapters after it exists to measure what comes out. Work on it first, on the bench, with the housing empty and the column untouched.

NOTE · Two stars, one at a time, same footprint and same driver: WHITE for bring-up, BLUE 470 nm for calibration and every real exposure. Build with the white one.

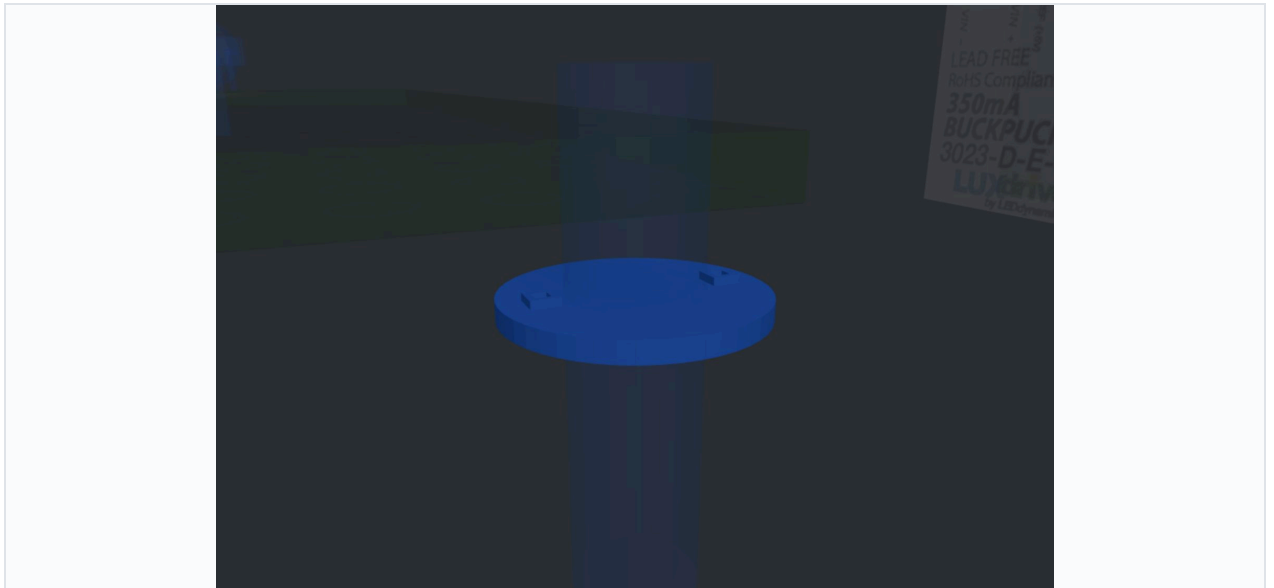


Fig 02.7-1 · 3D Visualizer · cree-xpe2-star

02.7.2



Find the two pads, and which way the die faces

cree-xpe2-emitter

The 3.5 mm die sits ON the board, and the board carries two pads: + ANODE and – CATHODE. Note which is which NOW, with the board bare and the markings visible. The die faces DOWN the column once this is assembled.

HAZARD · Once the star is seated on the ring the pad markings are underneath your own wiring and under the ring's lip. If you are going to read them, read them here.

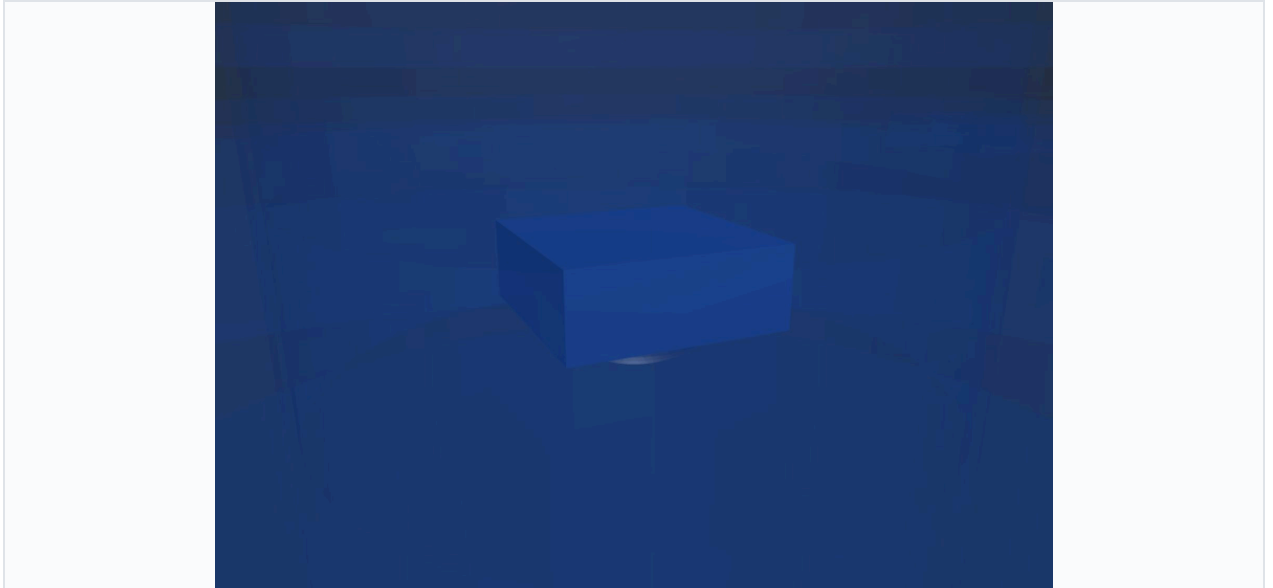


Fig 02.7-2 · 3D Visualizer · cree-xpe2-emitter

02.7.3



Solder the two legs -- and get the polarity right

cree-xpe2-star

Tin both pads, then land 18AWG on each: LED + from the BuckPuck to the ANODE pad, and the 350mA return from the CATHODE pad. Keep the iron on each pad only as long as it takes -- the board is aluminium and it pulls heat away from the tip, so a joint that looks bright can be cold underneath.

HAZARD · POLARITY IS NOT A CONVENTION HERE, it is the whole circuit. An LED is a diode: reversed, it does not light, and because the BuckPuck holds CURRENT rather than voltage nothing downstream complains about it. You get a dead lamp, a driver that looks healthy and a meter that reads plausibly -- and you will go looking for the fault everywhere except the two joints you just made.

NOTE · The polarity you set here is carried by every connection after it, through the WAGO ground star and back to the driver. Reverse it at the star and it is reversed for the whole build.

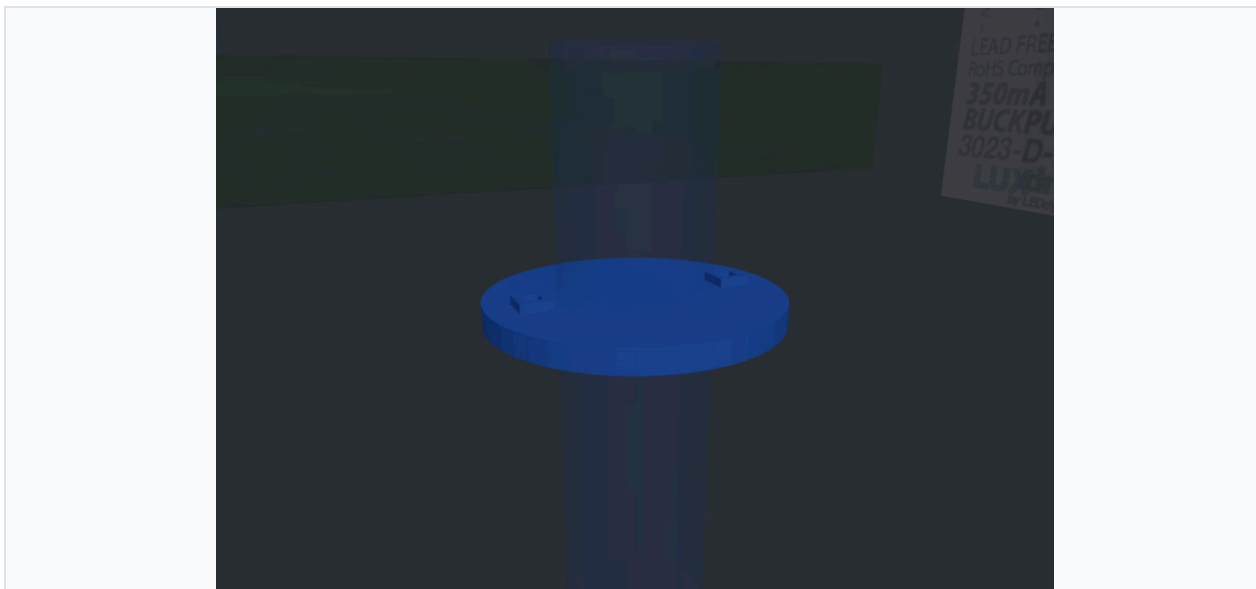


Fig 02.7-3 · 3D Visualizer · cree-xpe2-star

02.7.4



Train the leads before anything is seated

cree-xpe2-star

Dress the two legs away from the die, sleeve them, and wrap them along the housing so they leave on ONE side and in the direction they will travel -- which is UP, through the hollow centre of the mounting rod in the next step. Trained now, the star drops into place and the leads follow it. Untrained, you are feeding two stiff 18AWG legs up a 10 mm bore inside an assembled stack with one hand.

HAZARD · No lead may sit between the star and the ring. A lead under the board lifts one edge of it, and a star that does not sit flat puts the die off the plane the whole optical column is measured from.

NOTE · Leave a service loop. The lamphouse is the one place in this build where a lead that is exactly long enough is too short.

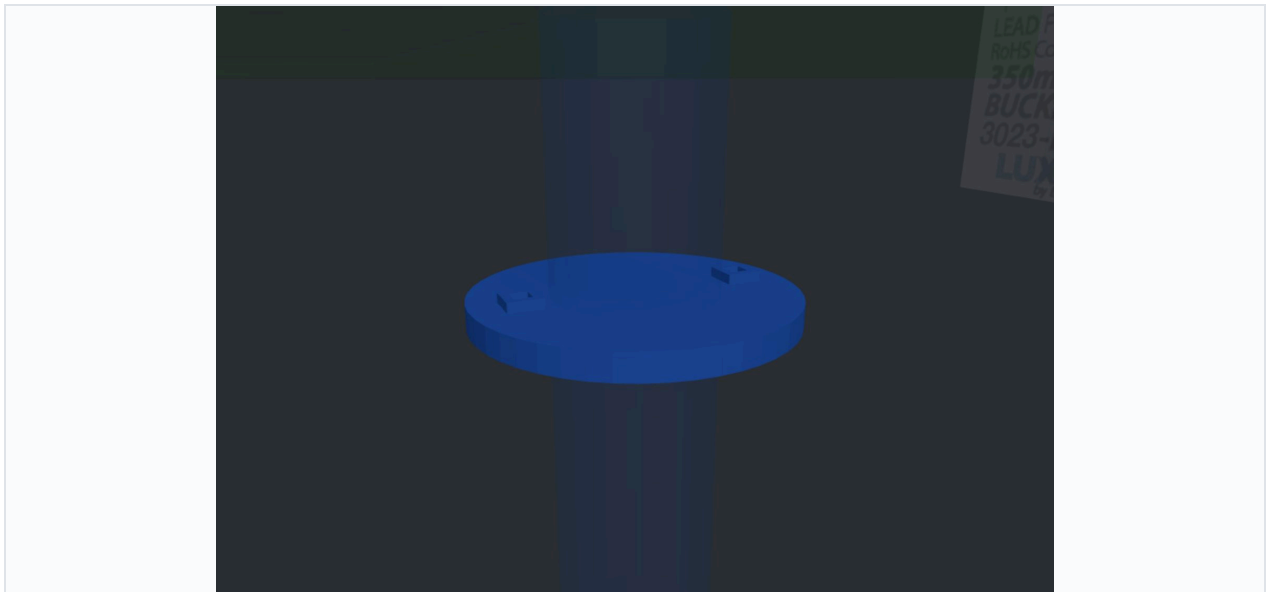


Fig 02.7-4 · 3D Visualizer · cree-xpe2-star

02.7.5



The diffuser goes in **FRONT** of the LED, and it is not optional

carclo-optic

Thread the Carclo 10140 into the front of the housing -- the face that will point down the column -- so every photon the die emits passes through it. It spreads the emitter into an even field at a known angle. Do it with the housing in your hand: it is an easy turn now and an awkward reach once the housing is on the spacer.

HAZARD · Assemble without it, or with it loose, and the die images as a SPOT rather than a field. That gets worse as the aperture closes: at f/22 the projection is sharper and a hot spot is exactly what you see.

NOTE · APERTURE, while you are thinking about it. Diffuse light is what this instrument wants, not sharp light. f/8 is the working aperture. f/11 is acceptable. f/4 and below tends to be too bright, and f/22 buys sharpness you did not want at the cost of the evenness you did.

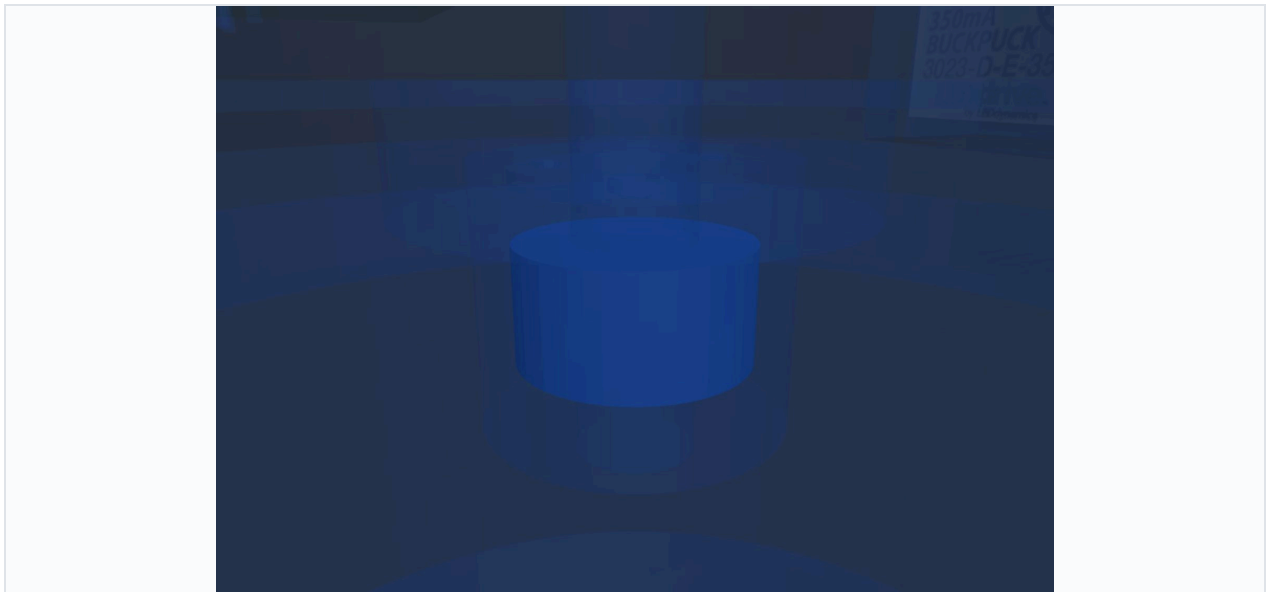


Fig 02.7-5 · 3D Visualizer · carclo-optic

02.7.6



Now it is one sub-assembly

dlh-1upeg

Housing, optic, star and two trained leads: that is the light element, finished, and it goes onto the column as a single piece. Everything from here is the column underneath it.

HAZARD · Handle the optic by its edge. A fingerprint on that face sits on the optical axis for the life of the build.



Fig 02.7-6 · 3D Visualizer · dlh-1upeg

02.7.7



The column starts at the seat

lightspacer-lower-ring

The spacer seat lands on the enlarger body. Get it flat and square before anything goes above it: every part after it inherits whatever tilt it has.

NOTE · It is the lowest part of the lamphouse at z437 and the emitter ends up at z541, so a degree of tilt here is a degree of tilt 100 mm higher up.

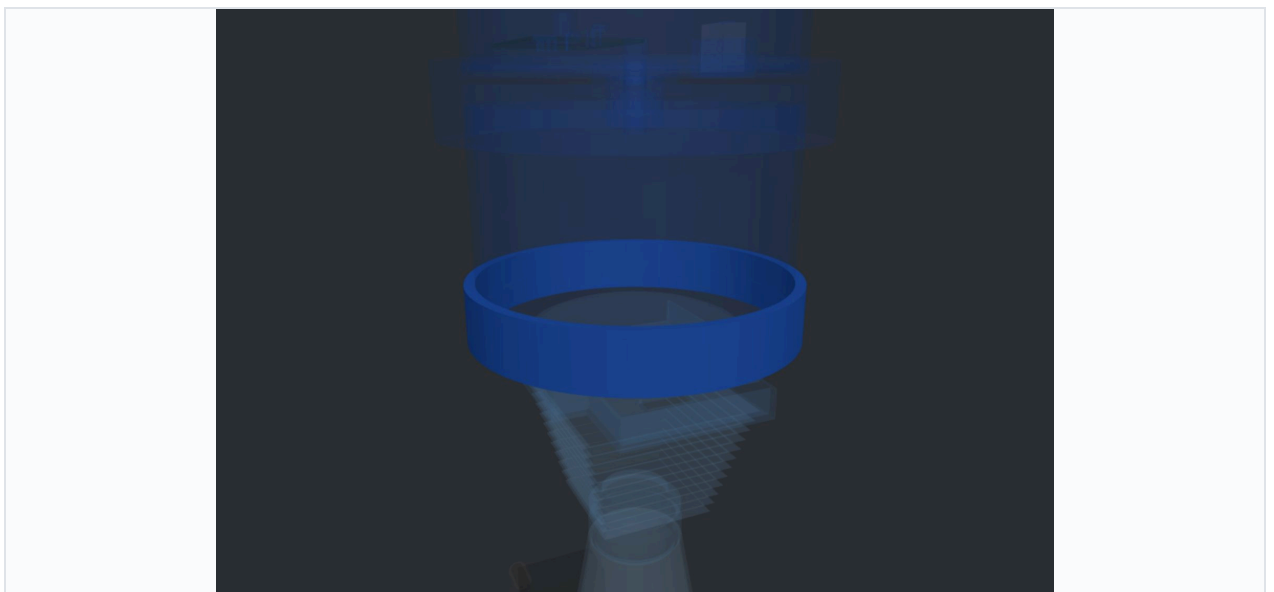


Fig 02.7-7 · 3D Visualizer · lightspacer-lower-ring

02.7.8



Spacer next -- it sets the drop

lightspacer-large

The light spacer goes onto the seat. Its height IS the emitter-to-optics distance, so it is fitted before anything optical and not adjusted afterwards.

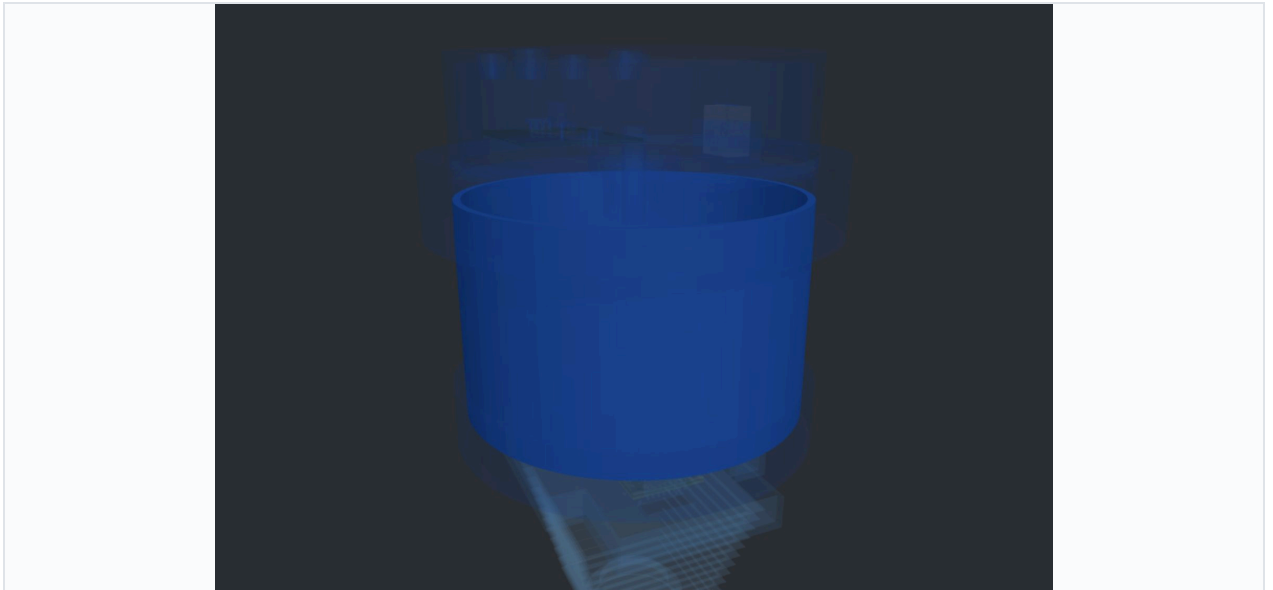


Fig 02.7-8 · 3D Visualizer · lightspacer-large

02.7.9



Land the light element on the spacer

dlh-1upeg

The sub-assembly goes down onto the spacer stack as one piece, leads leaving on the side you trained them to.



Fig 02.7-9 · 3D Visualizer · dlh-1upeg

02.7.10



Mount on the threaded rod -- lamp hardware, not printed hardware

lamp-rod

The assembly mounts to the light ring on a threaded HOLLOW metal rod -- the standard lamp-housing part, the kind that comes in any lamp kit. It protrudes directly through the CENTRE of the light ring, on the same axis the light takes, and takes a bolt and washer on the far side. That is the whole fixing: one pass-through, one bolt, one washer.

HAZARD · It also keeps the load and the heat OFF the plastic. The metal rod carries the assembly and isolates it from the PETG around it: a printed part asked to be the structural fixing in a warm lamphouse is a part that creeps, and a spacer that creeps moves the light. Let the metal take it.

NOTE · Hollow is the point. The trained leads run UP the centre of the rod, so the wiring leaves the lamphouse through the same part that holds it -- no second hole, no lead pinched between two faces, and the run is already pointing where it has to go next.

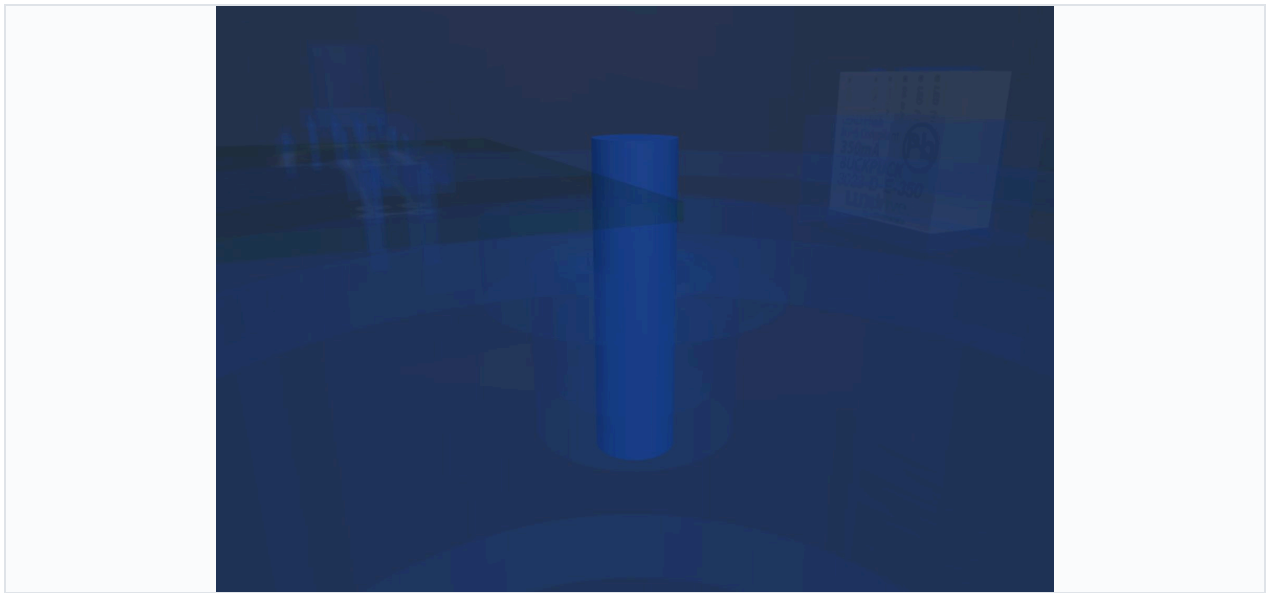


Fig 02.7-10 · 3D Visualizer · lamp-rod

02.7.11



Ring over the star -- and the view closes

cree-lightring

The light ring goes over the star and holds it on axis and on plane. The ring is a disc across the whole aperture: once it is on, nothing below it can be seen or reached without lifting it off again.

HAZARD · Last chance to check the die faces DOWN and both joints are sound. After this the only way back to them is back out through this ring.

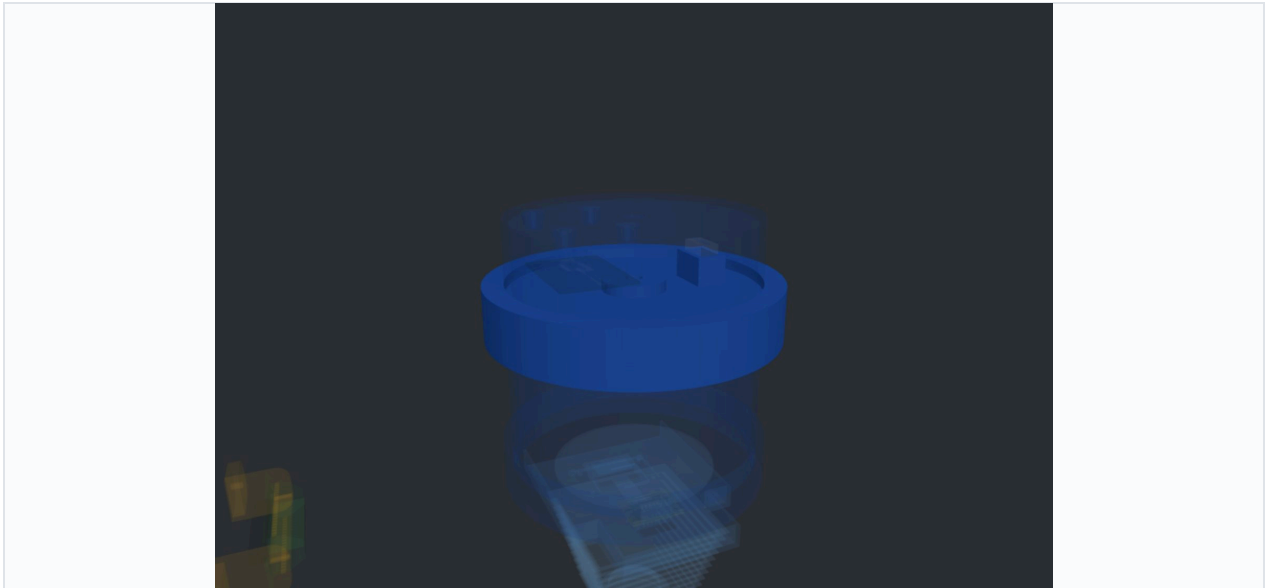


Fig 02.7-11 · 3D Visualizer · cree-lightring

02.7.12



The lid goes on last -- and not yet

light-circuit-hat

The circuit lid closes the top and carries the control board on its four standoffs. Leave it OFF. The emitter's leads are reachable from above only while it is off, they are landed in 02.8, and the lid is fitted in 02.9 once the wiring is proved.

HAZARD · Fit it now and the next two activities mean taking it off again. Nothing under this lid can be reached through it.



Fig 02.7-12 · 3D Visualizer · light-circuit-hat

02.8 Emitter and Control Side Wiring

ONLINE: ACTIVITY 02.8 · 3D VISUALIZER

The 350 mA legs. Both are direct between the driver and the star — the schematic says so twice.

02.8.1



Identify the star's pads

cree-xpe2-star

Identify the anode and cathode on the star — a single Cree XP-E2 on a 20 mm metal-core board. They sit on opposite sides of the top face, anode toward the driver.

NOTE · White for bring-up, blue 470 nm for calibration. Same board, same driver — what changes is what the photometry means.

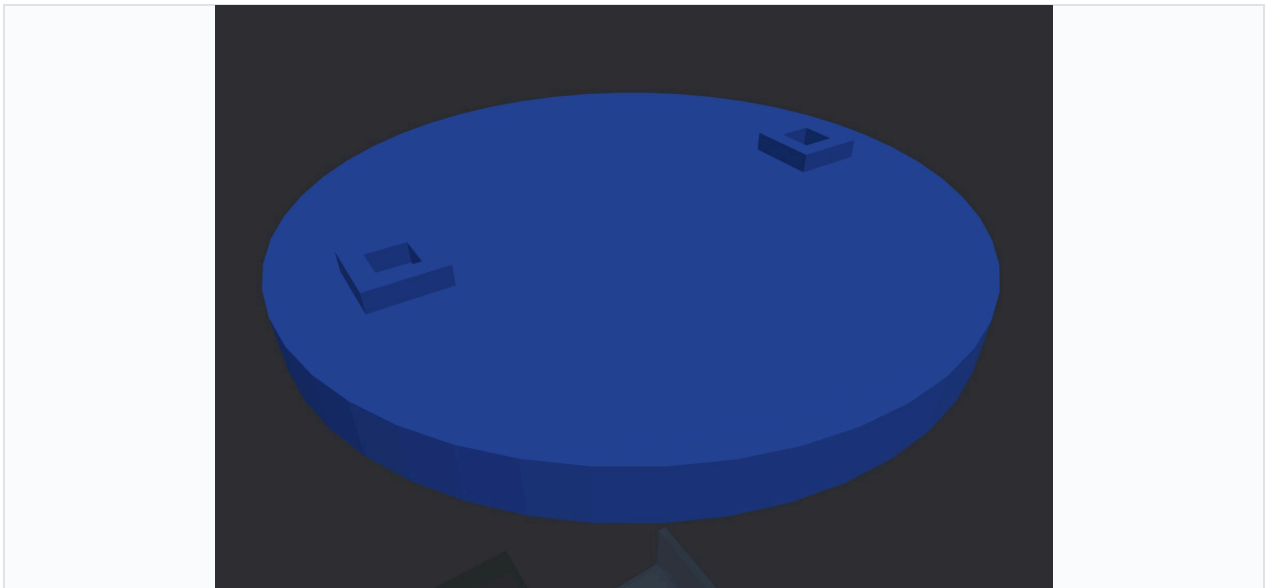


Fig 02.8-1 · 3D Visualizer · cree-xpe2-star

02.8.2



LED+ — driver to the star's anode

W.020

Run the ORANGE LED+ lead from the driver to the star's anode, dropping it through the light ring to reach the board. This leg carries 350 mA.

HAZARD · Direct, and only direct. Nothing may sit between the driver and the emitter on this leg.

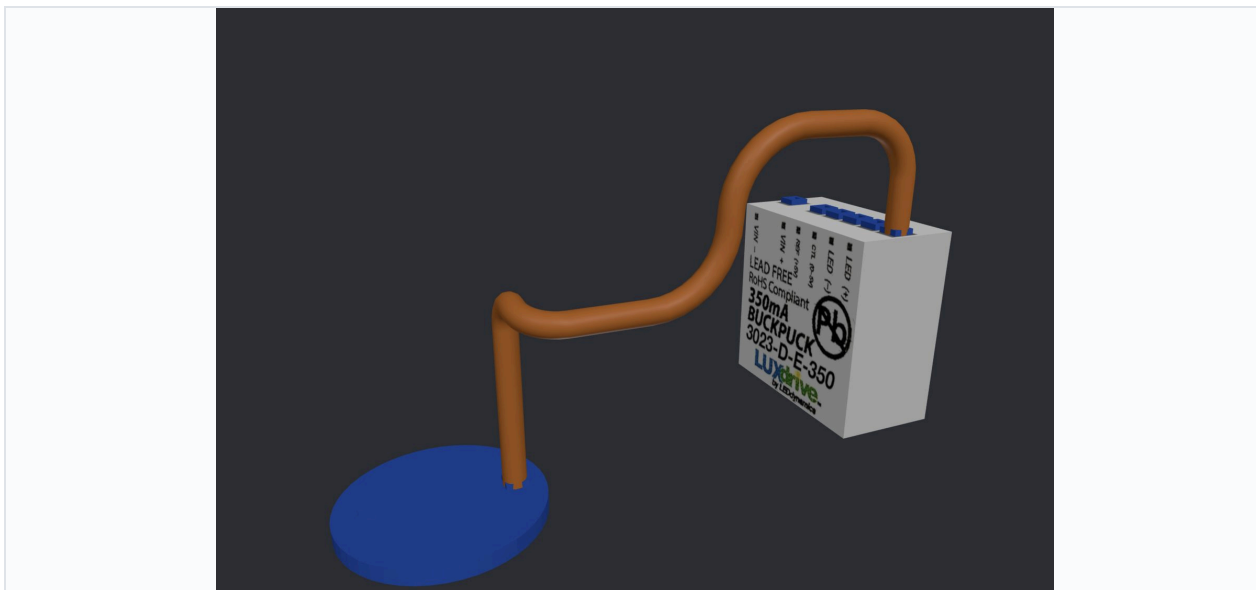


Fig 02.8-2 · 3D Visualizer · W.020

02.8.3



Fit the 3-way ground star

wago-led-gnd

Fit a WAGO 221-413 above the light ring. Three conductors meet here: the emitter's return, the driver's LED–, and the board's control-ground reference.

NOTE · One splice, so the reference and the return are at the same potential by construction. Meeting at two places would let the control voltage move with the LED current.

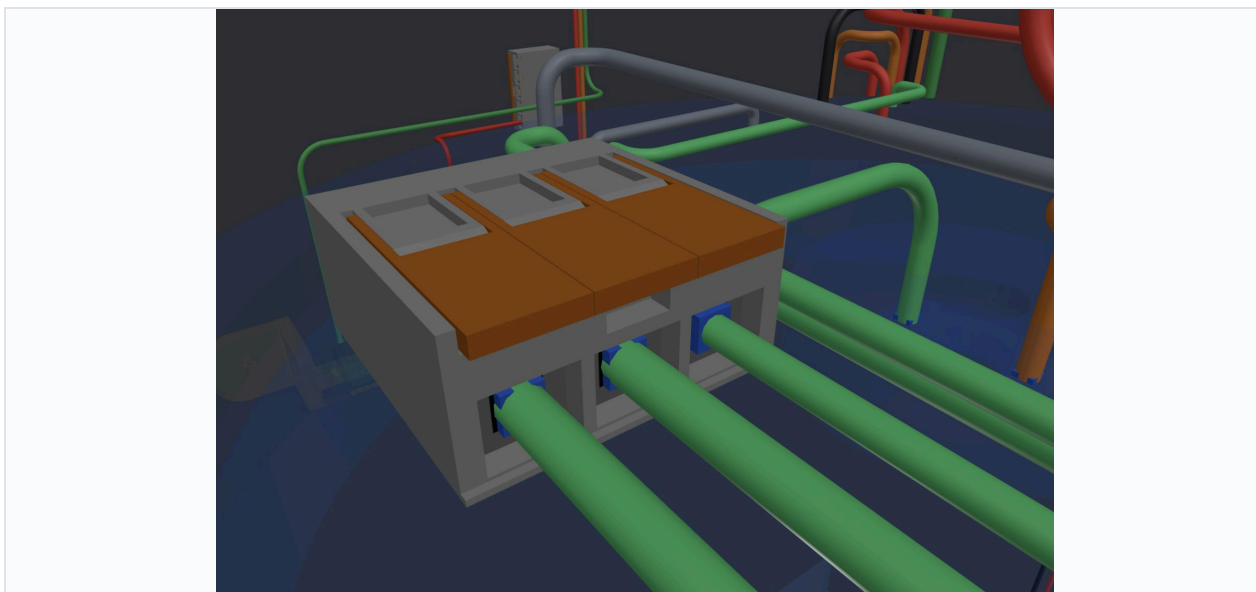


Fig 02.8-3 · 3D Visualizer · wago-led-gnd

02.8.4



Star cathode → ground star, way 1

W.021

Run a GREEN lead from the star's cathode back up through the light ring into way 1 of the ground star. This is the 350 mA return.

HAZARD · It carries the full drive current. Seat it properly — this splice wants a lever on all three, not a hand-twist.

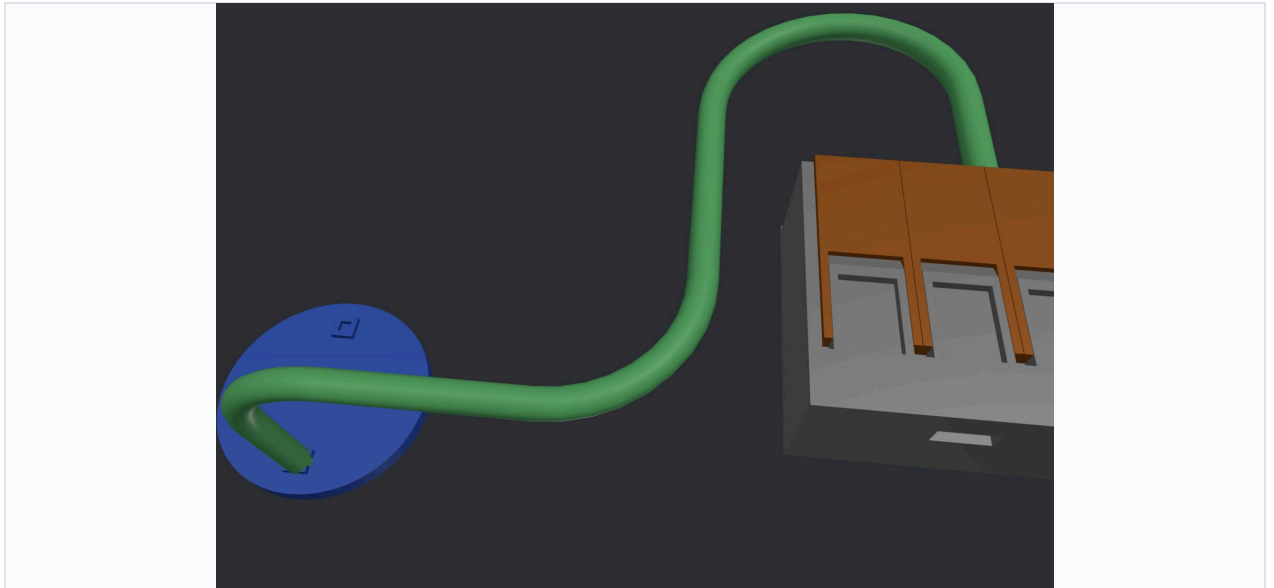


Fig 02.8-4 · 3D Visualizer · W.021

02.8.5



Ground star way 2 → driver LED-

W.022

Run a GREEN lead from way 2 of the ground star to the BuckPuck's LED-. This completes the second half of the return.

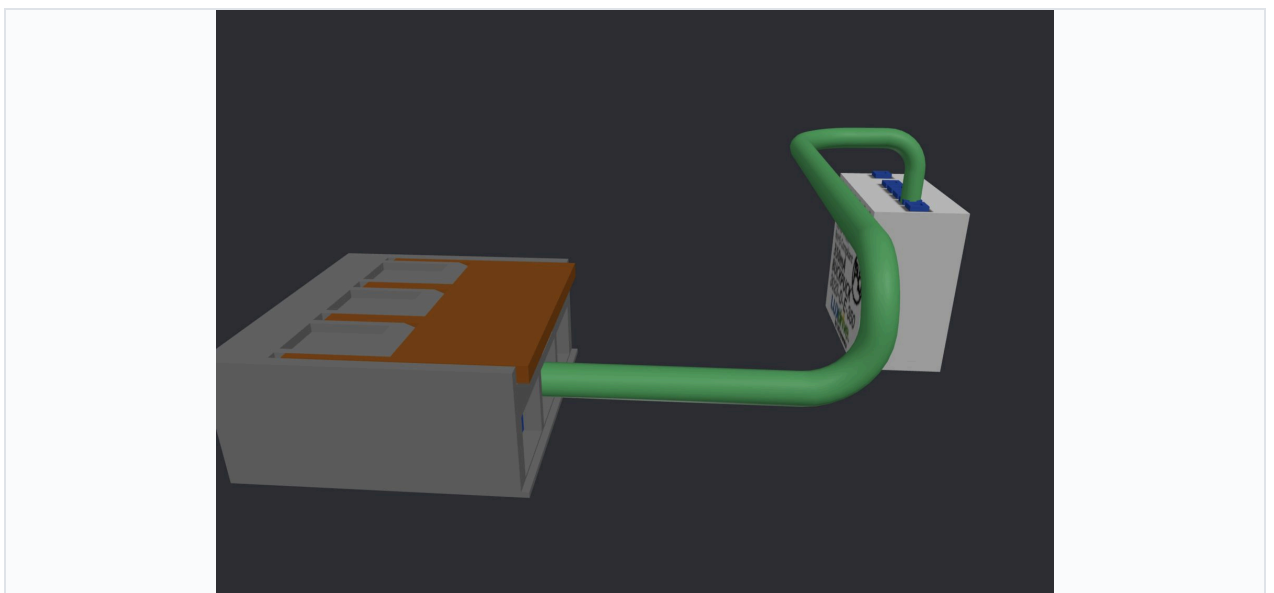


Fig 02.8-5 · 3D Visualizer · W.022

Two wires out of the board's OUT header.

02.8.6



T8 → BuckPuck CTRL

W.013

Run a GRAY lead from T8 on the OUT header to the BuckPuck's CTRL input, carrying the board's collector output to the driver.

HAZARD · Open circuit here = full brightness. This wire failing is not a dark lamp.

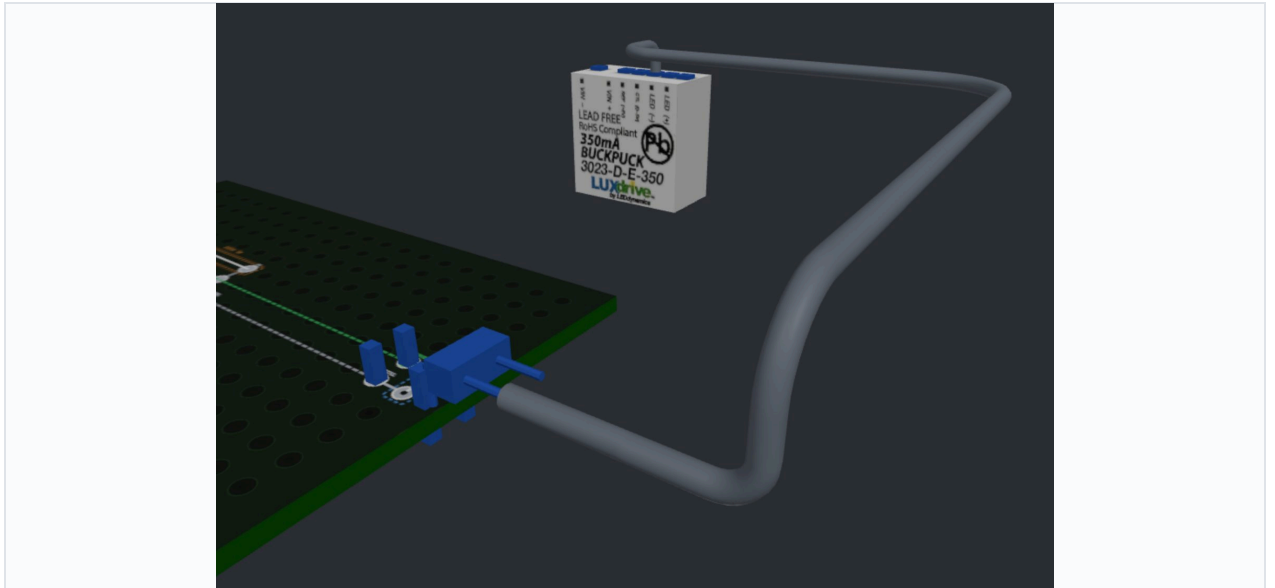


Fig 02.8-6 · 3D Visualizer · W.013

02.8.7



T9 → ground star, way 3

W.014

Run a GREEN lead from T9 to way 3 of the ground star, joining the control-ground reference to the same splice as the emitter return.

NOTE · Shares a node with 350 mA, never a path. That distinction is the whole reason the splice exists.

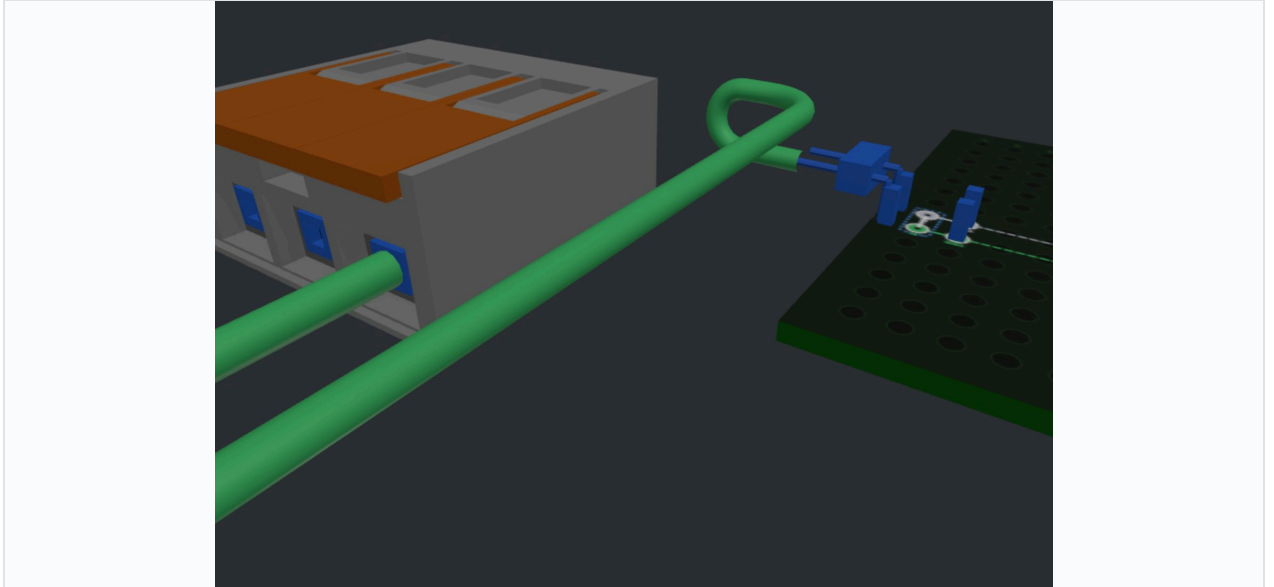


Fig 02.8-7 · 3D Visualizer · W.014

02.9 Into the Machine, Then Power, Then Prove

ONLINE: ACTIVITY 02.9 · 3D VISUALIZER

Three feeds from the GPIO breakout. All of them leave the light circuit through one hole.

02.9.1



Find the bore

light-circuit-hat

Locate the 12 mm bore dead centre in the hat's top plate. All five wires leaving the light circuit pass through it.

NOTE · The other holes in that plate are the board's standoffs. Do not thread anything through those.

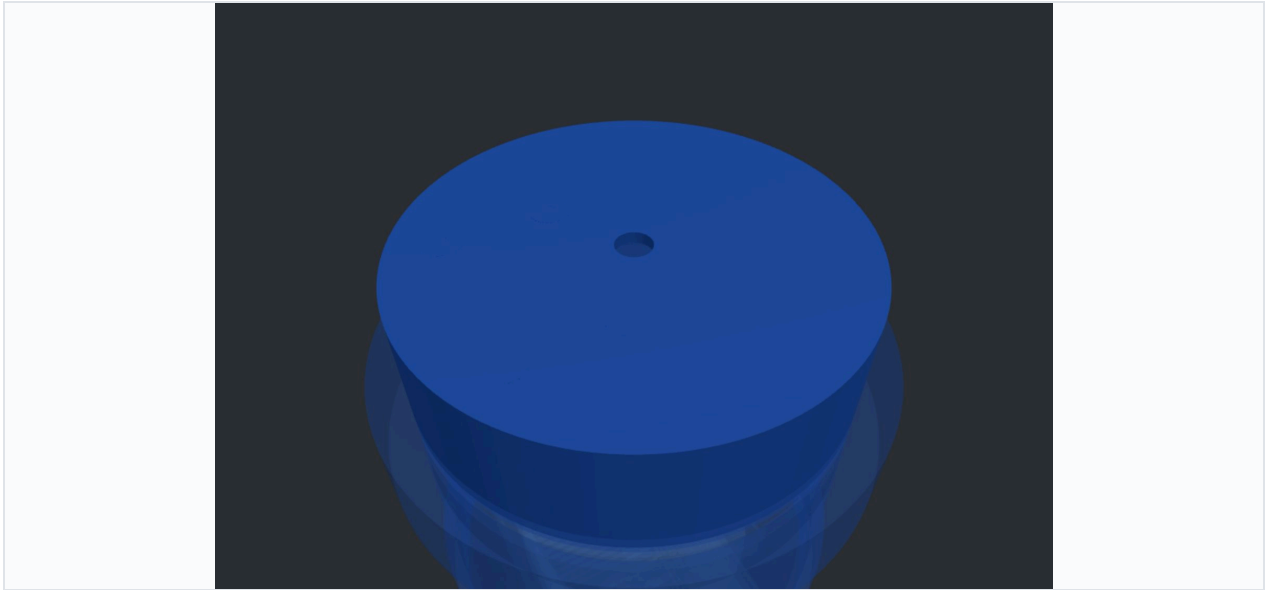


Fig 02.9-1 · 3D Visualizer · light-circuit-hat

02.9.2



+5 V — the breakout end (Pin 2)

W.010

Push the RED lead onto Pin 2 of the GPIO breakout. This is the board's supply, and the only conductor on this run that can source current.

NOTE · Leave 2–3 in of straight off the header before the first bend. A terminal reached by a cable that turns immediately is under permanent tension.

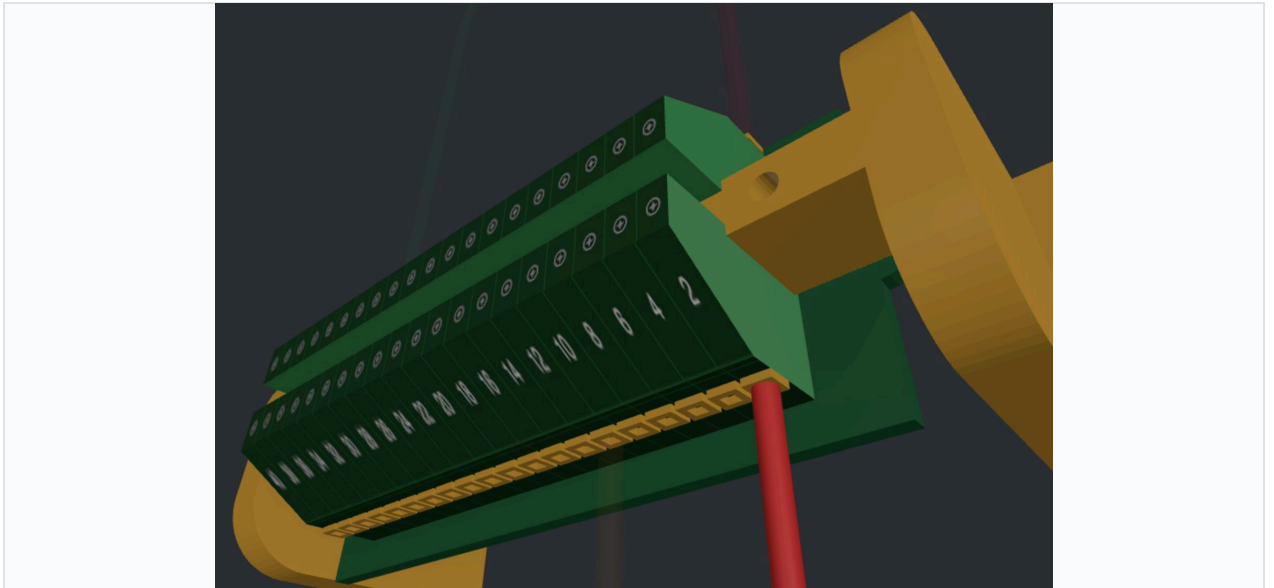


Fig 02.9-2 · 3D Visualizer · W.010

02.9.3



+5 V — the board end (A7)

W.010

Land the other end in A7 on the IN header. A7 is the board's +5 V pad, and it bridges to B7 where the +5 V jumper starts.

NOTE · The run goes up the left wall, across the ceiling and down through the hat's bore. All three Pi runs share that lane — dress them together.

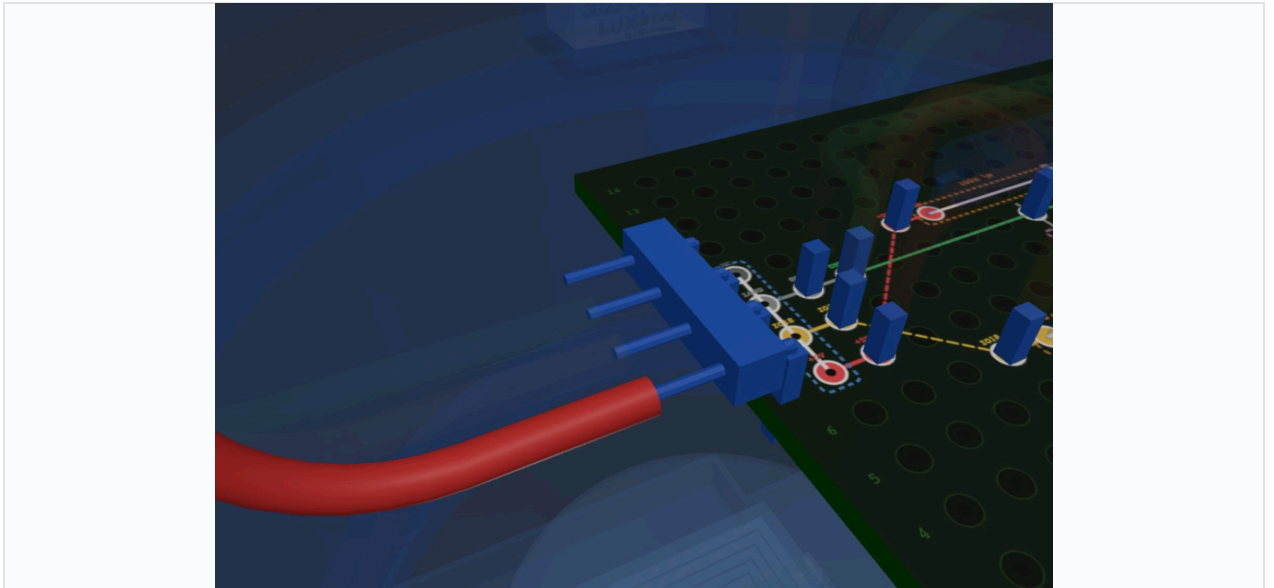


Fig 02.9-3 · 3D Visualizer · W.010

02.9.4



GPIO18 — the breakout end (Pin 12)

W.011

Push the ORANGE lead onto Pin 12 of the GPIO breakout. GPIO18 is the hardware-PWM pin: this is the wire the duty cycle travels down.

HAZARD · Meter the breakout before you trust its labels. A ribbon fitted 180° out mirrors all 40 pins, and + 5 V arriving on a 3.3 V input will not announce itself.

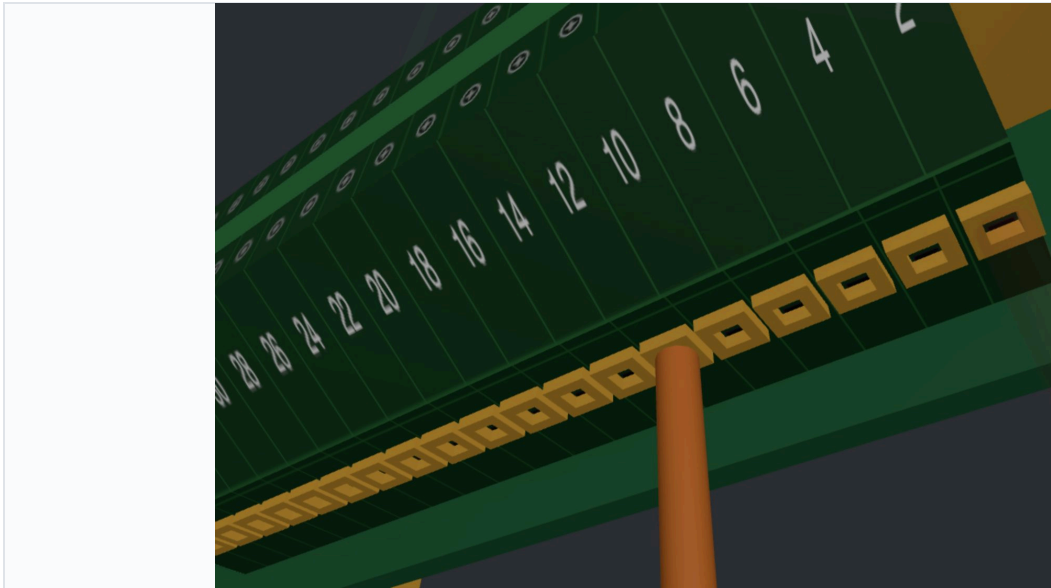


Fig 02.9-4 · 3D Visualizer · W.011

02.9.5



GPIO18 — the board end (A8)

W.011

Land the other end in A8, in its own lane in the bundle. A8 bridges to B8, and the orange jumper carries it from there to R1.

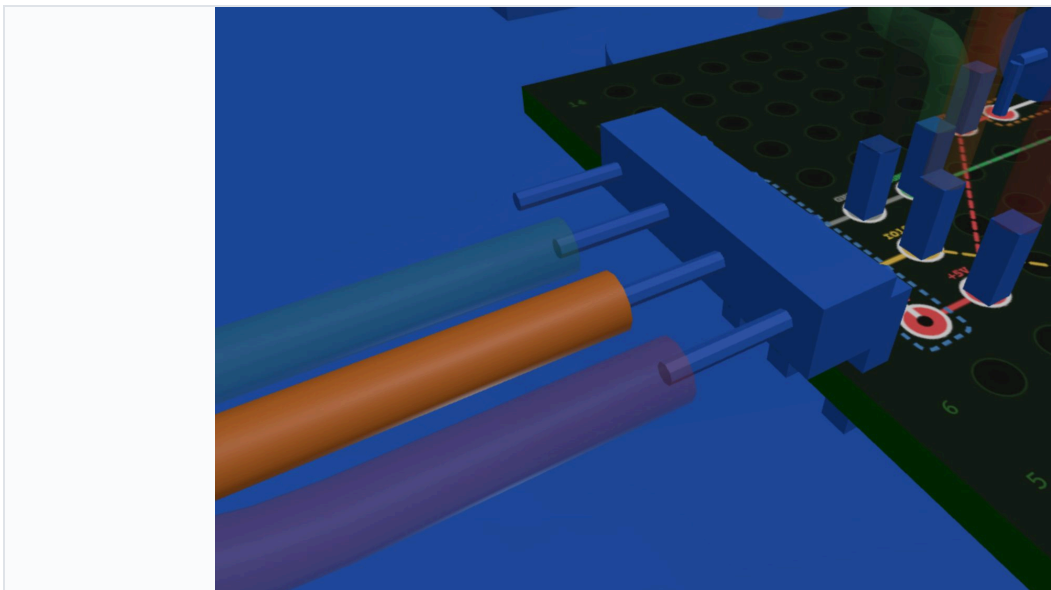


Fig 02.9-5 · 3D Visualizer · W.011

02.9.6



Logic ground — the breakout end (Pin 39)

W.012

Push the GREEN lead onto Pin 39. This is the reference the whole control circuit is measured against.

NOTE · Pin 39 is a top-row pin, so it leaves upward. Pins 2 and 12 are bottom row and leave downward — approach an even pin from below, never across the top row.

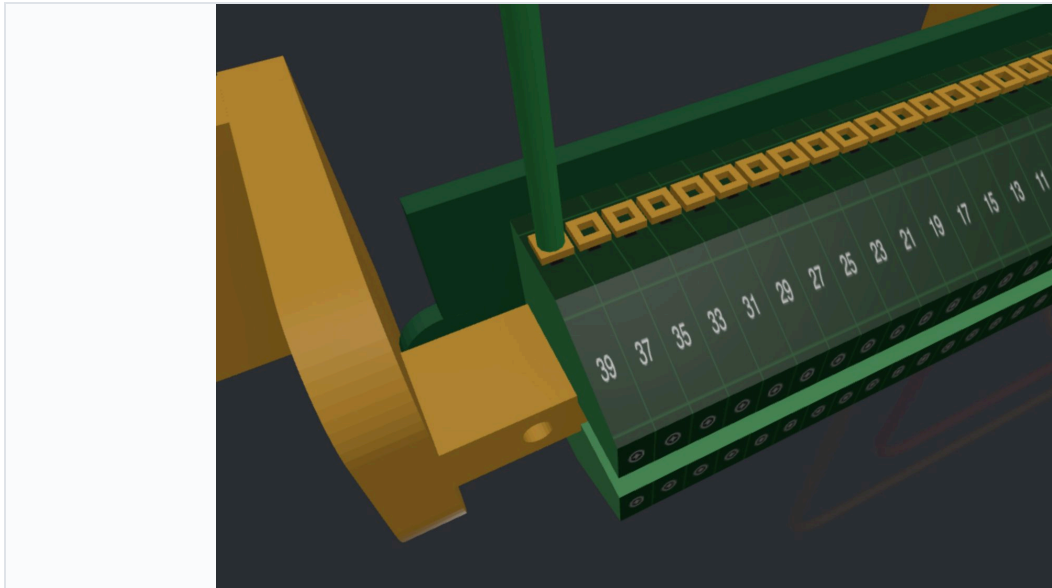


Fig 02.9-6 · 3D Visualizer · W.012

02.9.7



Logic ground — the board end (A9)

W.012

Land the other end in A9. A9 bridges to B9, which feeds both the ground jumper and the control-ground reference leaving on the OUT header.

NOTE · Without this one lead the board has no common with the Pi and nothing switches. It does not half-work — it does not work.

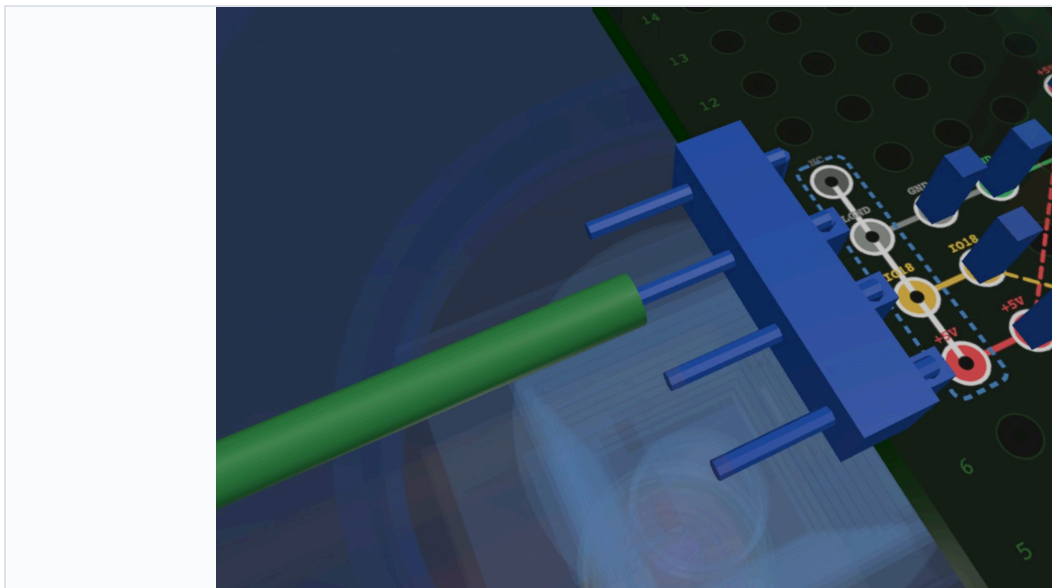


Fig 02.9-7 · 3D Visualizer · W.012

02.9.8



Clip the bundle to the enclosure

cable-clip-01

Clip the bundle to the enclosure: one clip a short way off each bend, and one at the middle of each straight between them.

NOTE · A clip **ON** a corner fights the cable's own spring and works its screw loose. Beside the corner, it takes the load in shear.

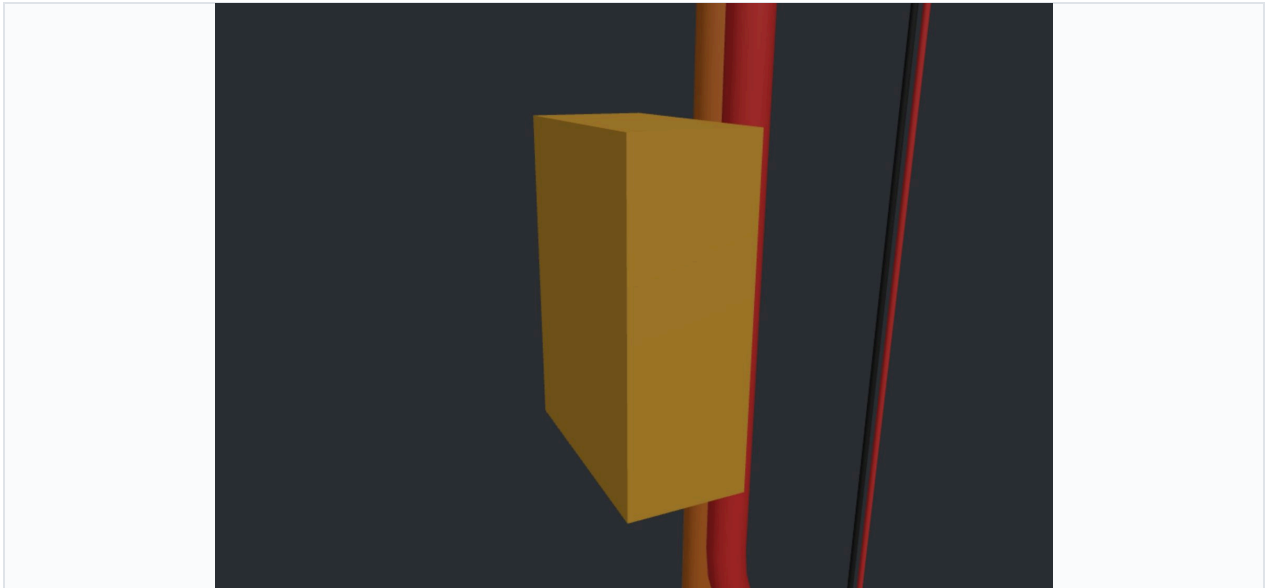


Fig 02.9-8 · 3D Visualizer · cable-clip-01

Nothing here is energised until the final step, and the two gates sit upstream of the driver.

02.9.9



Know the supply before you plug it in

psu15

Check the supply before you plug anything in: PSU15, +15 V, ≥ 0.75 A, centre-positive. It is the only part of the device that lives outside the enclosure. 15 V is a choice, not just a legal value: the 3-series string is ~ 10.5 V plus ~ 2 V of BuckPuck dropout, so the floor is ~ 12.5 V, and 24 V burns ~ 5 W as heat where 15 V burns ~ 2 W for the same light.

HAZARD · Check polarity before it goes near the jack. The driver runs on 13–24 V, so a substituted brick can be within spec for the driver and badly wrong for a resistor chosen at 15 V.

NOTE · Leave it unplugged. It is not energised until the last step of this section.

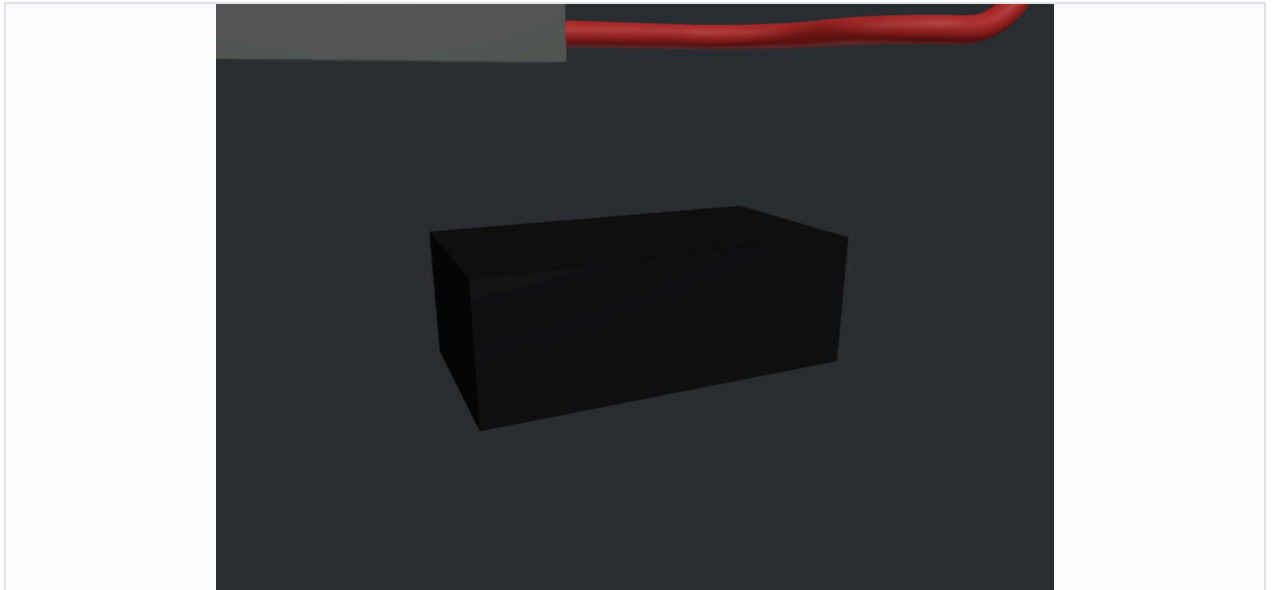


Fig 02.9-9 · 3D Visualizer · psu15

02.9.10



Fit the inlet

barrel-jack

Mount the barrel jack inside its bushing in the back panel, so the supply plug pushes in from outside without opening the cabinet.

NOTE · This is the breakaway point — the one connection meant to be pulled. The bushing takes the tug, not the jack's tails.



Fig 02.9-10 · 3D Visualizer · barrel-jack

02.9.11



Jack + → inlet splice

W.019

Place the RED lead from the jack into a 2-way WAGO. The wire should sit ~51 mm off the floor, at the back.

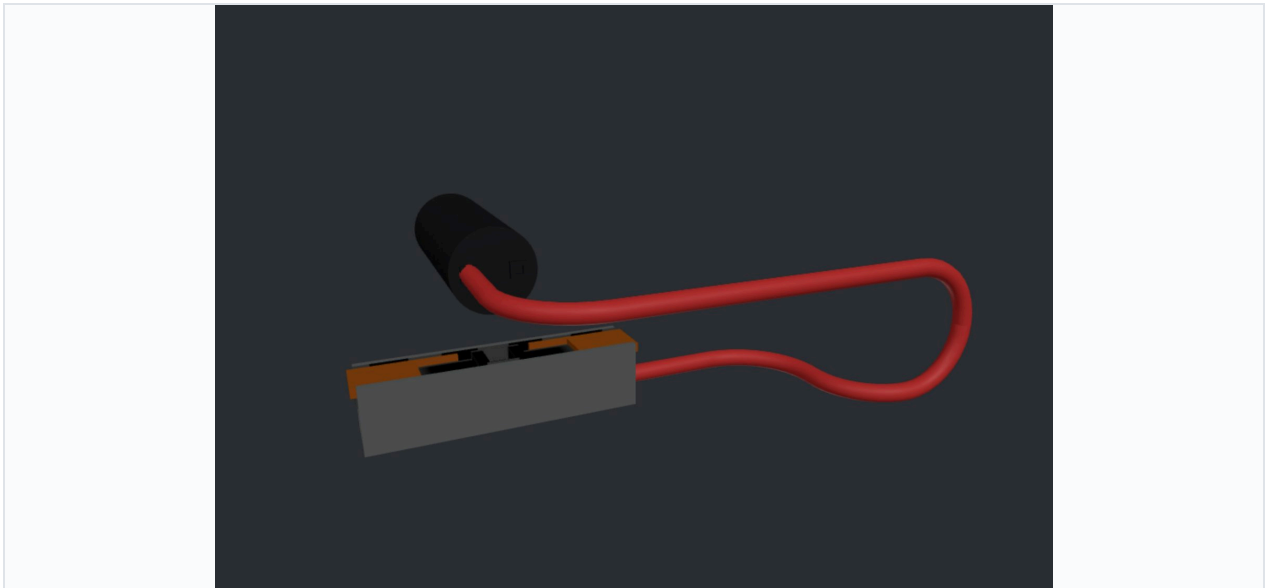


Fig 02.9-11 · 3D Visualizer · W.019

☐

W.018

NOTE · Up the back wall and along the ceiling from here. This is the longest single run in the device — measure it in place before you cut.

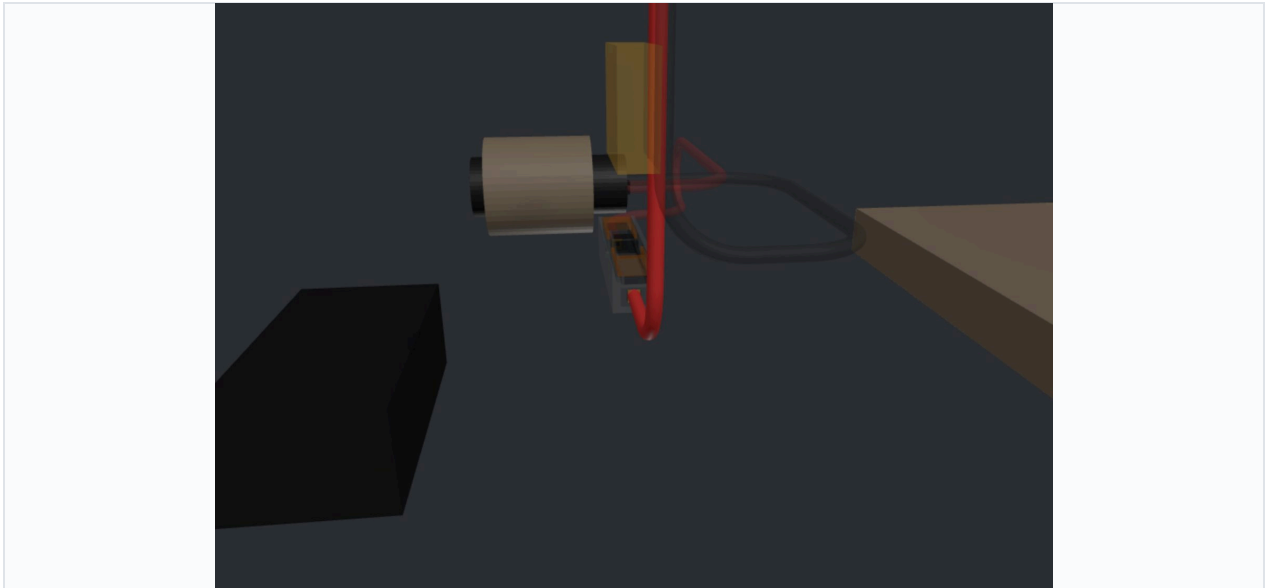


Fig 02.9-12 · 3D Visualizer · W.018

02.9.13



Supply rail — the SW1 end (upper lug)

W.018

Land the other end on SW1's upper lug — the rail side. Bring it in through the ceiling of the switchbox so lifting the panel does not strain the joint.

NOTE · It shares one hole with the run coming back down, 12 mm apart. Two holes would be two chances to leak light.

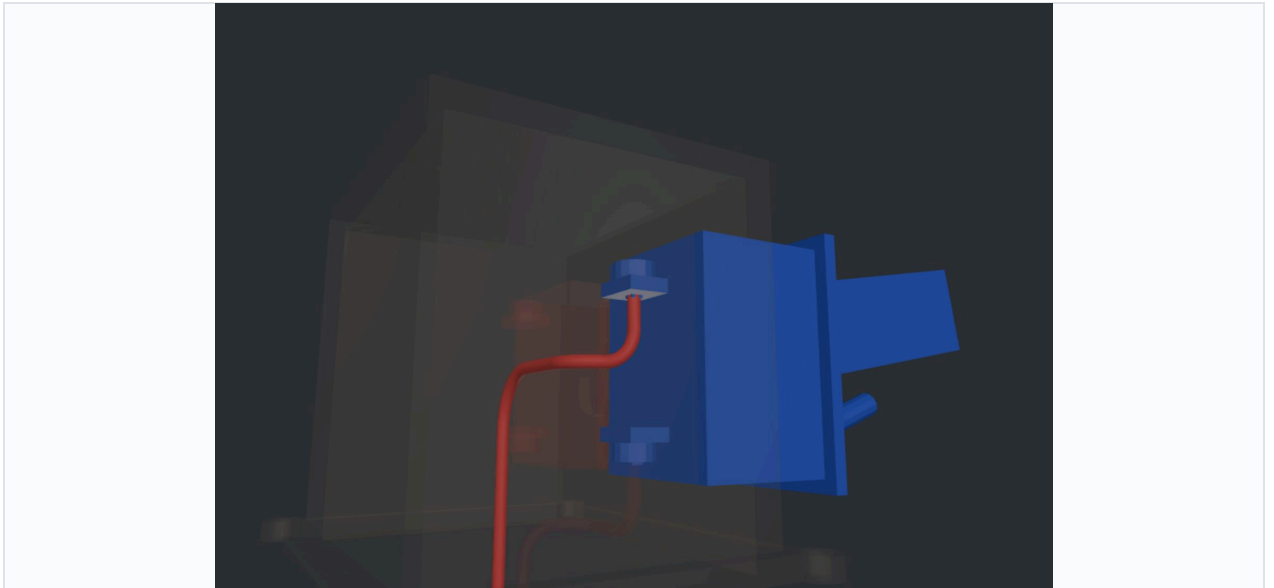


Fig 02.9-13 · 3D Visualizer · W.018

02.9.14



SW1 lower lug → door interlock terminal 3

W.017

Run a RED lead from SW1's lower lug to terminal 3 of the door interlock — back down through the ceiling and along to the interlock.

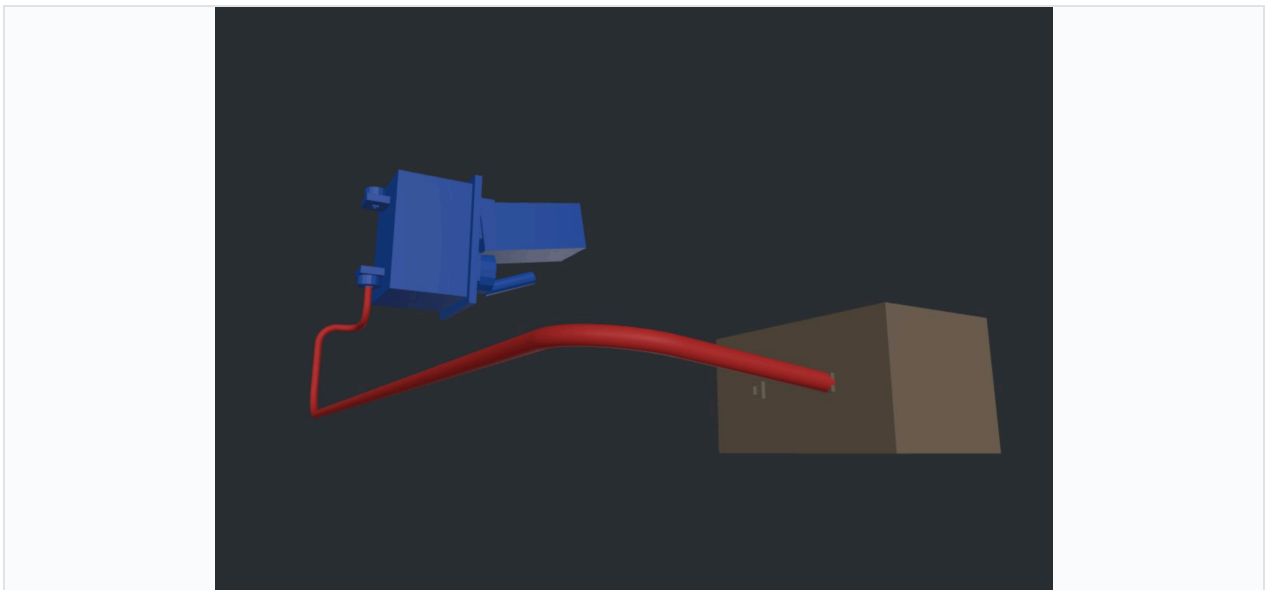


Fig 02.9-14 · 3D Visualizer · W.017

02.9.15



Gated rail — the interlock end (terminal 4)

W.015

Land the RED lead on terminal 4 of the door interlock — the NO pair.

HAZARD · On terminals 1 and 2 instead — the NC pair — it does the exact inverse: dark with the door shut, live with it open. It would pass any casual test.

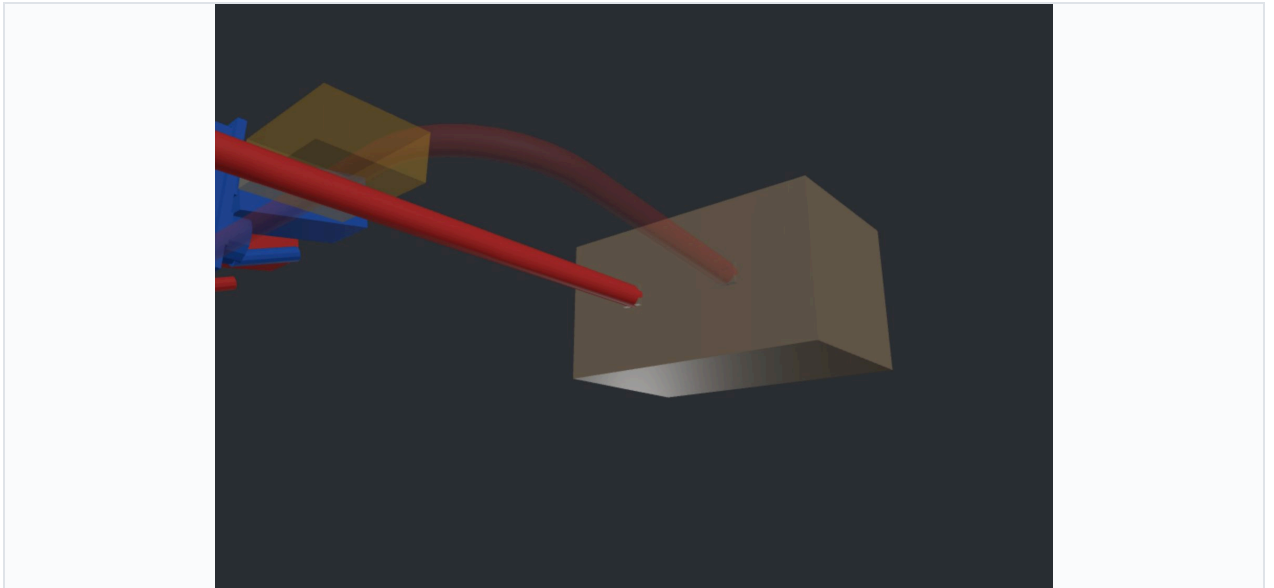


Fig 02.9-15 · 3D Visualizer · W.015

02.9.16



Gated rail — the BuckPuck end (Vin+)

W.015

Land the other end on the BuckPuck's Vin+. This is the last leg of the gated supply: everything upstream of this terminal is switched, everything downstream is the driver's to manage.

NOTE · Both gates are now upstream of the driver. An open circuit anywhere in this leg leaves the rail dark, which is the correct direction to fail.



Fig 02.9-16 · 3D Visualizer · W.015

02.9.17



Return — the BuckPuck end (Vin-)

W.016

Land the BLACK lead on the BuckPuck's Vin-.

HAZARD · The return is **NOT** switched. Cutting the rail at SW1 or at the door does not isolate this conductor — it is referenced to the supply whenever the brick is plugged in.



Fig 02.9-17 · 3D Visualizer · W.016

02.9.18



Return — the jack end (- sleeve)

W.016

Land the other end on the jack's - sleeve terminal. It runs beside the red for the whole drop.

NOTE · Beside, not twisted with. The pair is easier to trace and easier to replace when each lead keeps its own path.

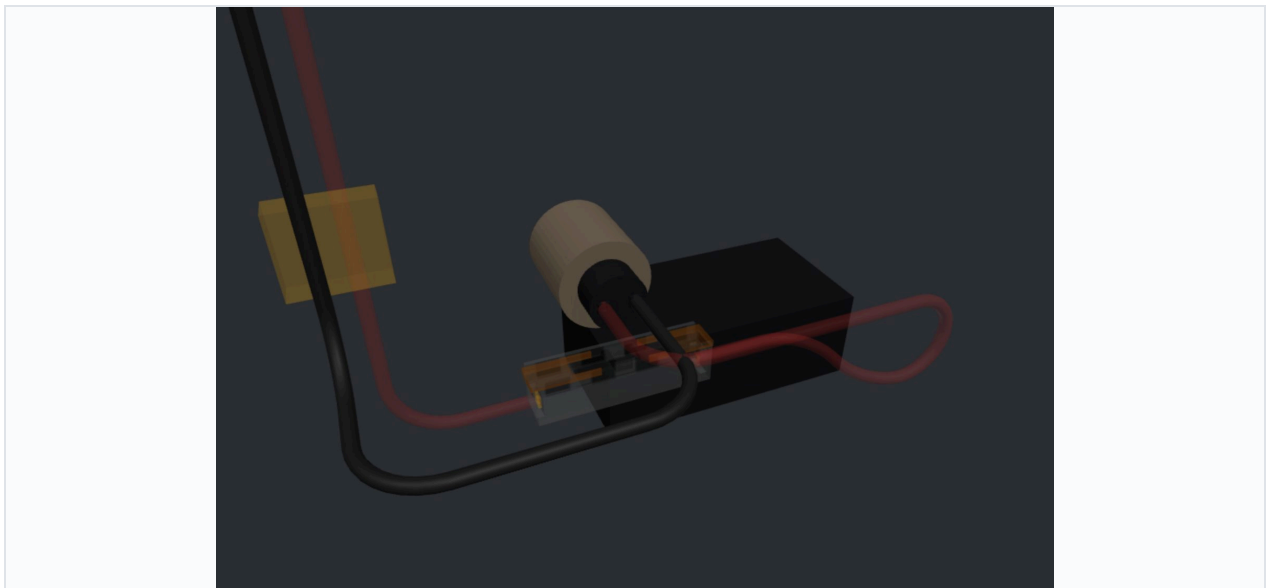


Fig 02.9-18 · 3D Visualizer · W.016

Two checks before the first power, in this order. Both must fail dark.

02.9.19

Door open must be dark



With SW1 closed and the brick plugged in, open the door. Confirm the emitter goes out.

HAZARD · If it stays lit, the interlock is on the NC pair. Stop and rewire it to 3 and 4.

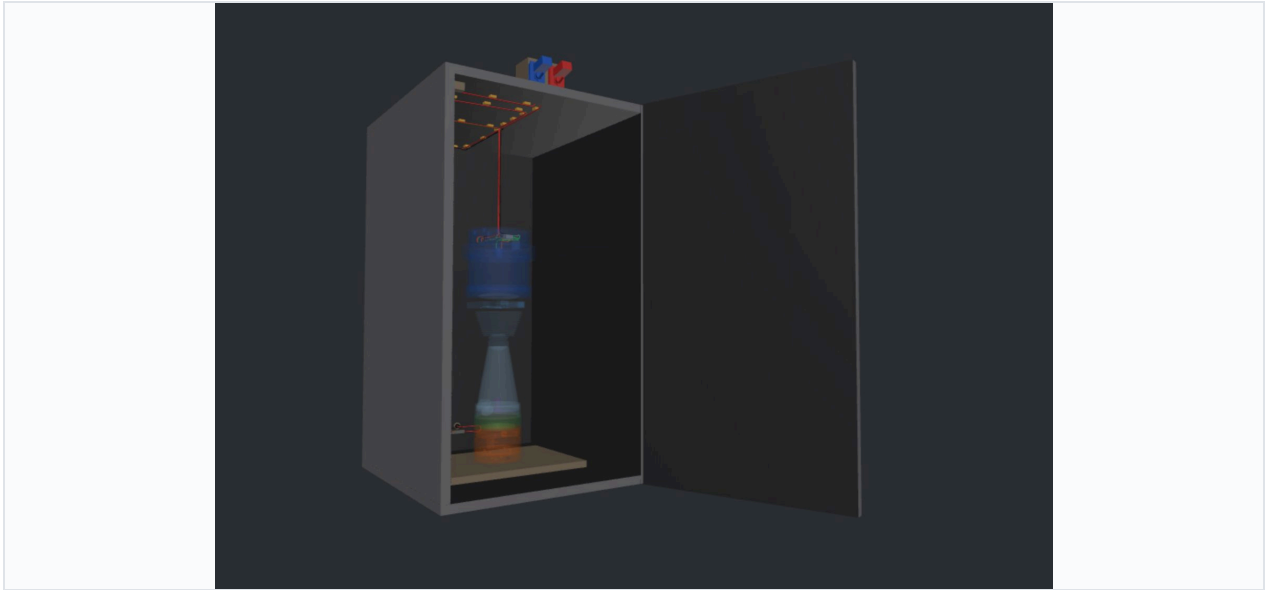


Fig 02.9-19 · 3D Visualizer ·

02.9.20

SW1 off must be dark



sw1-light

With the door shut and the brick in, open SW1. Confirm the emitter is out.

NOTE · Only once both gates have been shown to work is it reasonable to let the Pi drive CTRL for the first time.

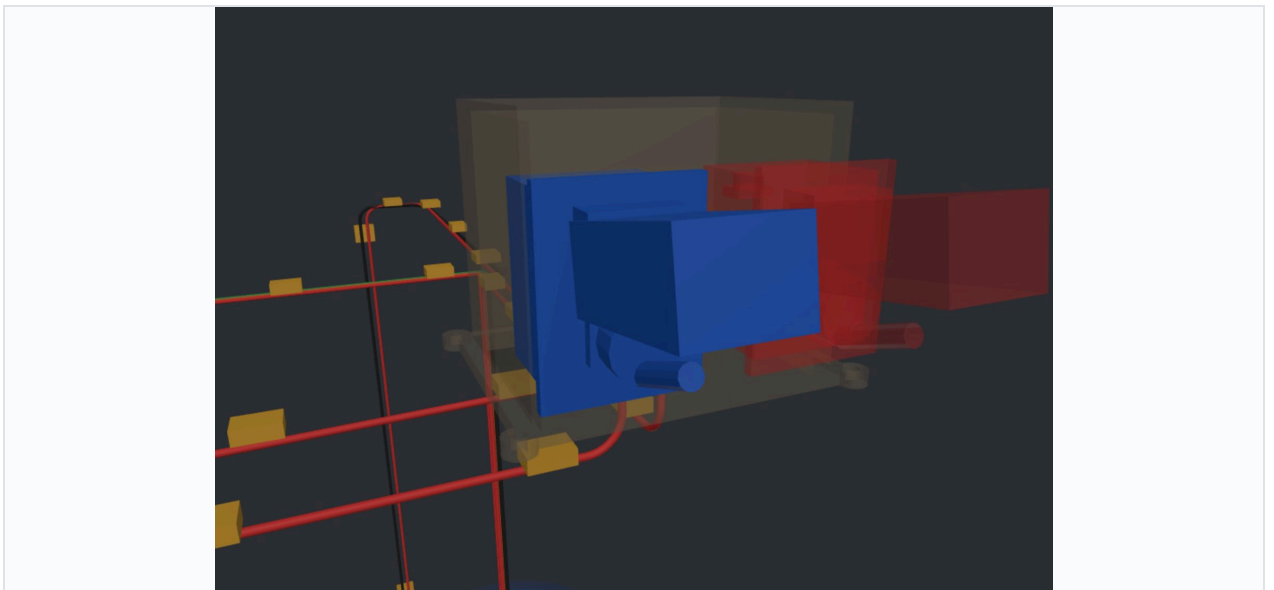


Fig 02.9-20 · 3D Visualizer · sw1-light

On cable management

This guide gives you no cut lengths, on purpose. Leave a service loop where a run crosses a hinge, clip a bundle where it would otherwise hang on a connector, and dress it so the door can open through it without pulling. Look at how Photoplasma 1.0 was wired: those decisions were made, lived with, and are worth copying or improving on.

Stop here

Both checks above must fail dark before the Pi is allowed to drive CTRL for the first time. If the emitter stays lit on either, the interlock is on the wrong contact pair. Stop and rewire it.